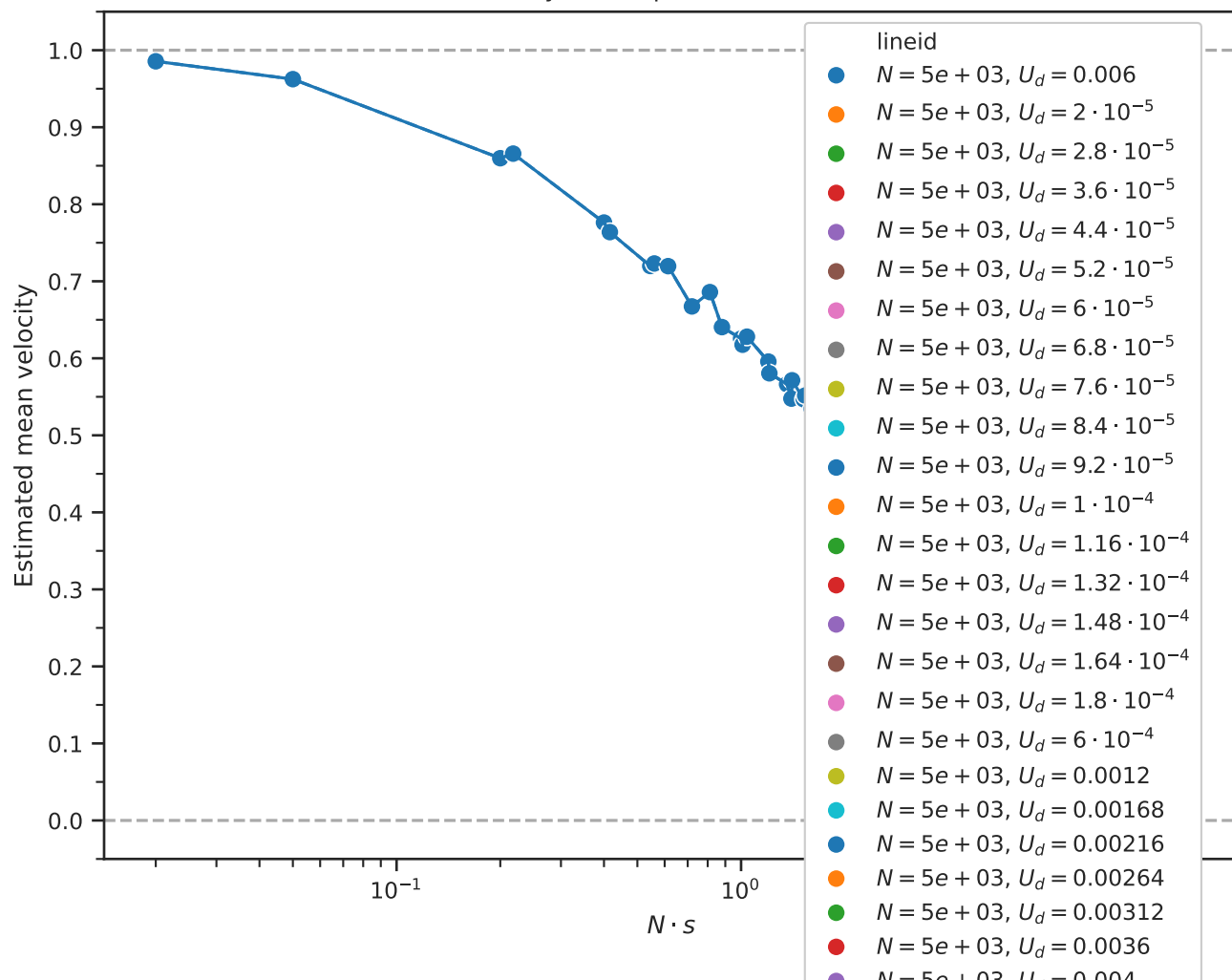
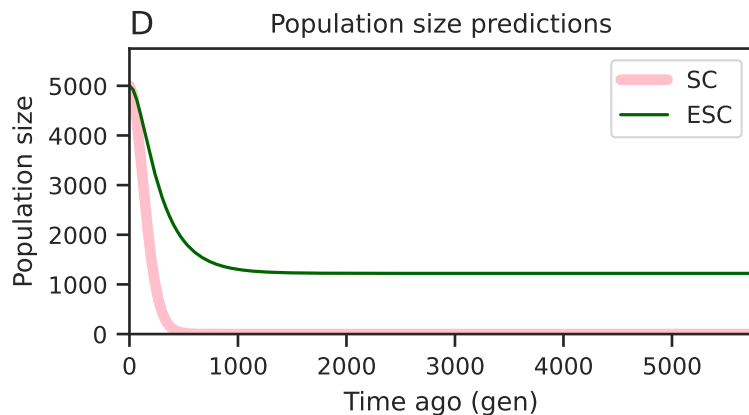
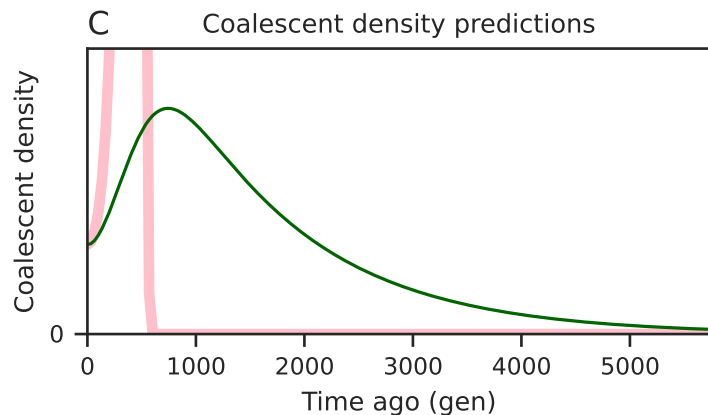
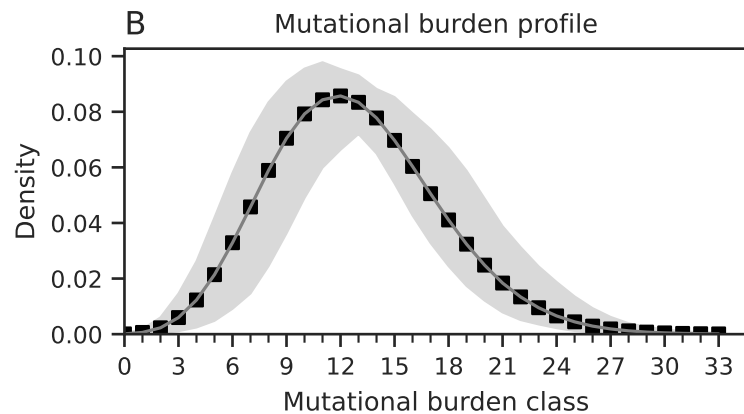
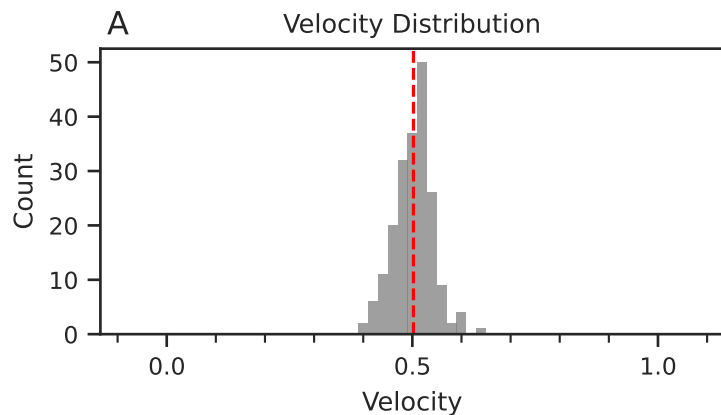


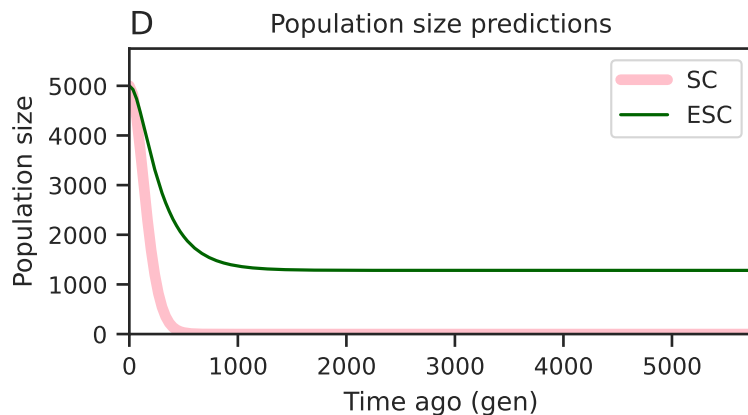
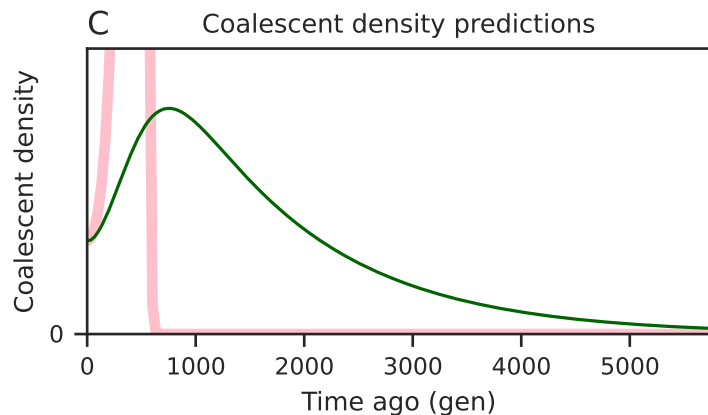
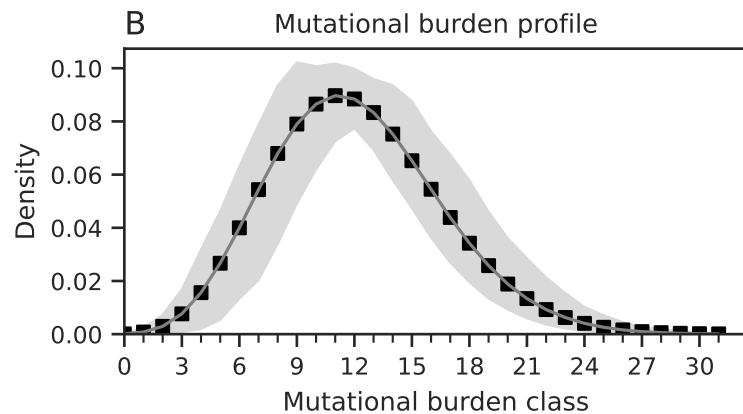
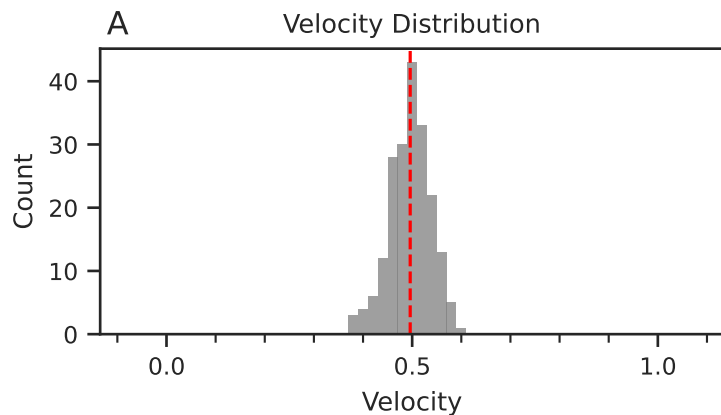
Mean velocity across parameter combinations



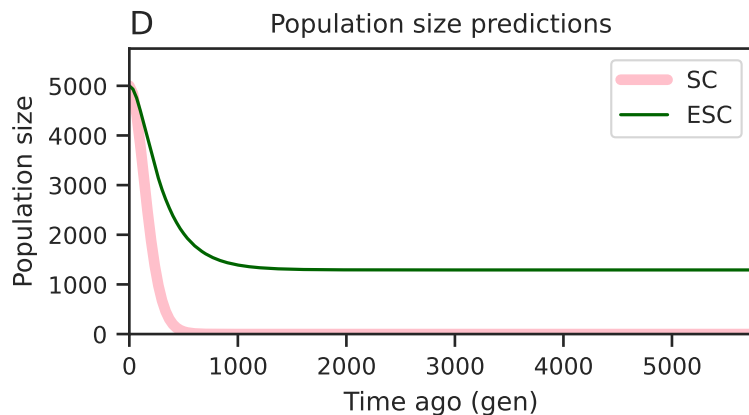
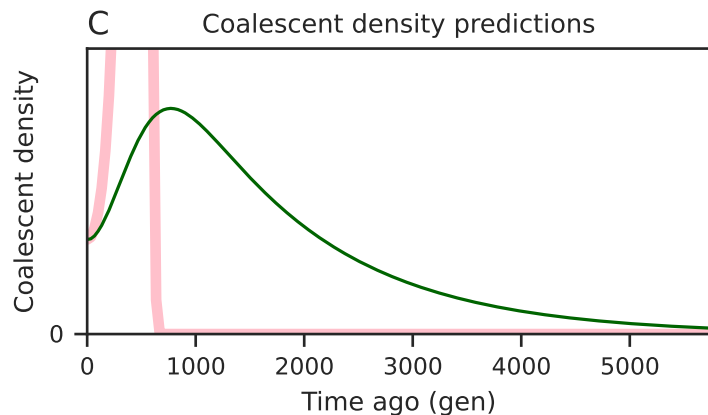
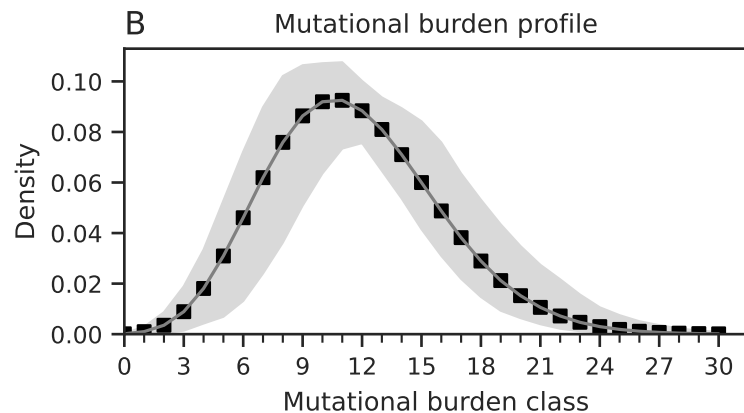
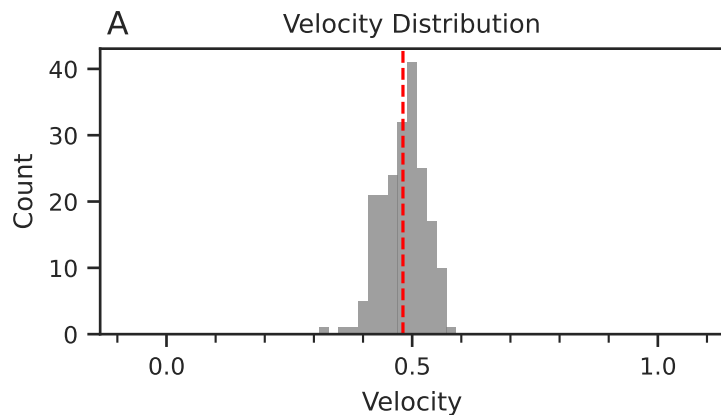
$N = 5e + 03$, $s = 0.001$, $U_d = 0.036$, $\phi = 1.16 \cdot 10^{-15}$
Mean Velocity= 0.5022 ± 0.0393 , $n = 200$ simulations



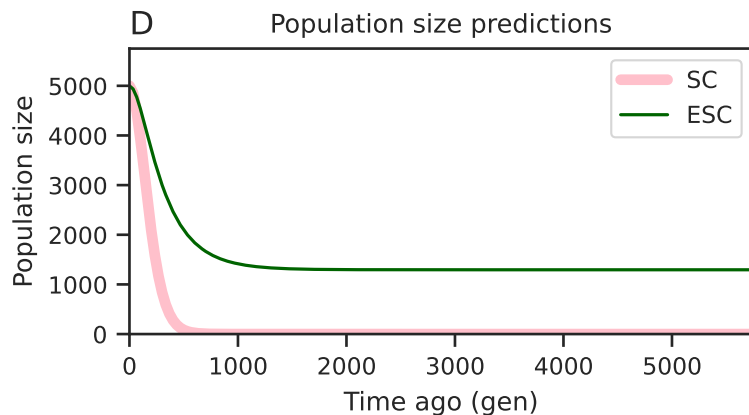
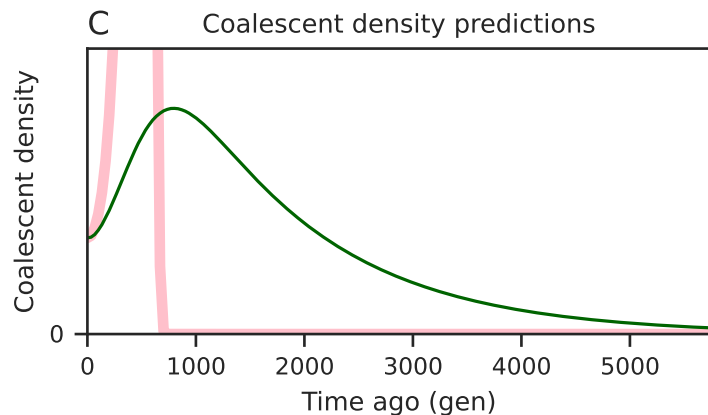
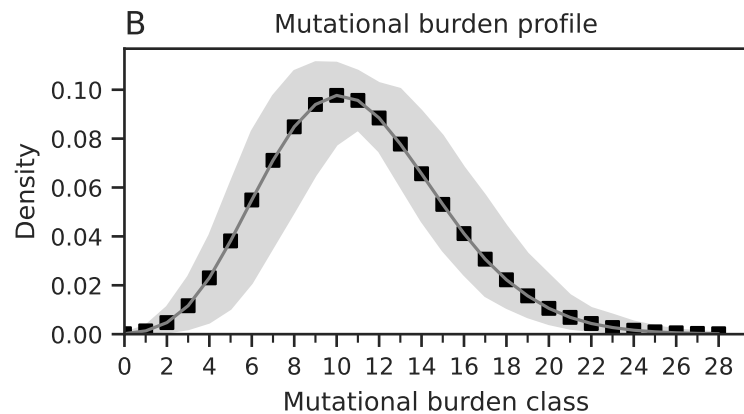
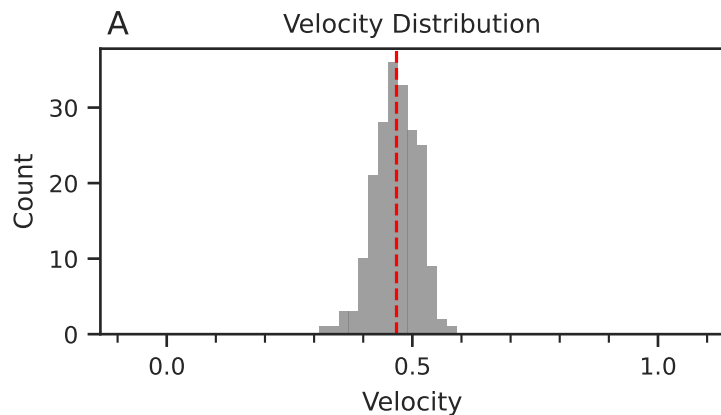
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0328$, $\phi = 2.85 \cdot 10^{-14}$
Mean Velocity= 0.4960 ± 0.0412 , $n = 200$ simulations



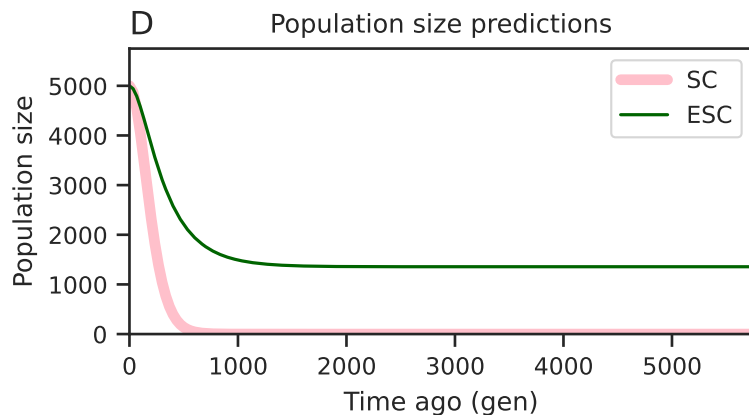
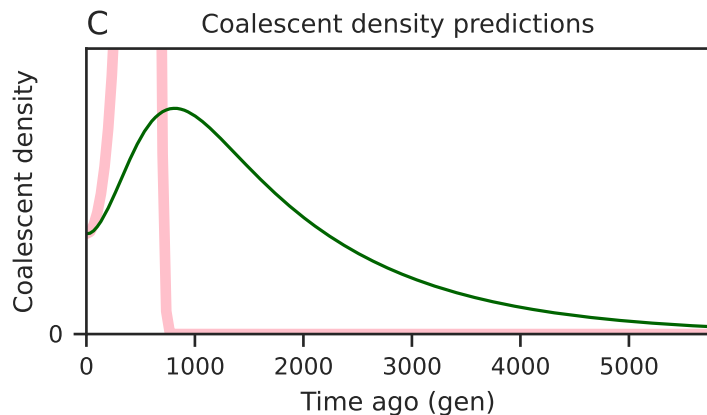
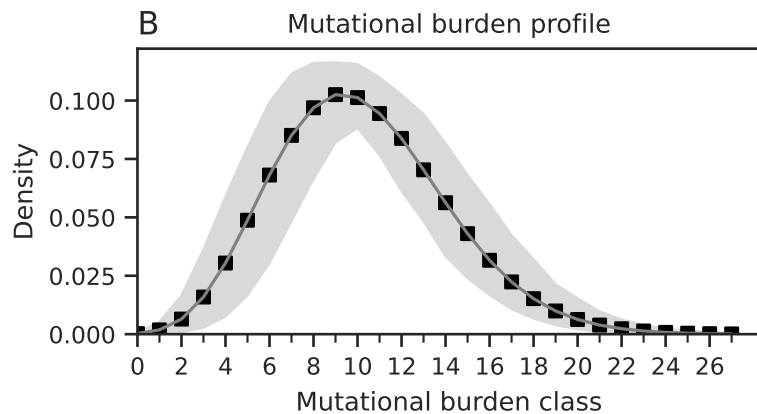
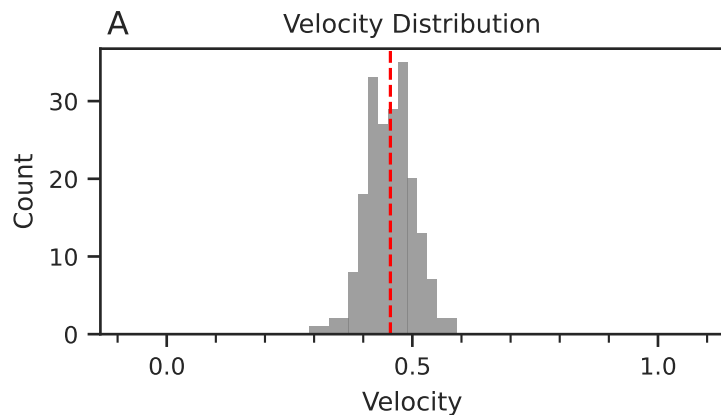
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0296$, $\phi = 6.98 \cdot 10^{-13}$
Mean Velocity= 0.4813 ± 0.0440 , $n = 200$ simulations



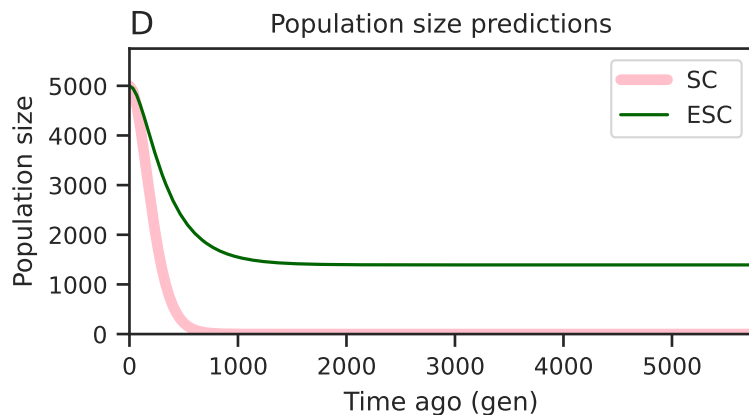
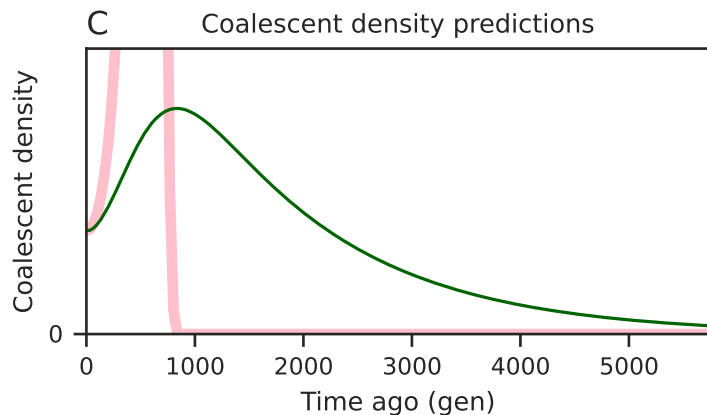
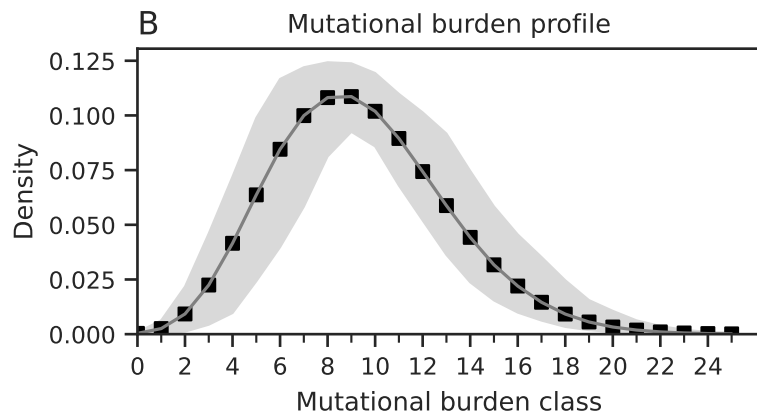
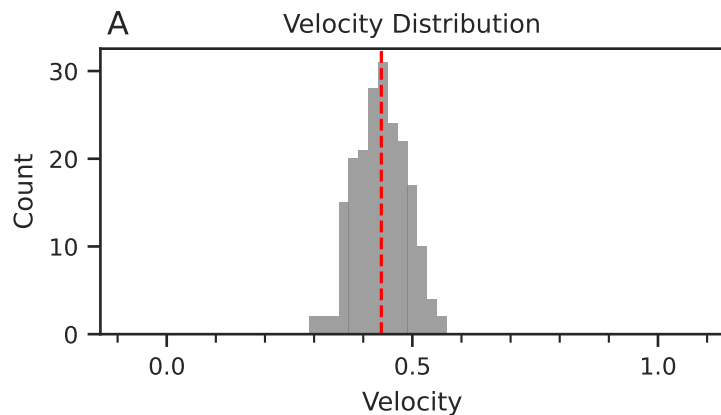
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0264$, $\phi = 1.71 \cdot 10^{-11}$
Mean Velocity= 0.4677 ± 0.0434 , $n = 200$ simulations



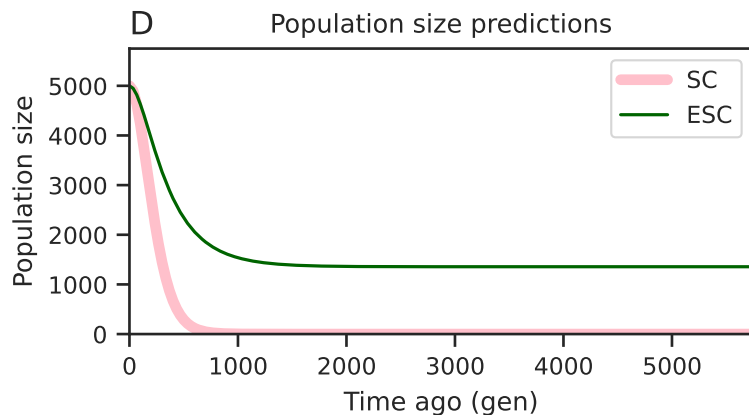
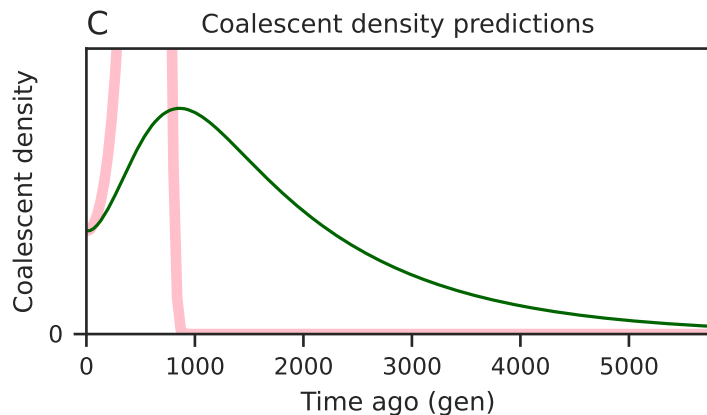
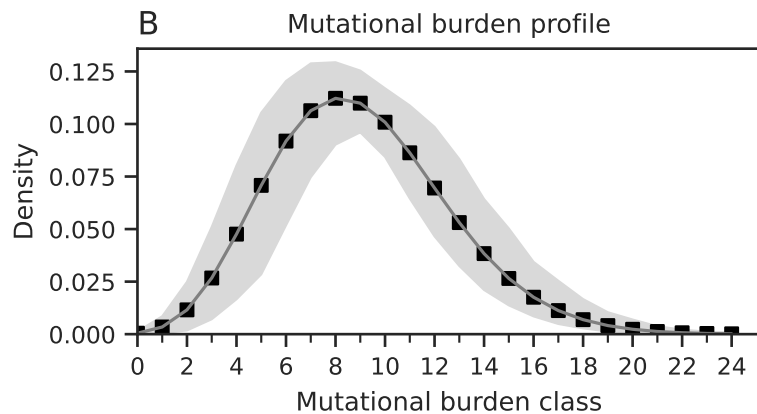
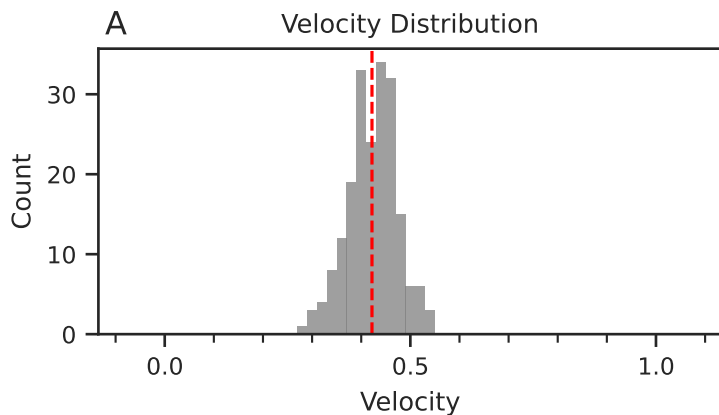
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0232$, $\phi = 4.2 \cdot 10^{-10}$
Mean Velocity= 0.4553 ± 0.0473 , $n = 200$ simulations



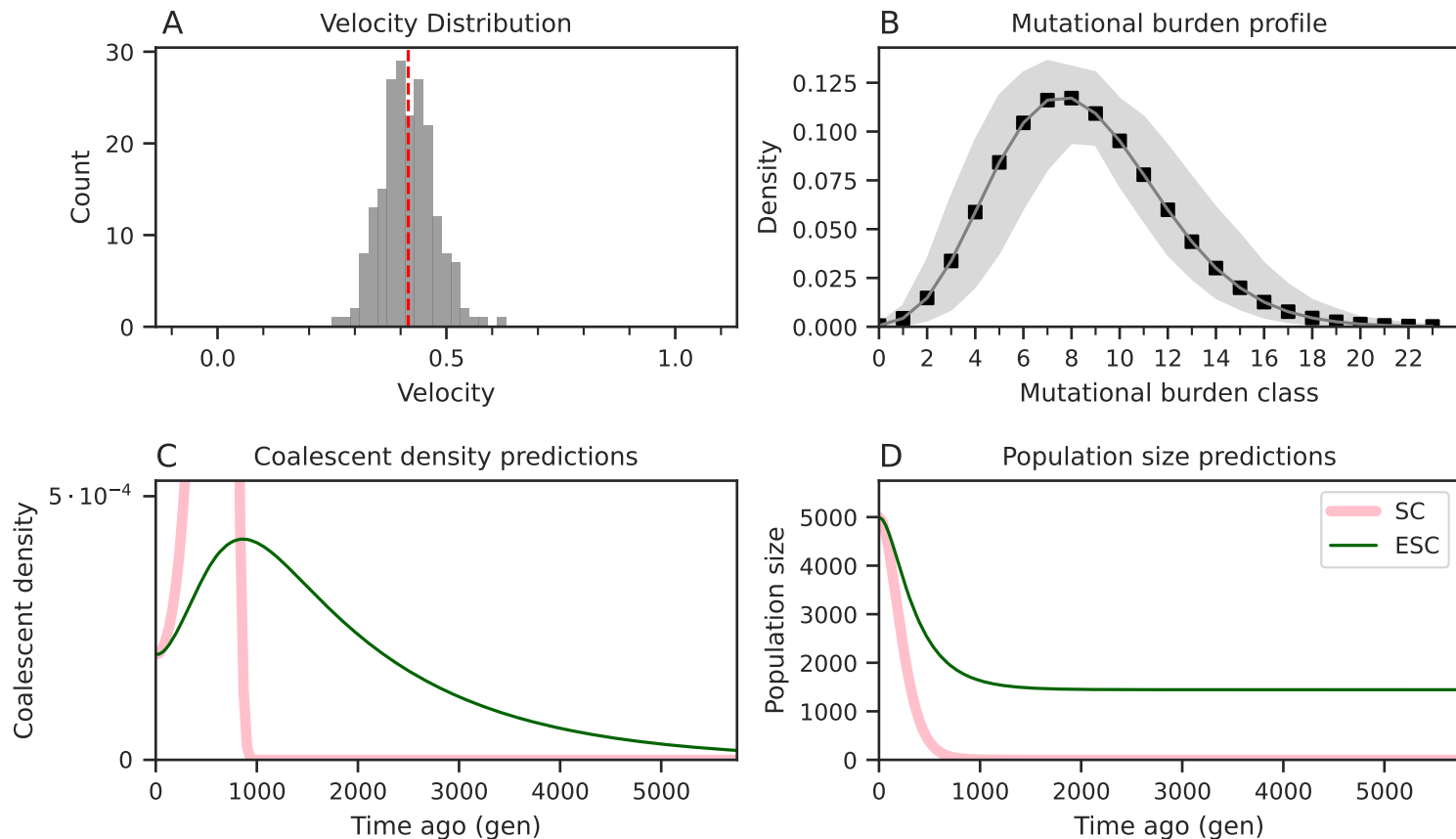
$N = 5e + 03$, $s = 0.001$, $U_d = 0.02$, $\phi = 1.03 \cdot 10^{-8}$
Mean Velocity= 0.4369 ± 0.0516 , $n = 200$ simulations



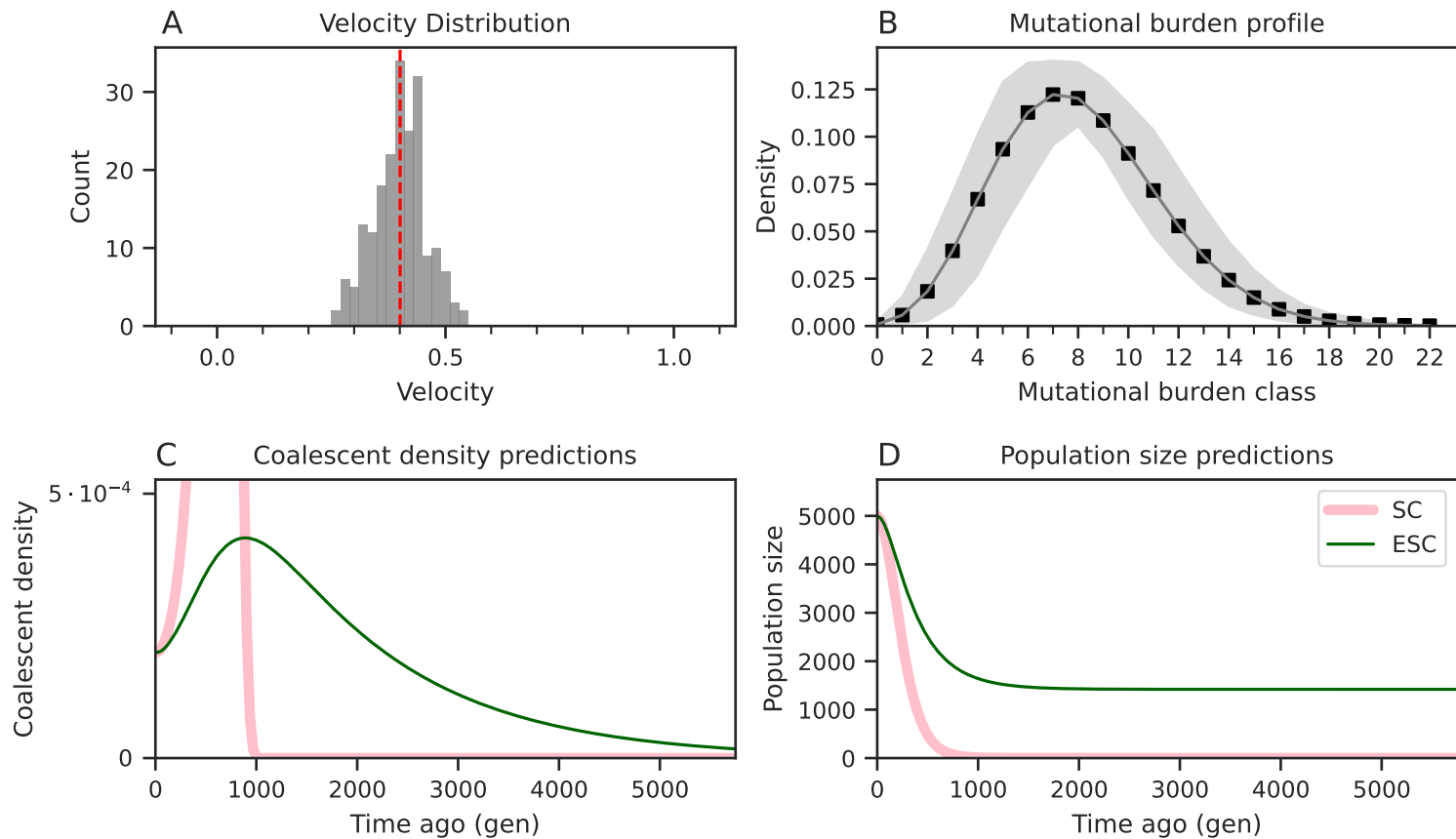
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0184$, $\phi = 5.1 \cdot 10^{-8}$
Mean Velocity= 0.4219 ± 0.0498 , $n = 200$ simulations



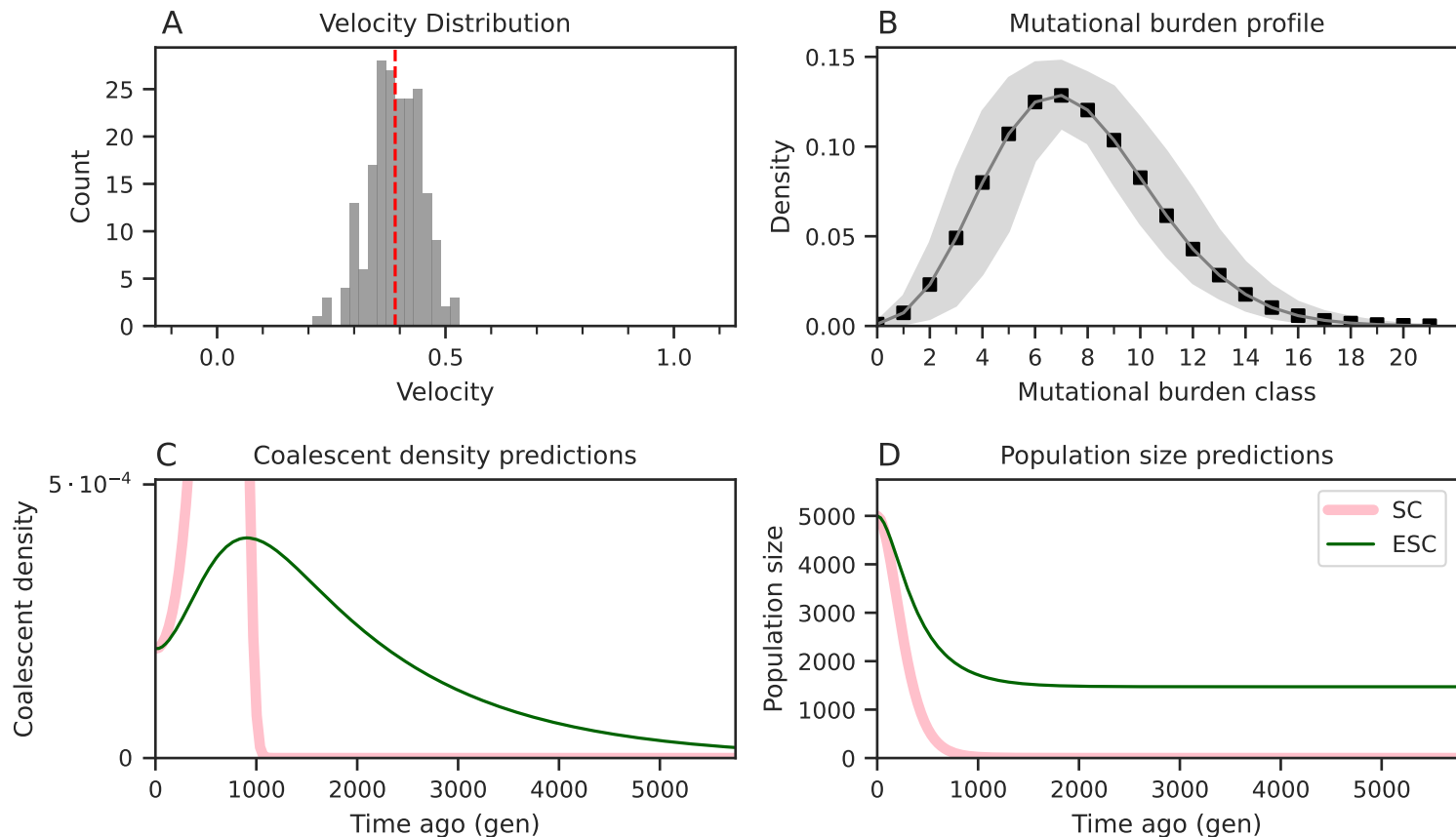
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0168$, $\phi = 2.53 \cdot 10^{-7}$
Mean Velocity= 0.4167 ± 0.0571 , $n = 200$ simulations



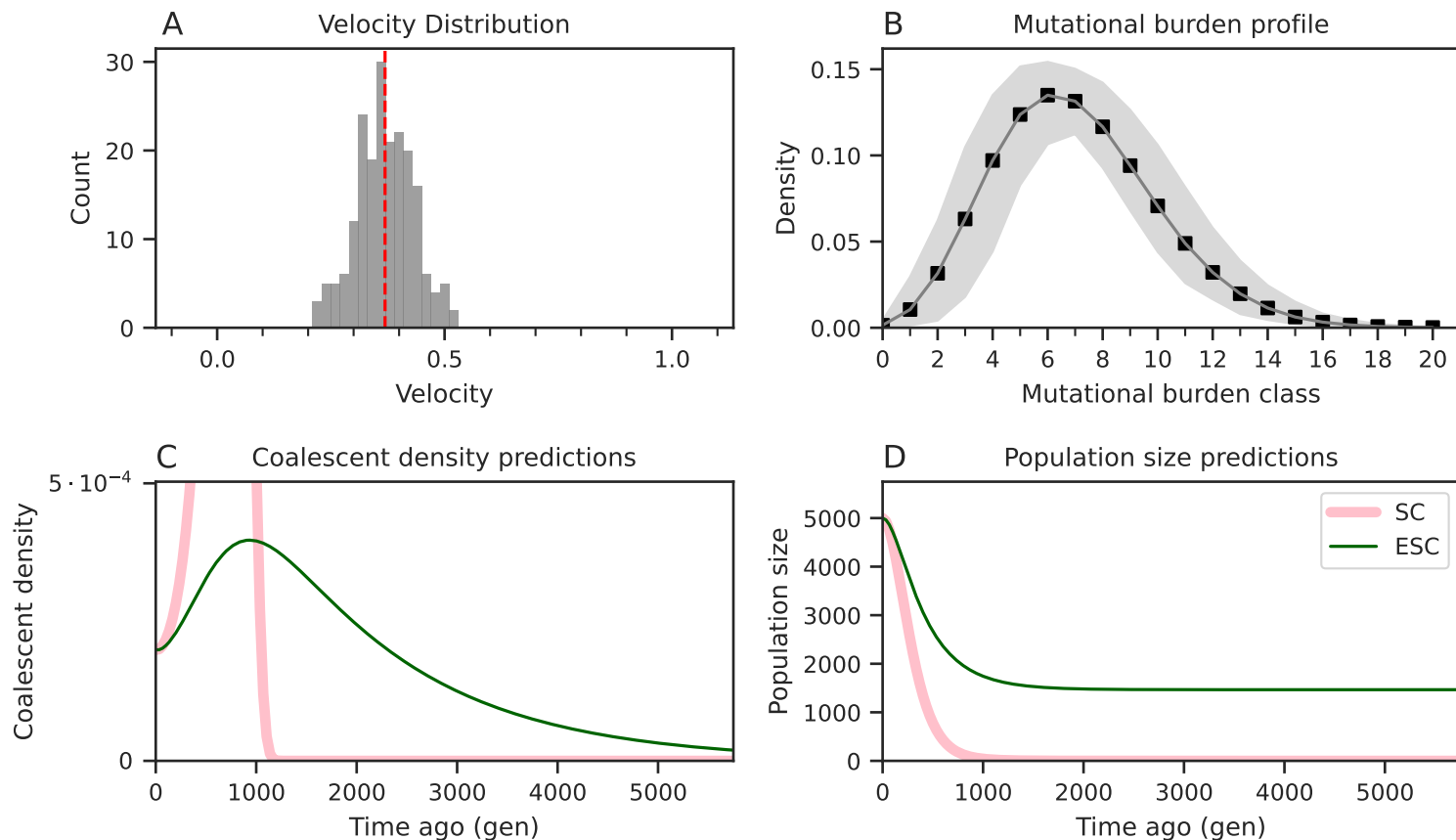
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0152$, $\phi = 1.25 \cdot 10^{-6}$
Mean Velocity= 0.4006 ± 0.0572 , $n = 200$ simulations



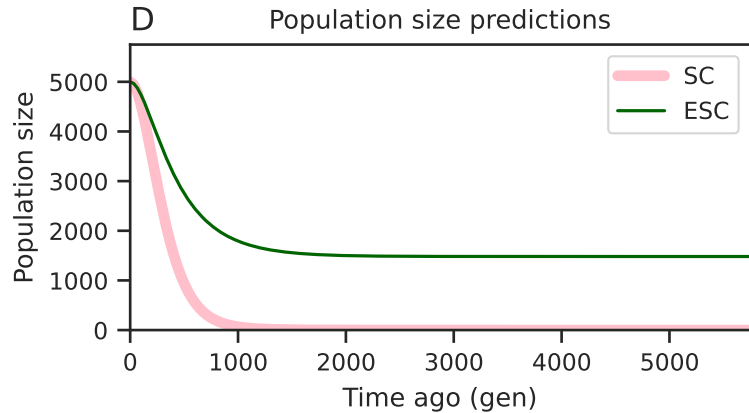
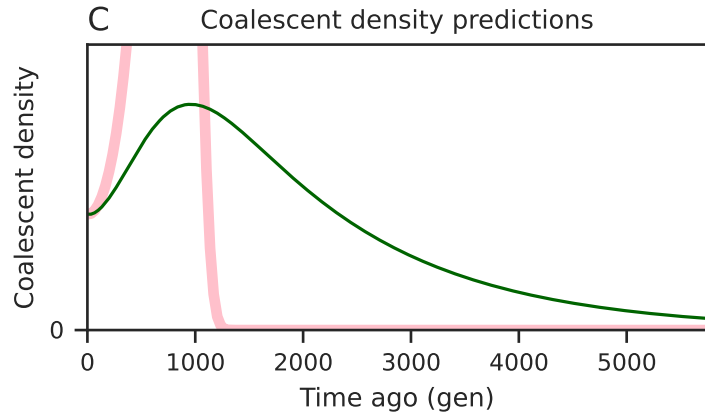
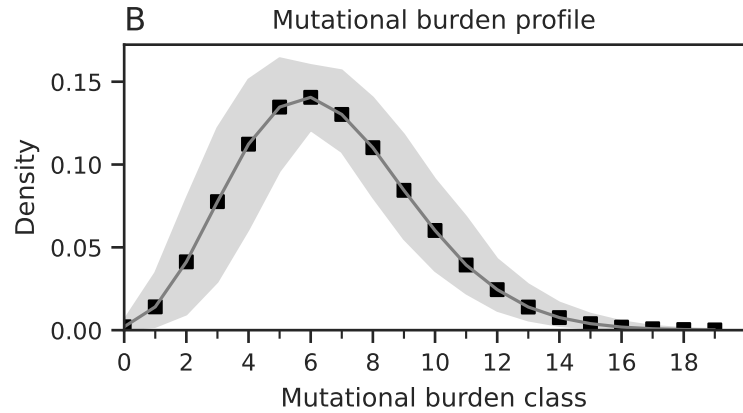
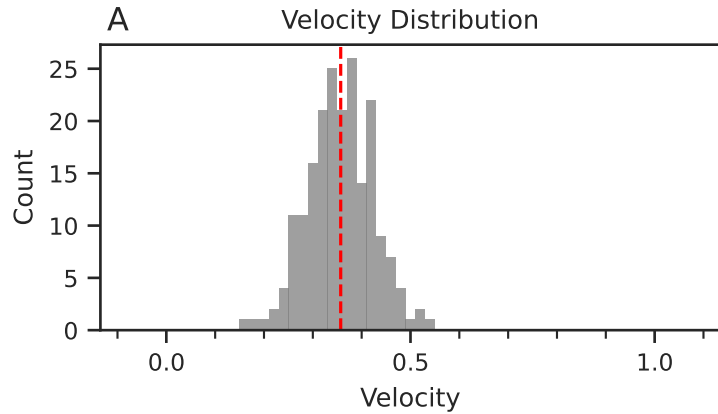
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0136$, $\phi = 6.2 \cdot 10^{-6}$
Mean Velocity = 0.3898 ± 0.0572 , $n = 200$ simulations



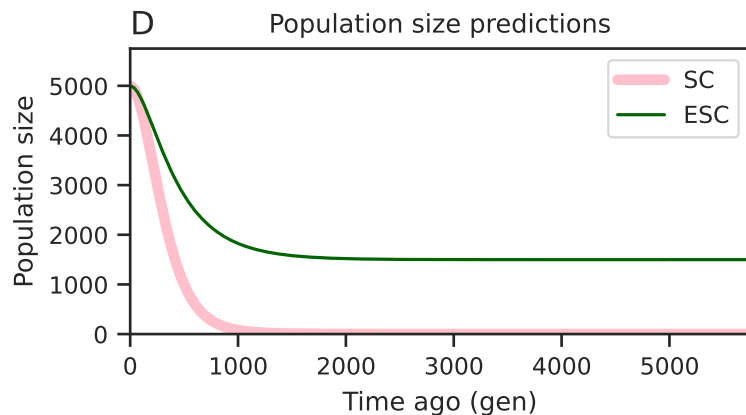
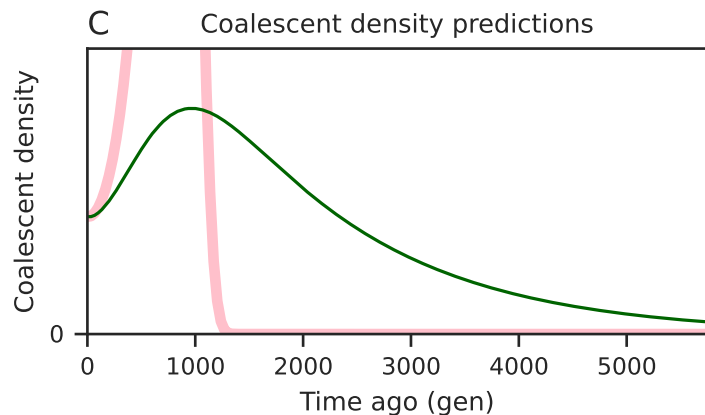
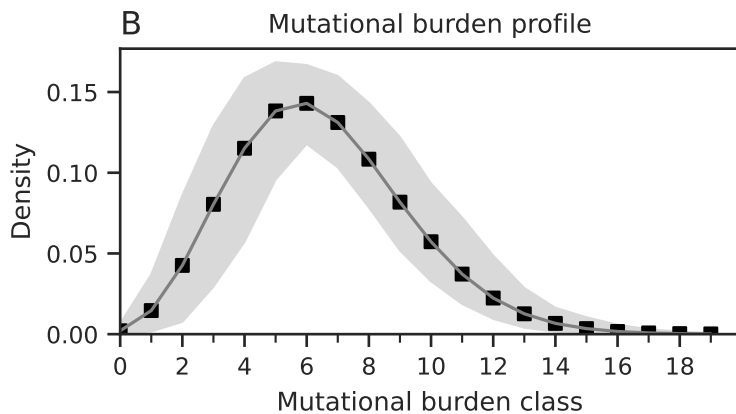
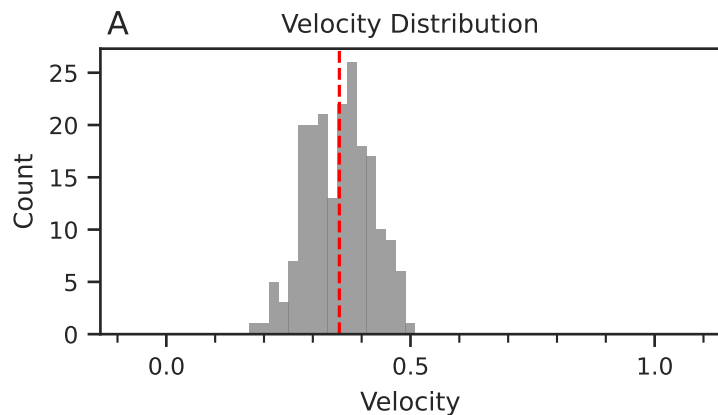
$N = 5e + 03$, $s = 0.001$, $U_d = 0.012$, $\phi = 3.07 \cdot 10^{-5}$
Mean Velocity= 0.3691 ± 0.0623 , $n = 200$ simulations



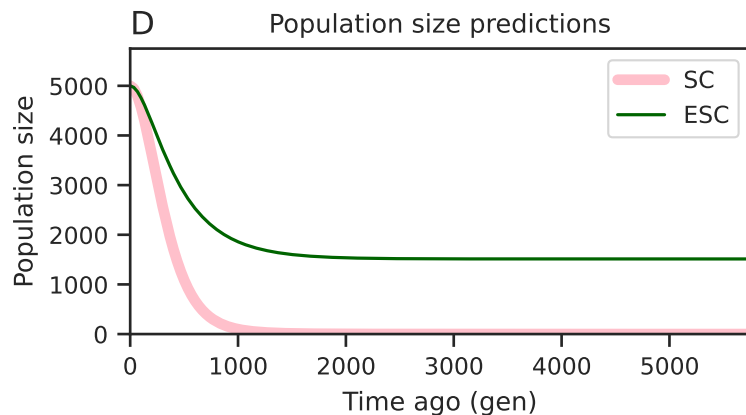
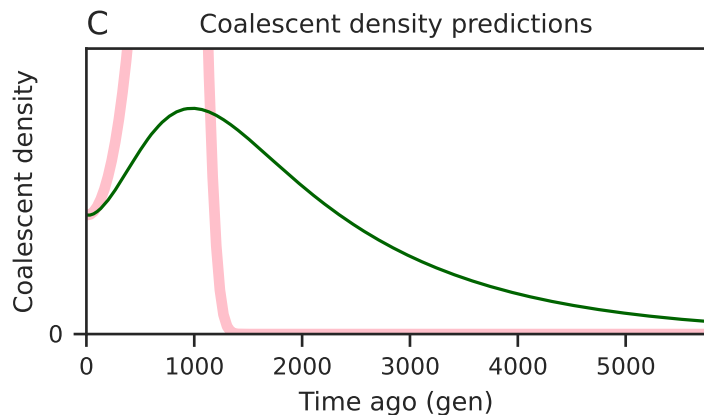
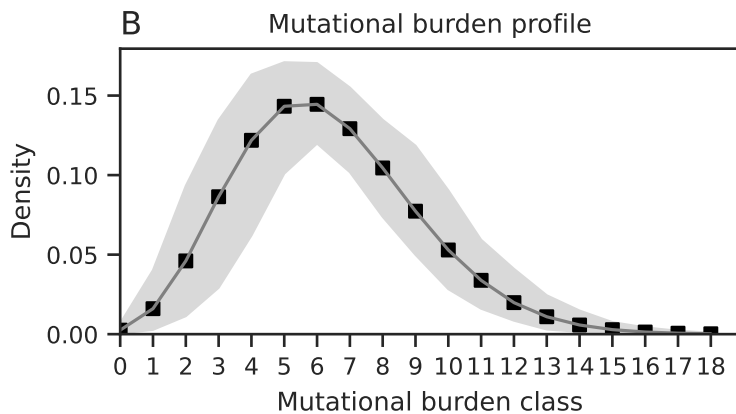
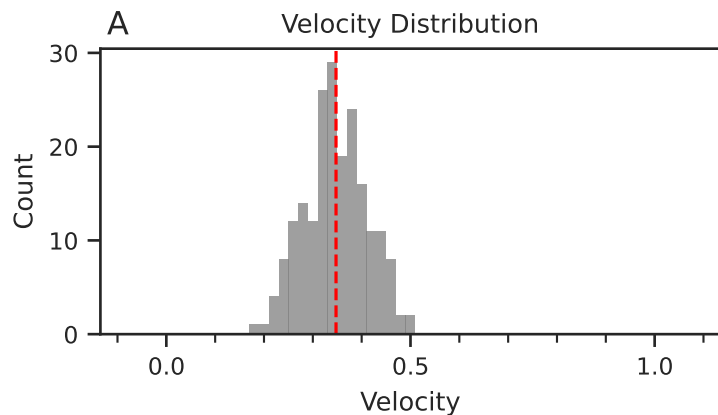
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0108$, $\phi = 1.02 \cdot 10^{-4}$
Mean Velocity= 0.3571 ± 0.0659 , $n = 200$ simulations



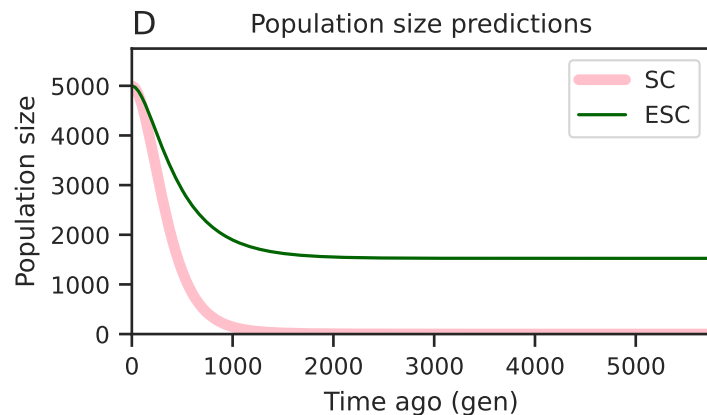
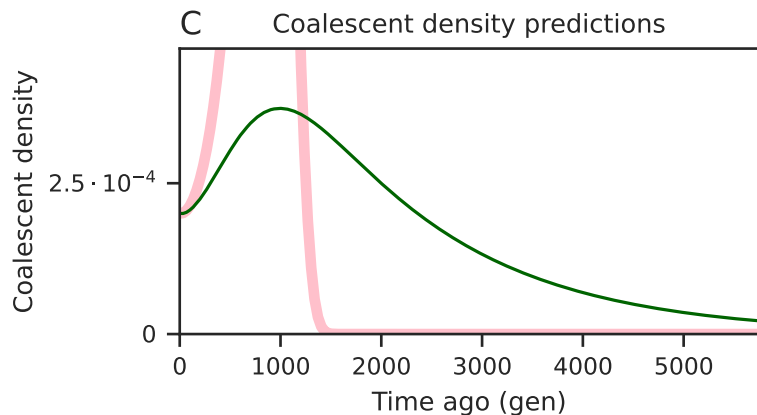
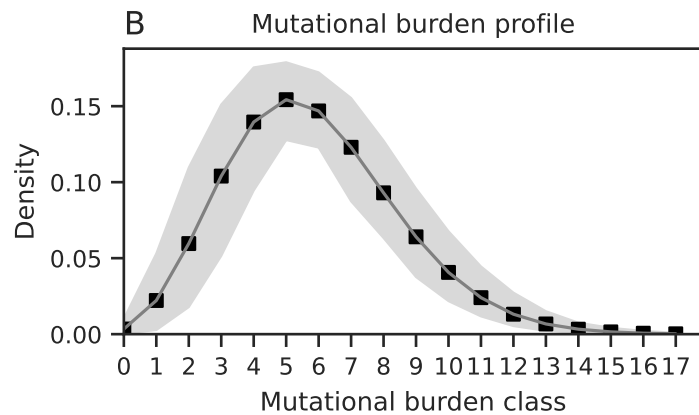
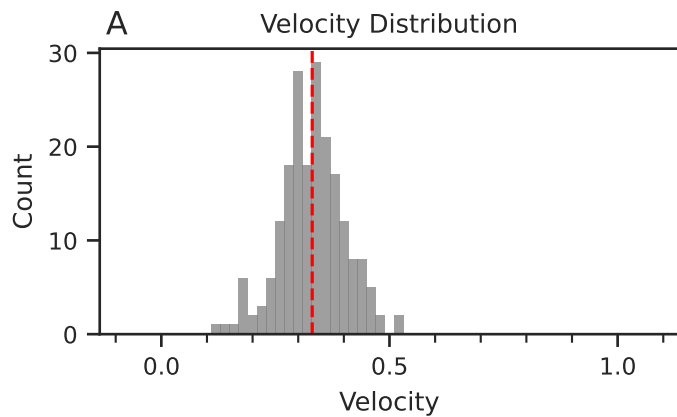
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0104$, $\phi = 1.52 \cdot 10^{-4}$
Mean Velocity= 0.3543 ± 0.0652 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 0.00984$, $\phi = 2.66 \cdot 10^{-4}$
Mean Velocity= 0.3475 ± 0.0632 , $n = 200$ simulations

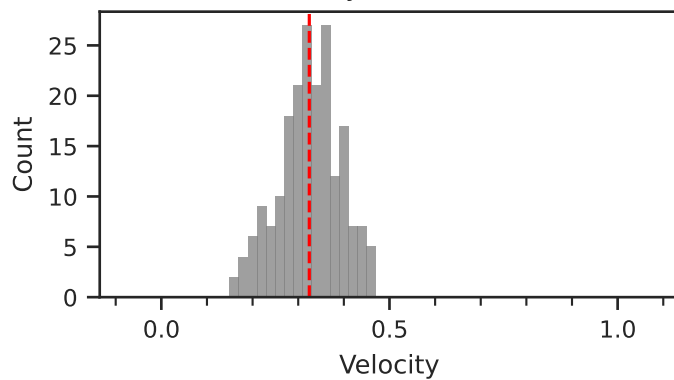


$N = 5e + 03$, $s = 0.001$, $U_d = 0.00888$, $\phi = 6.96 \cdot 10^{-4}$
Mean Velocity= 0.3306 ± 0.0695 , $n = 200$ simulations

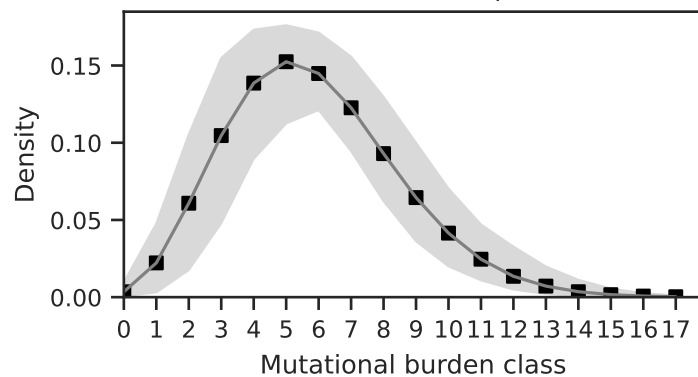


$N = 5e + 03$, $s = 0.001$, $U_d = 0.0088$, $\phi = 7.54 \cdot 10^{-4}$
Mean Velocity= 0.3243 ± 0.0659 , $n = 200$ simulations

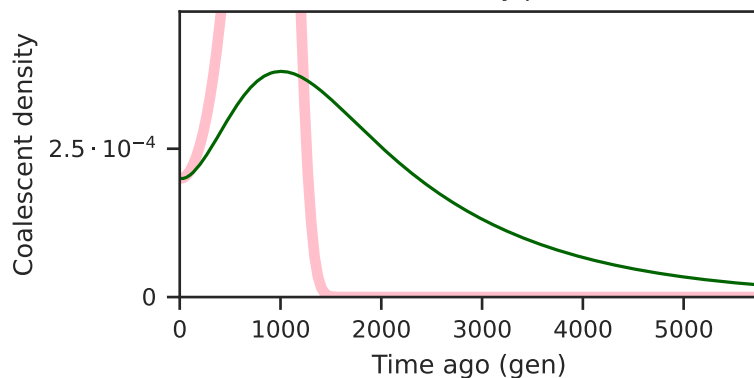
A Velocity Distribution



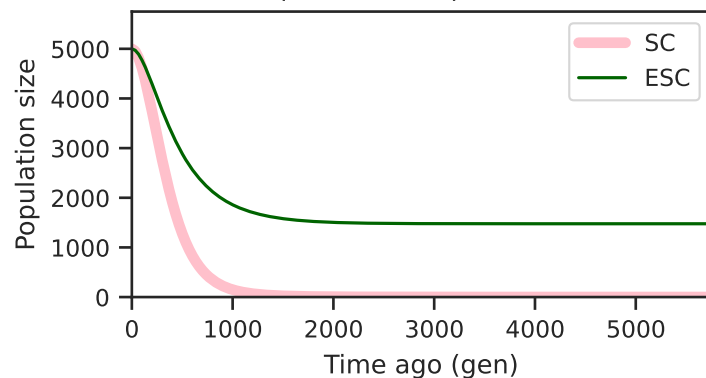
B Mutational burden profile



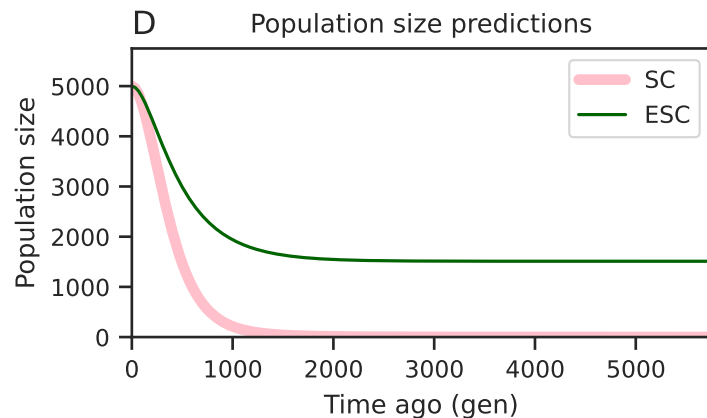
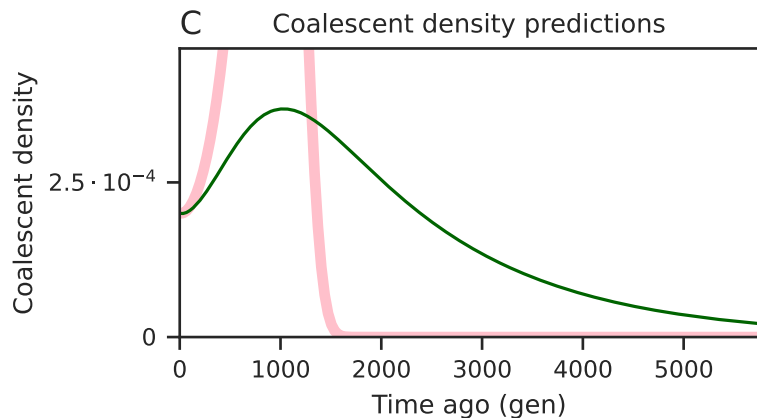
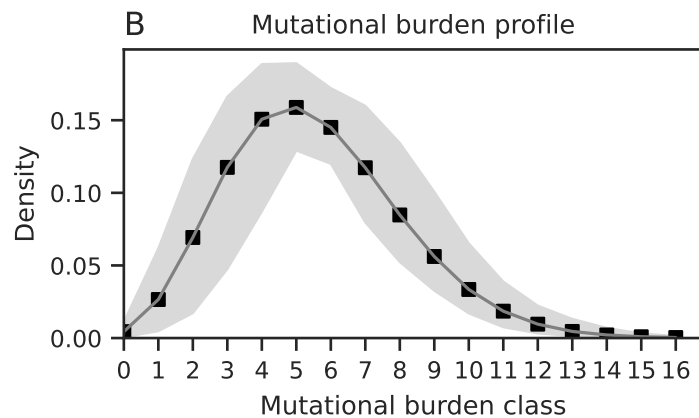
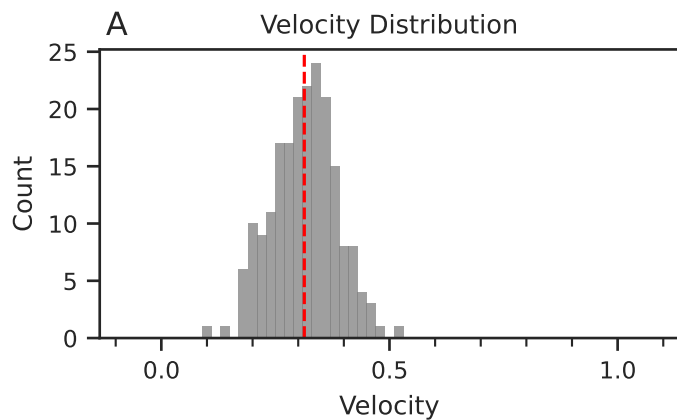
C Coalescent density predictions



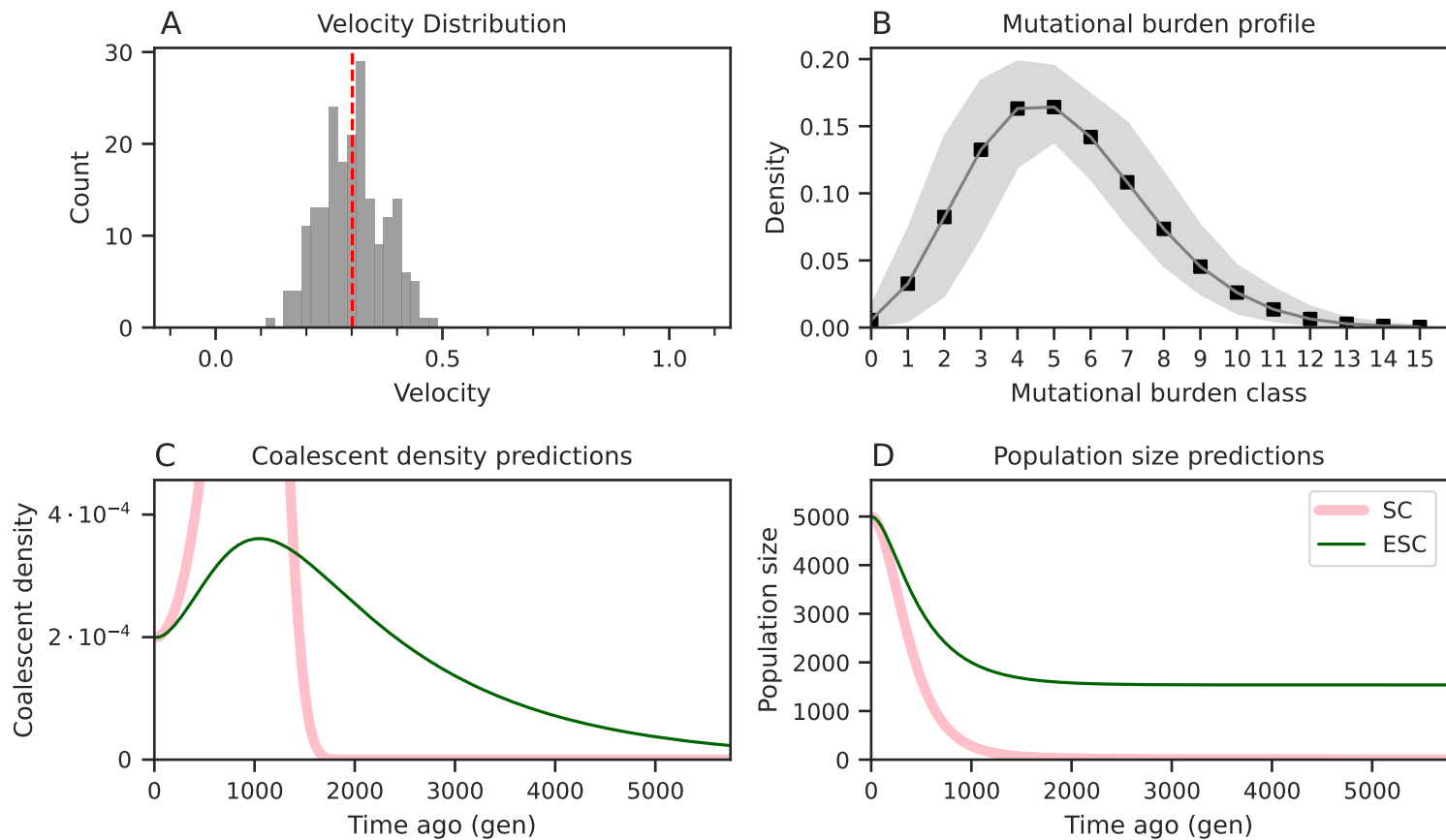
D Population size predictions



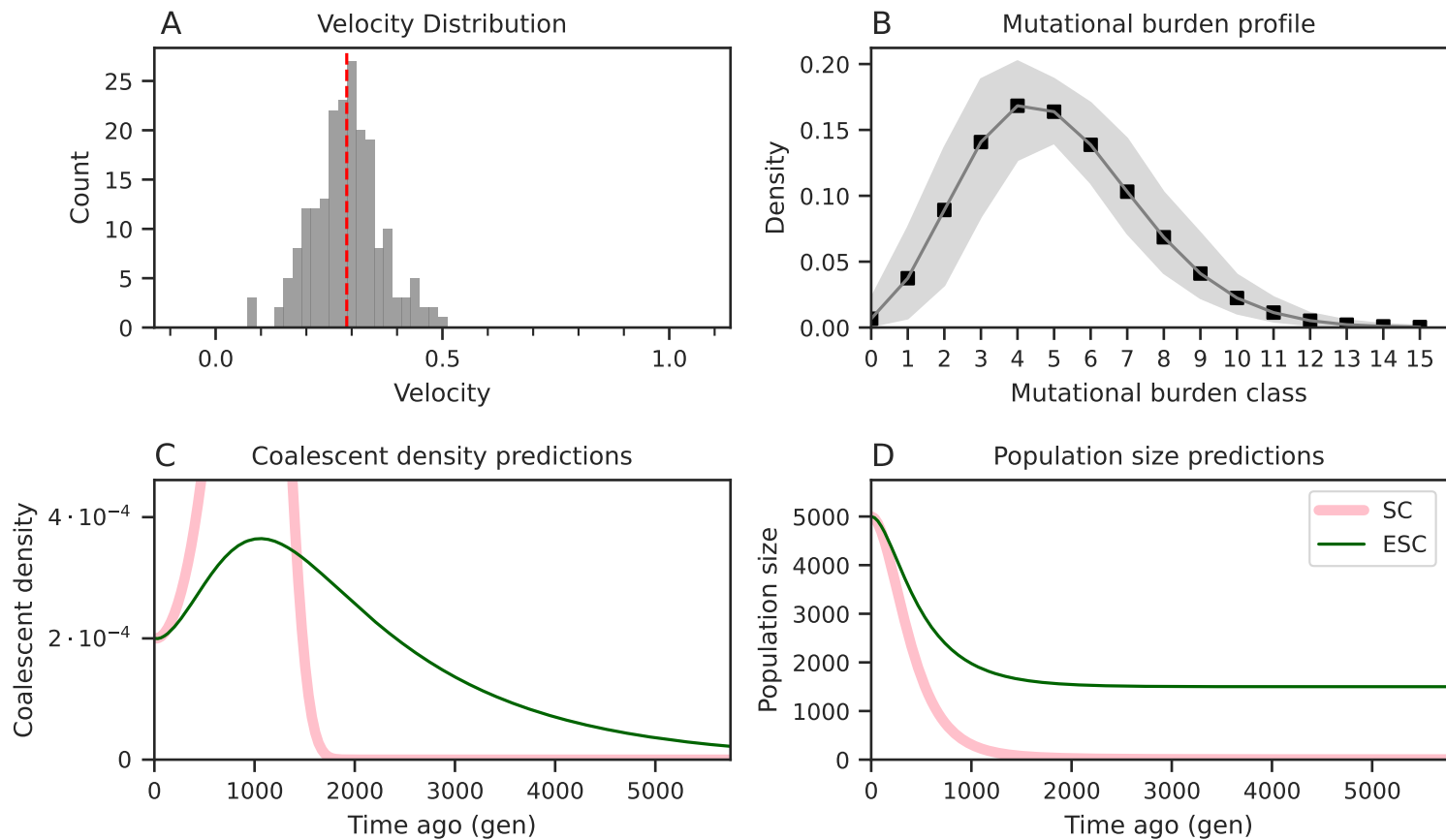
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00792$, $\phi = 0.00182$
Mean Velocity= 0.3133 ± 0.0708 , $n = 200$ simulations



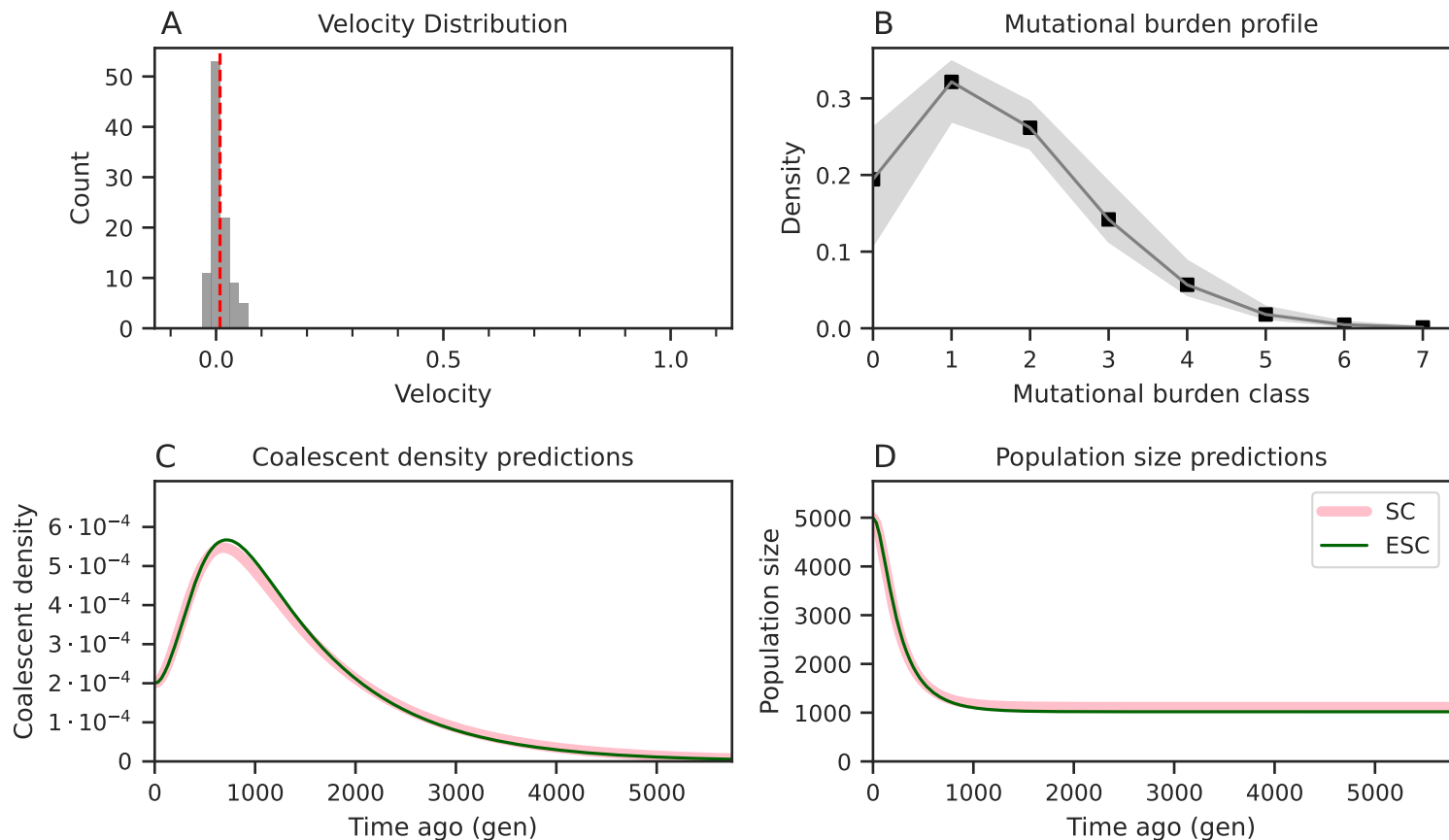
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0072$, $\phi = 0.00373$
Mean Velocity= 0.3014 ± 0.0701 , $n = 200$ simulations



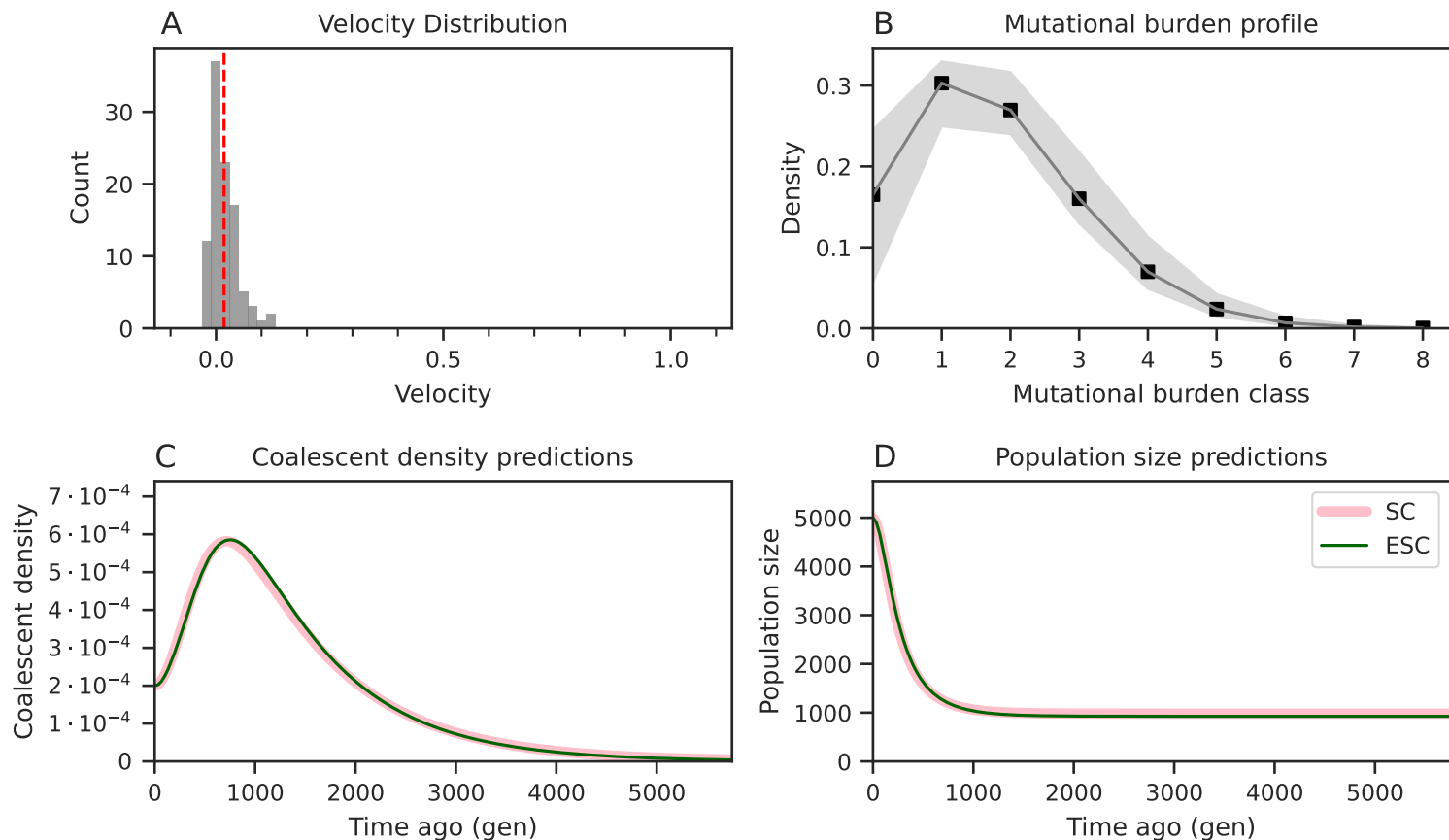
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00696$, $\phi = 0.00475$
Mean Velocity= 0.2891 ± 0.0753 , $n = 200$ simulations



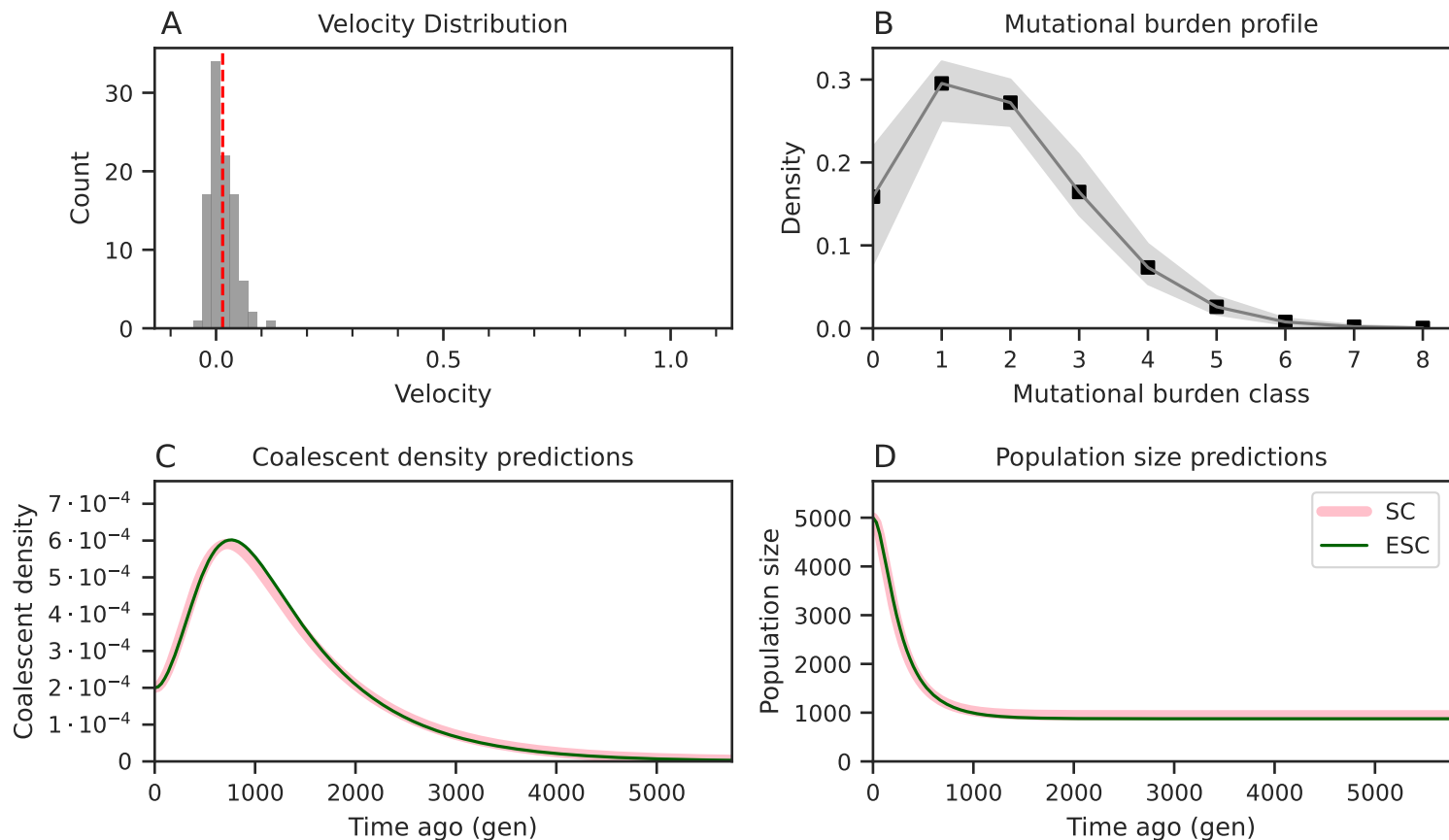
$N = 5e + 03$, $s = 0.004$, $U_d = 0.006$, $\phi = 4.46$
Mean Velocity= 0.0087 ± 0.0186 , $n = 100$ simulations



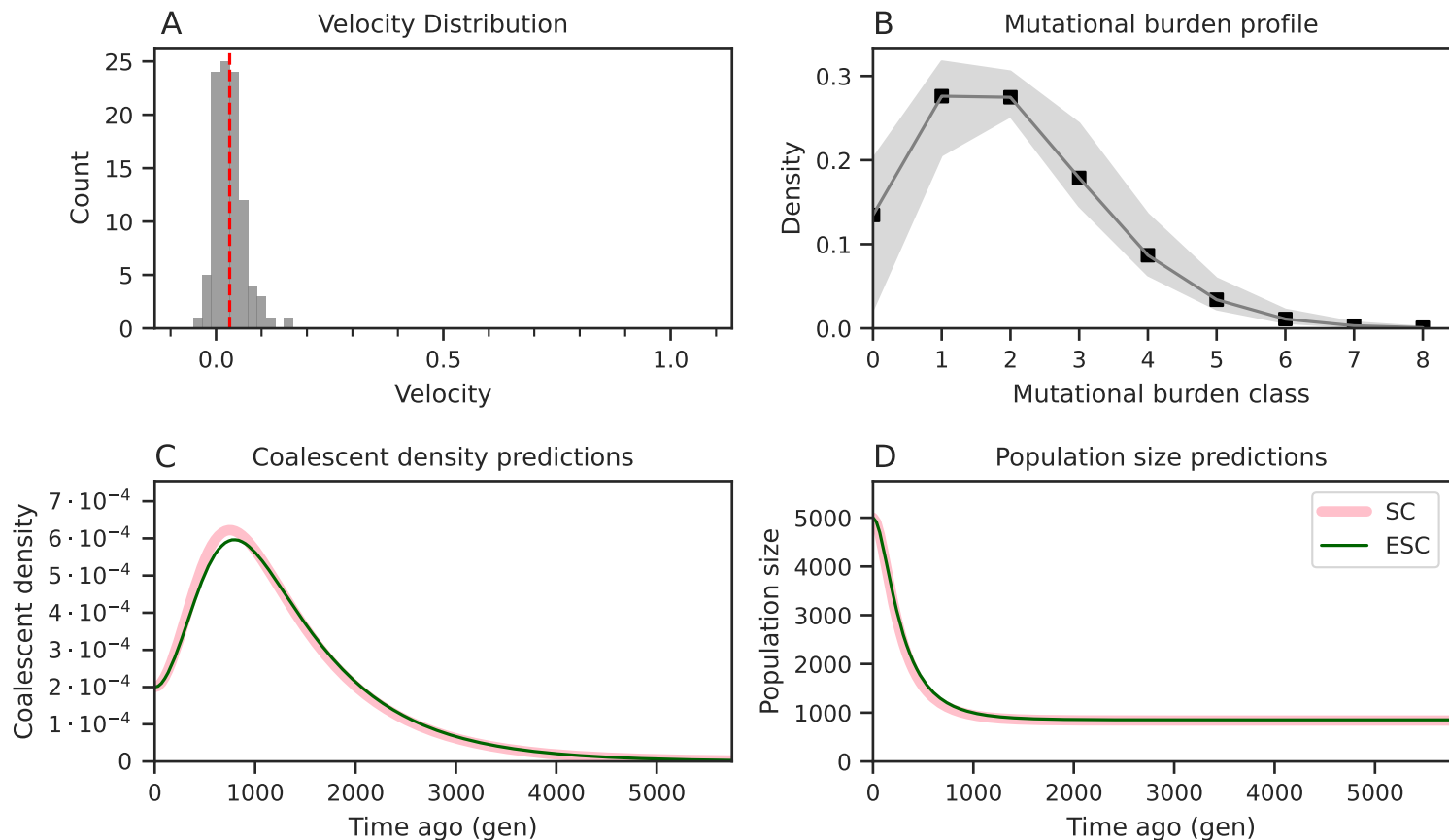
$N = 5e + 03$, $s = 0.00368$, $U_d = 0.006$, $\phi = 3.6$
Mean Velocity= 0.0176 ± 0.0281 , $n = 100$ simulations



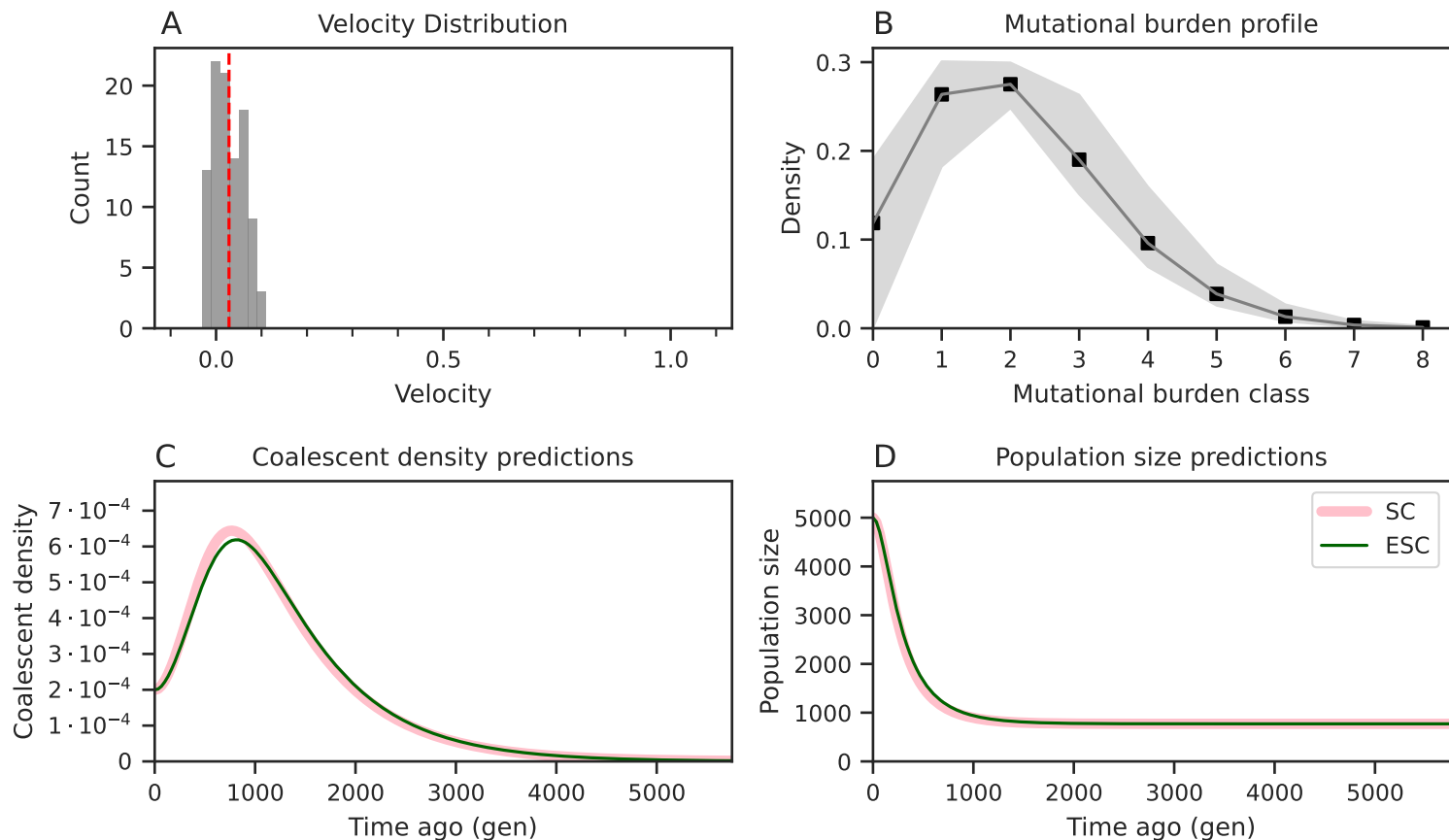
$N = 5e + 03$, $s = 0.0036$, $U_d = 0.006$, $\phi = 3.41$
Mean Velocity= 0.0145 ± 0.0275 , $n = 100$ simulations



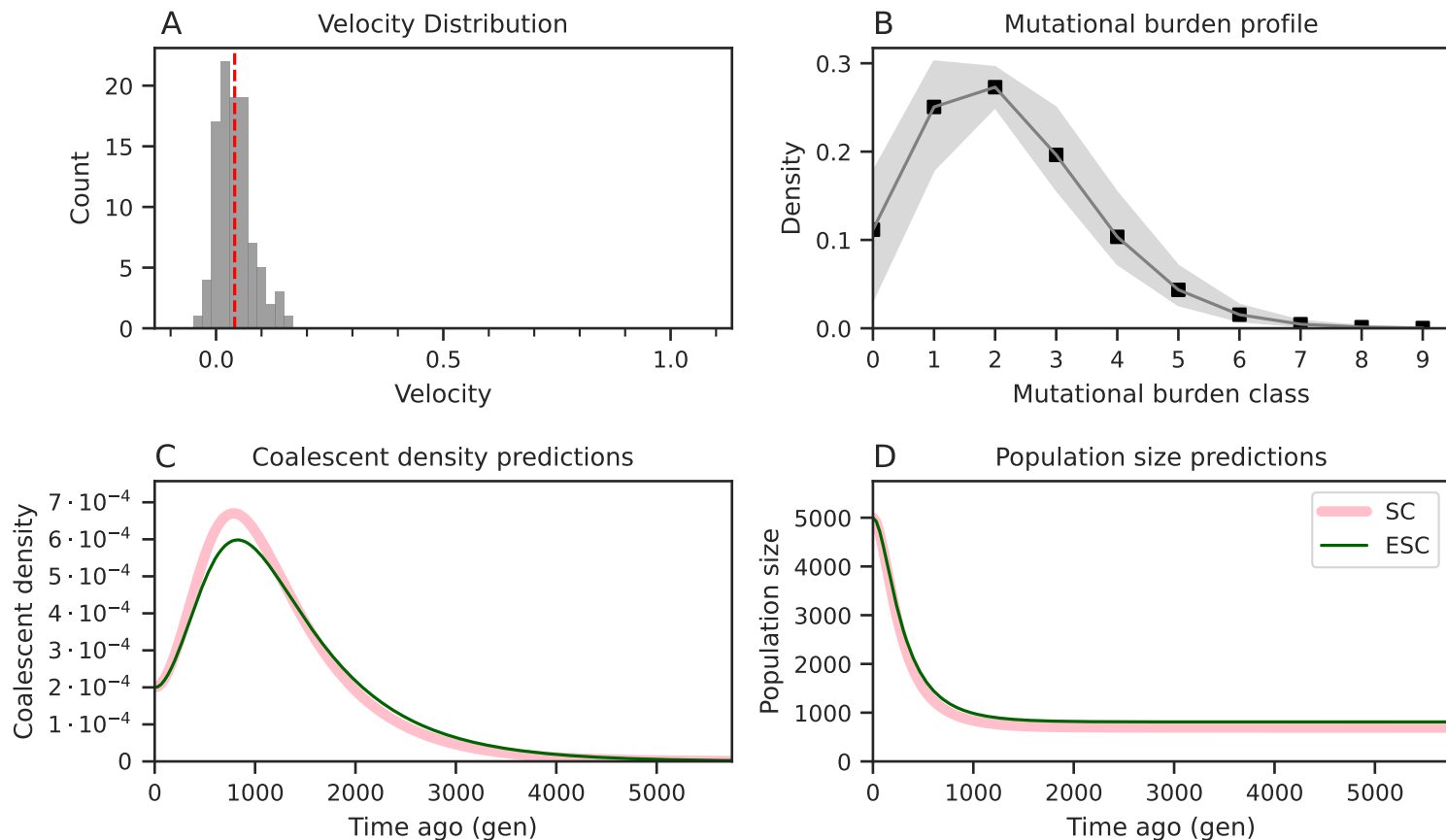
$N = 5e + 03$, $s = 0.00336$, $U_d = 0.006$, $\phi = 2.82$
Mean Velocity= 0.0300 ± 0.0319 , $n = 100$ simulations



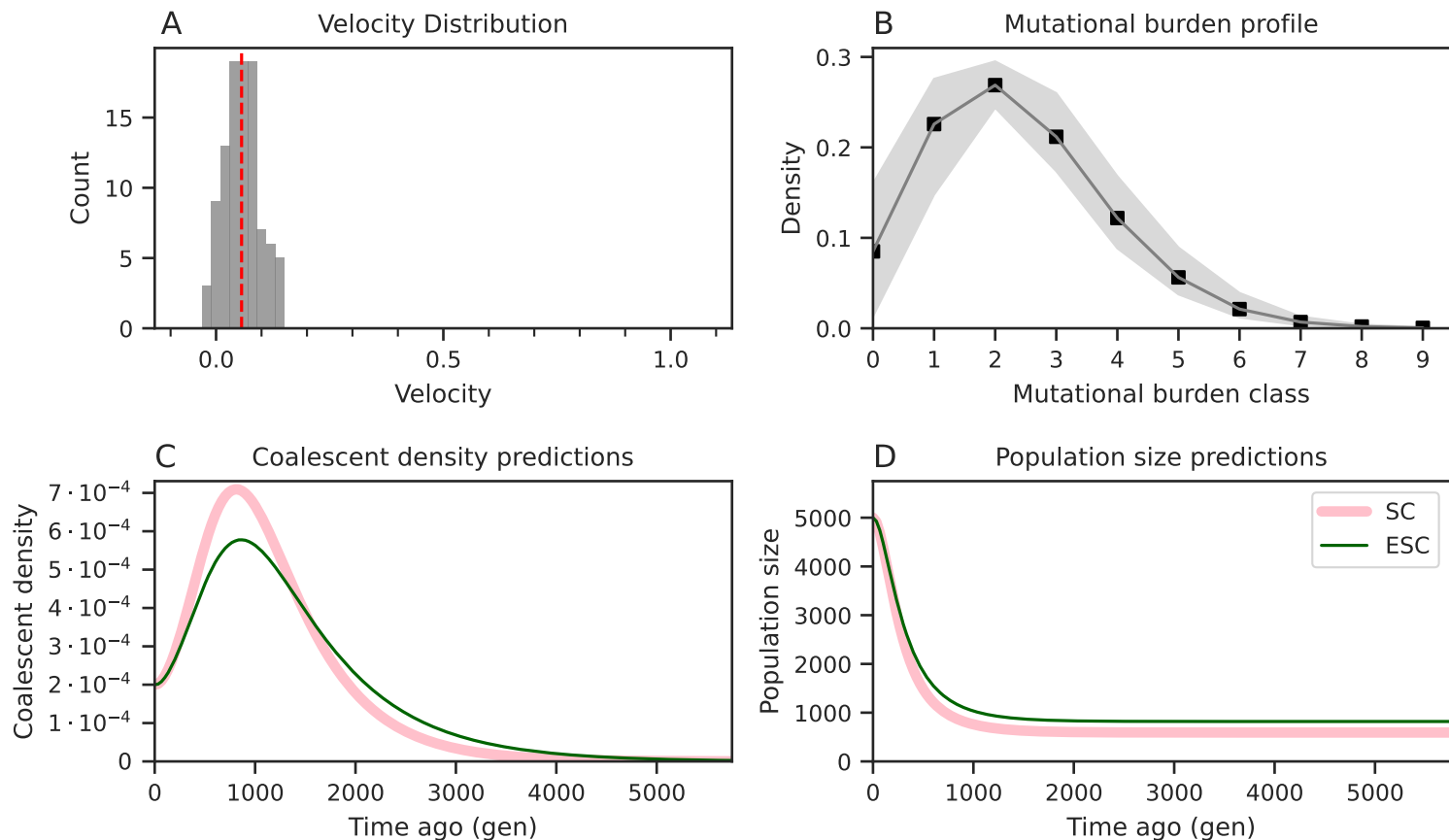
$N = 5e + 03$, $s = 0.00321$, $U_d = 0.006$, $\phi = 2.47$
Mean Velocity = 0.0285 ± 0.0321 , $n = 100$ simulations



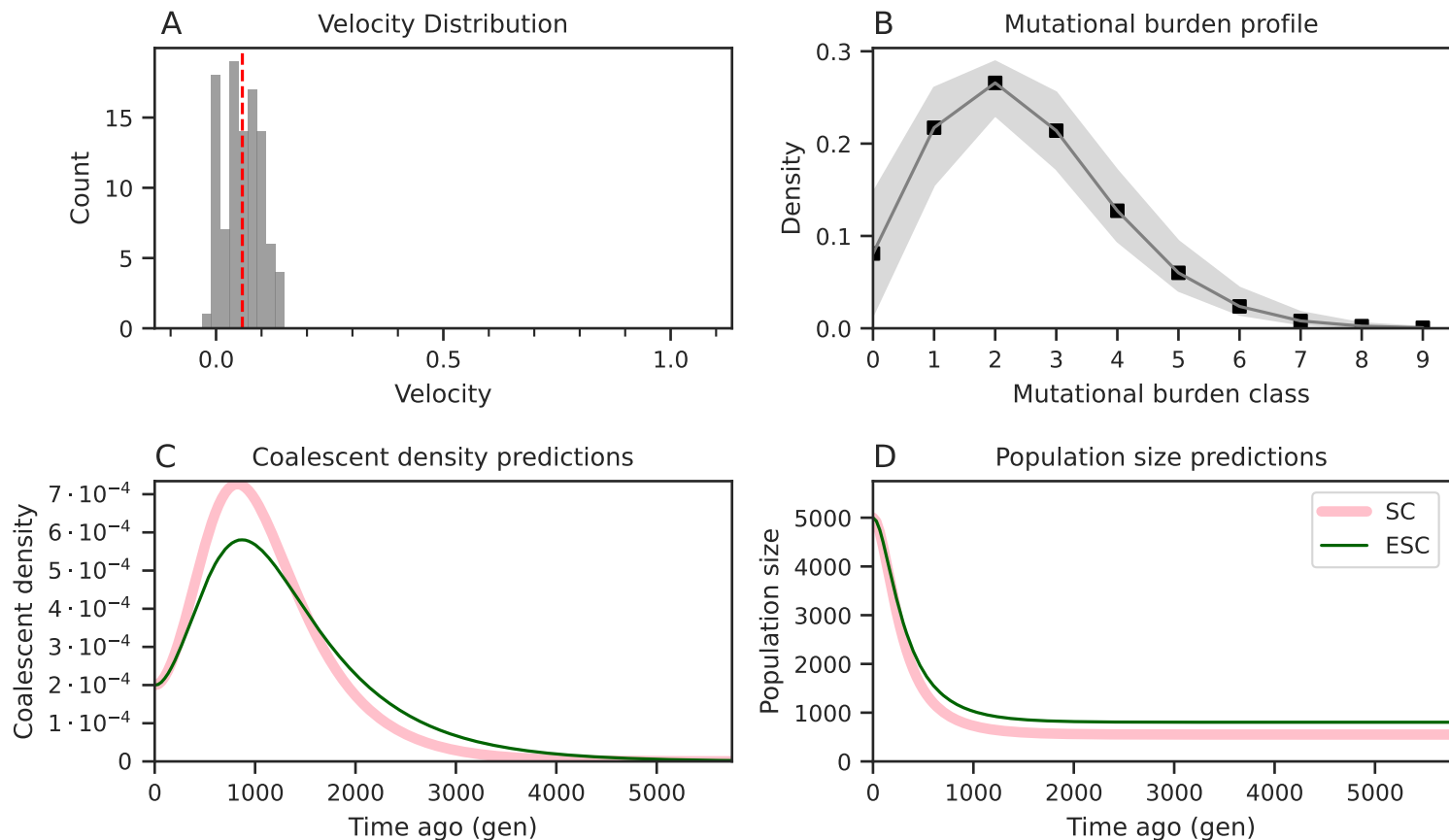
$N = 5e + 03$, $s = 0.00304$, $U_d = 0.006$, $\phi = 2.11$
Mean Velocity= 0.0409 ± 0.0380 , $n = 100$ simulations



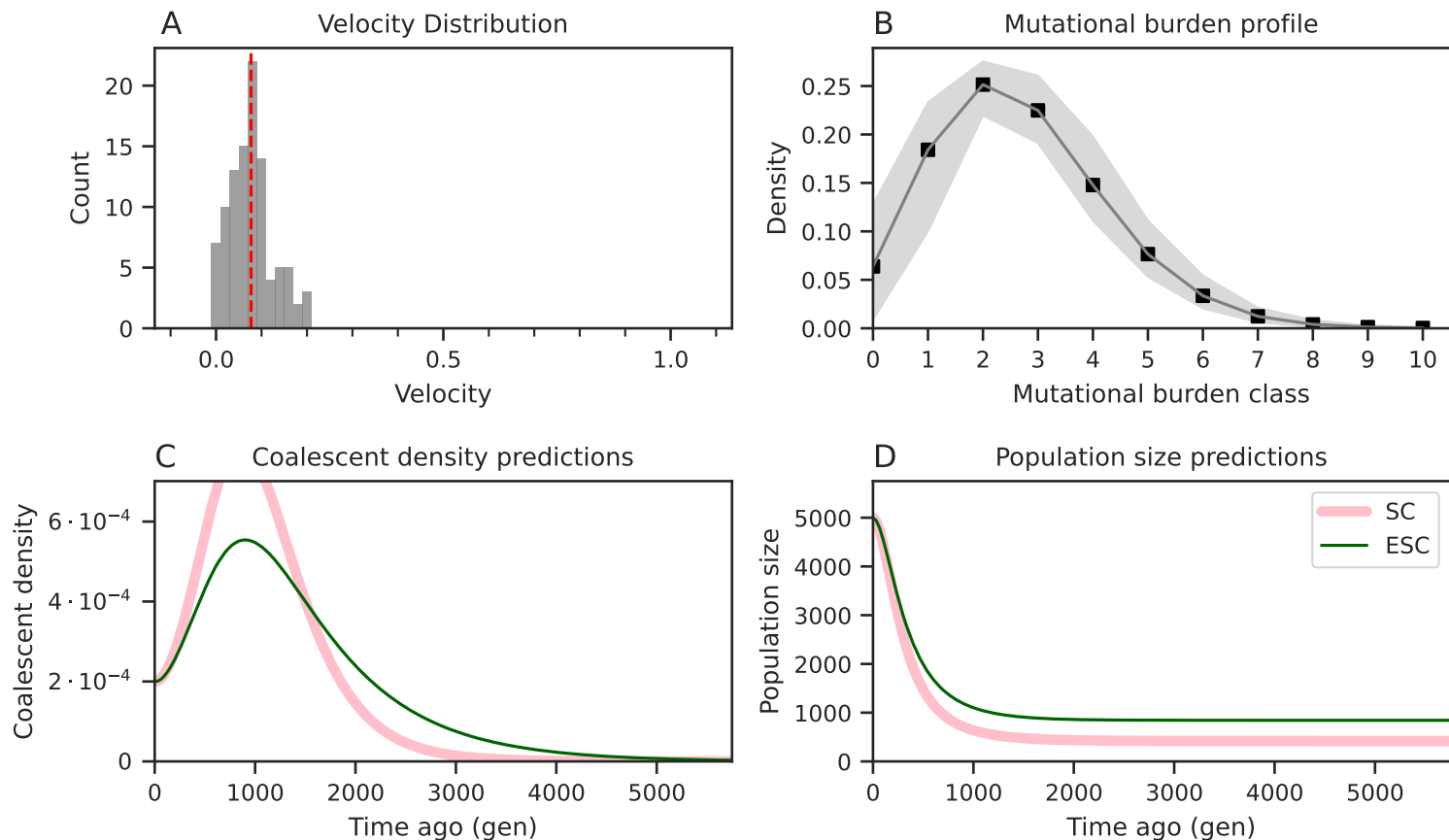
$N = 5e + 03$, $s = 0.00281$, $U_d = 0.006$, $\phi = 1.66$
Mean Velocity= 0.0562 ± 0.0389 , $n = 100$ simulations



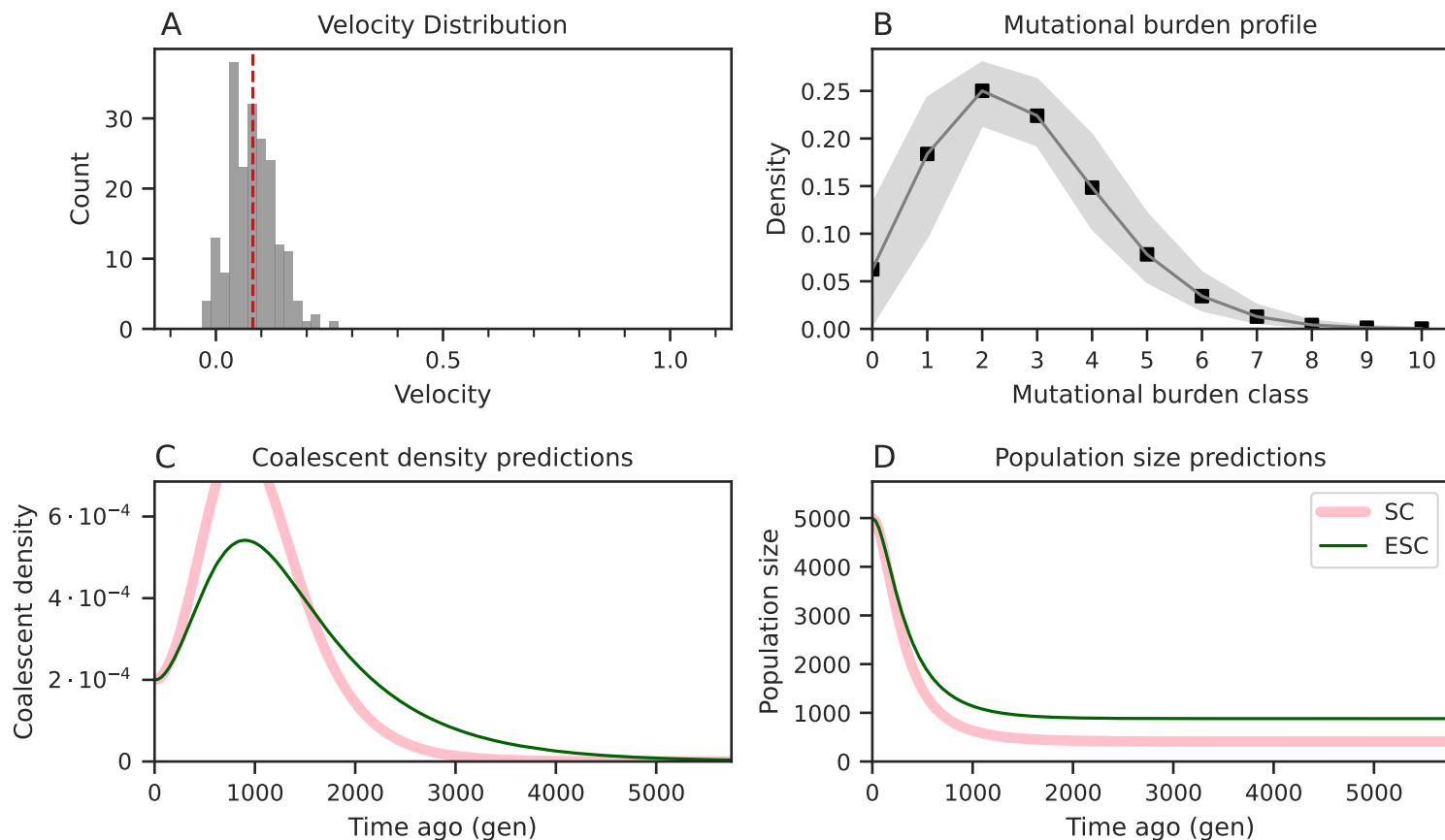
$N = 5e + 03$, $s = 0.00272$, $U_d = 0.006$, $\phi = 1.5$
Mean Velocity = 0.0578 ± 0.0409 , $n = 100$ simulations



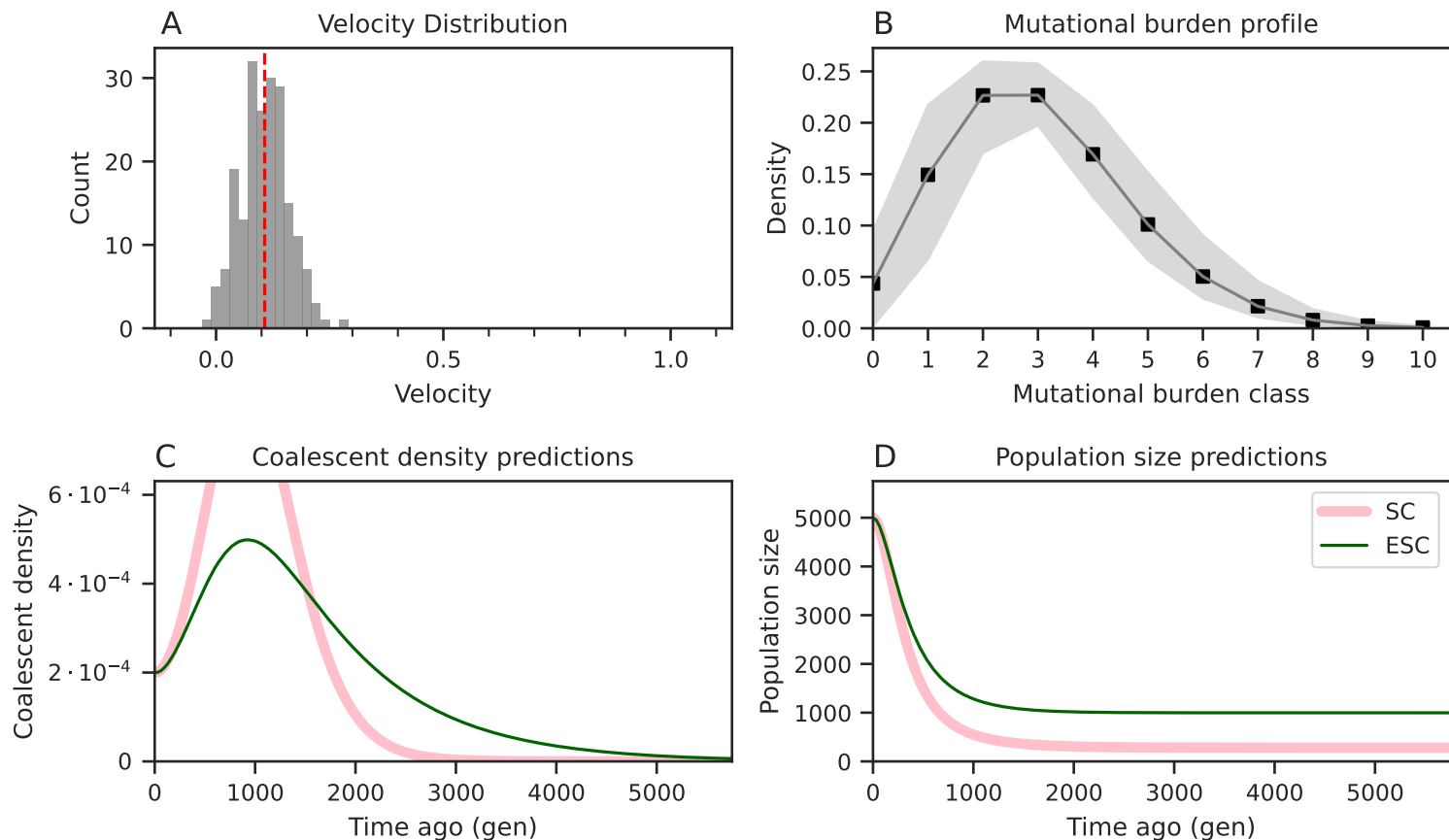
$N = 5e + 03$, $s = 0.00242$, $U_d = 0.006$, $\phi = 1.01$
Mean Velocity= 0.0769 ± 0.0482 , $n = 100$ simulations



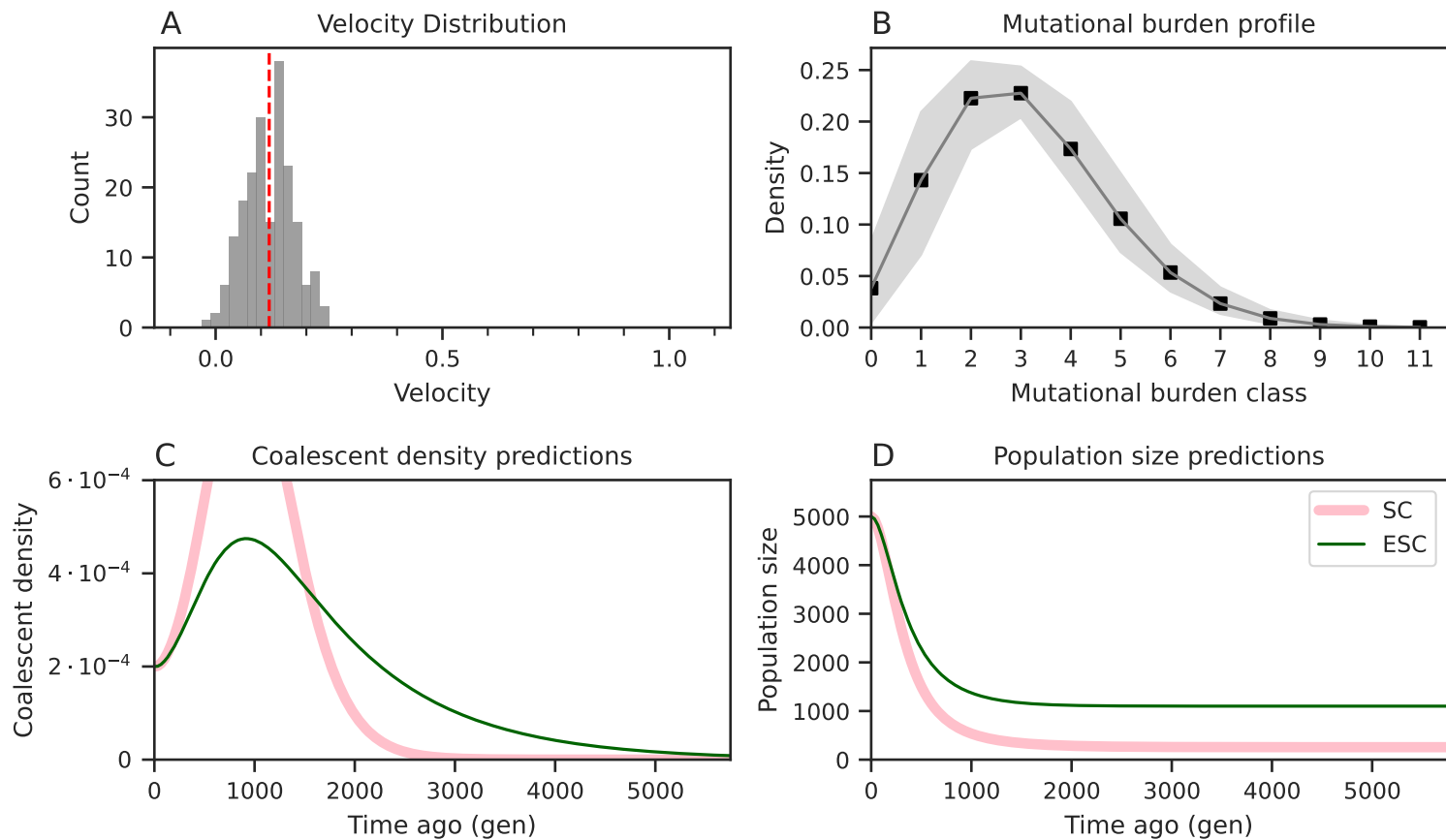
$N = 5e + 03$, $s = 0.0024$, $U_d = 0.006$, $\phi = 0.985$
Mean Velocity= 0.0814 ± 0.0501 , $n = 200$ simulations



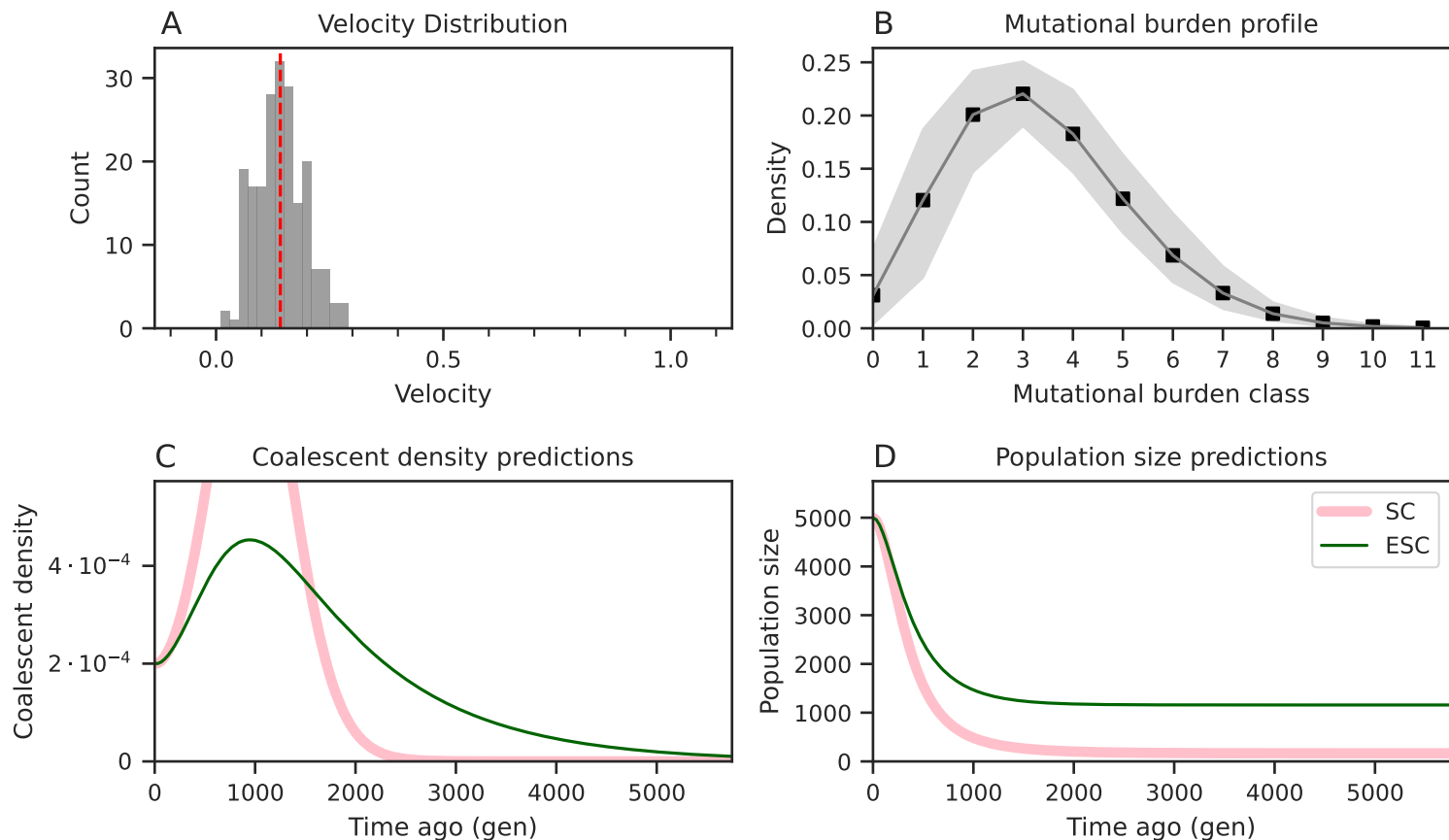
$N = 5e + 03$, $s = 0.00208$, $U_d = 0.006$, $\phi = 0.581$
Mean Velocity = 0.1070 ± 0.0527 , $n = 200$ simulations



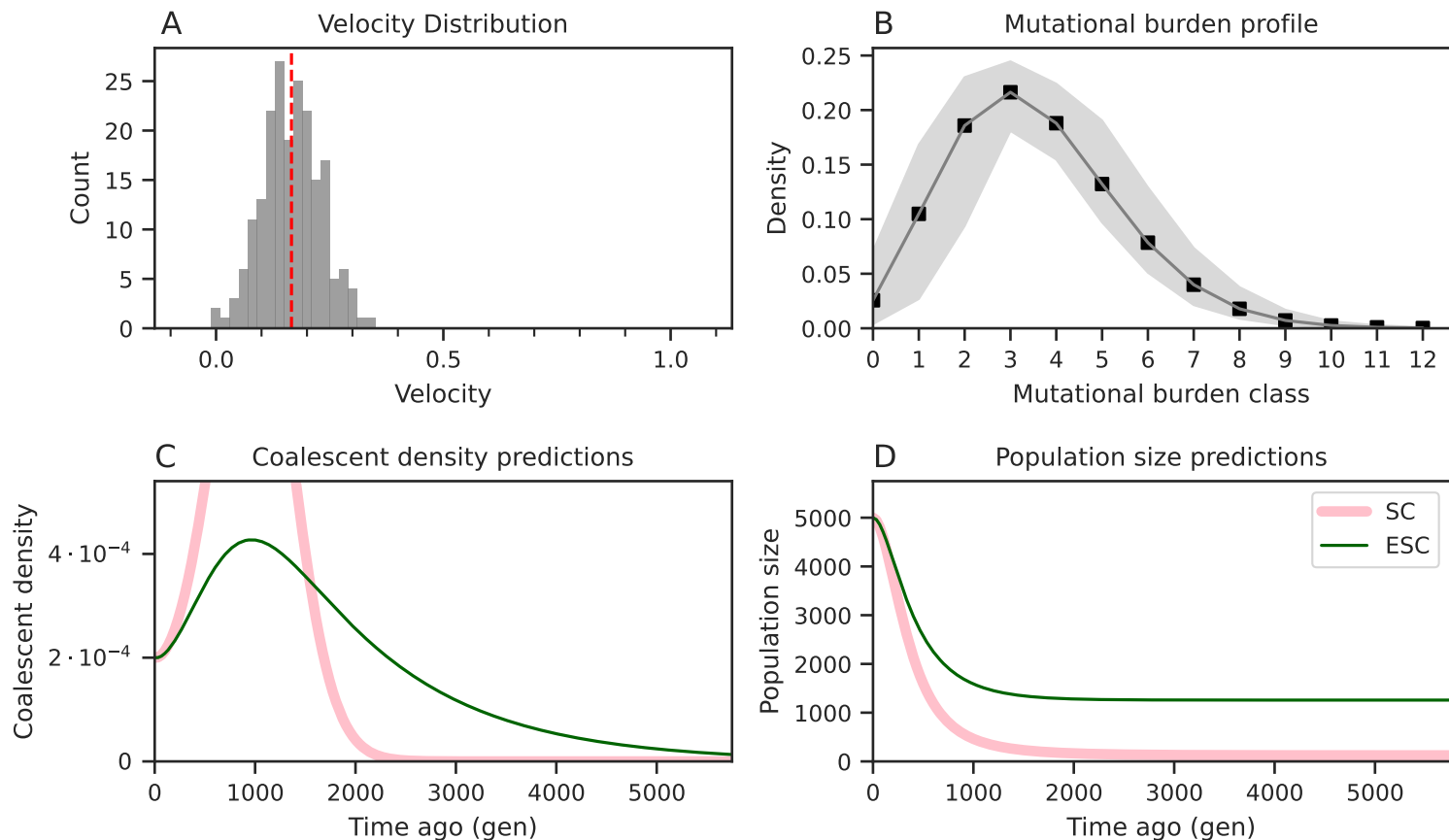
$N = 5e + 03$, $s = 0.00202$, $U_d = 0.006$, $\phi = 0.518$
Mean Velocity= 0.1181 ± 0.0533 , $n = 200$ simulations



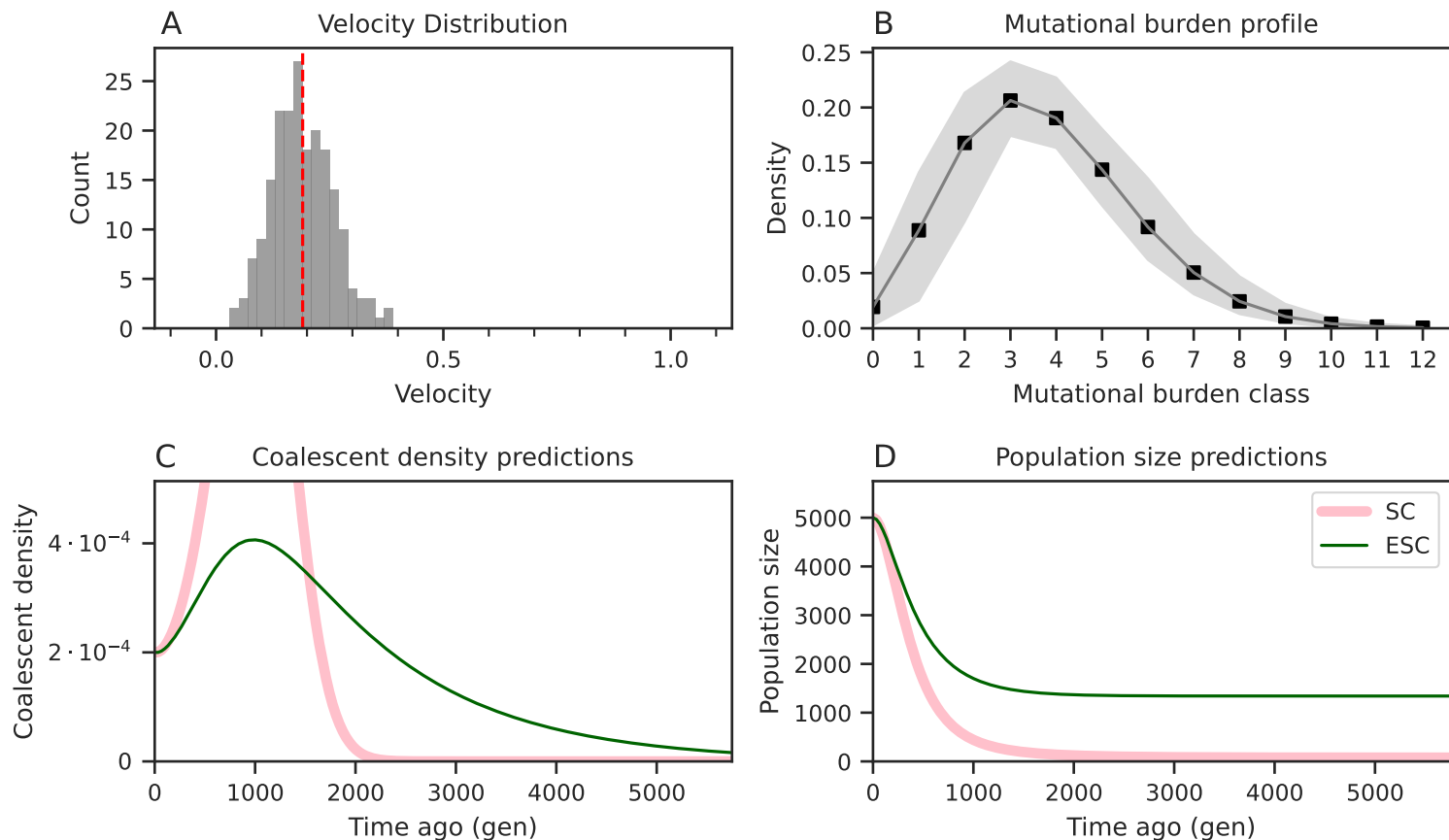
$N = 5e + 03$, $s = 0.00176$, $U_d = 0.006$, $\phi = 0.291$
Mean Velocity= 0.1414 ± 0.0542 , $n = 200$ simulations



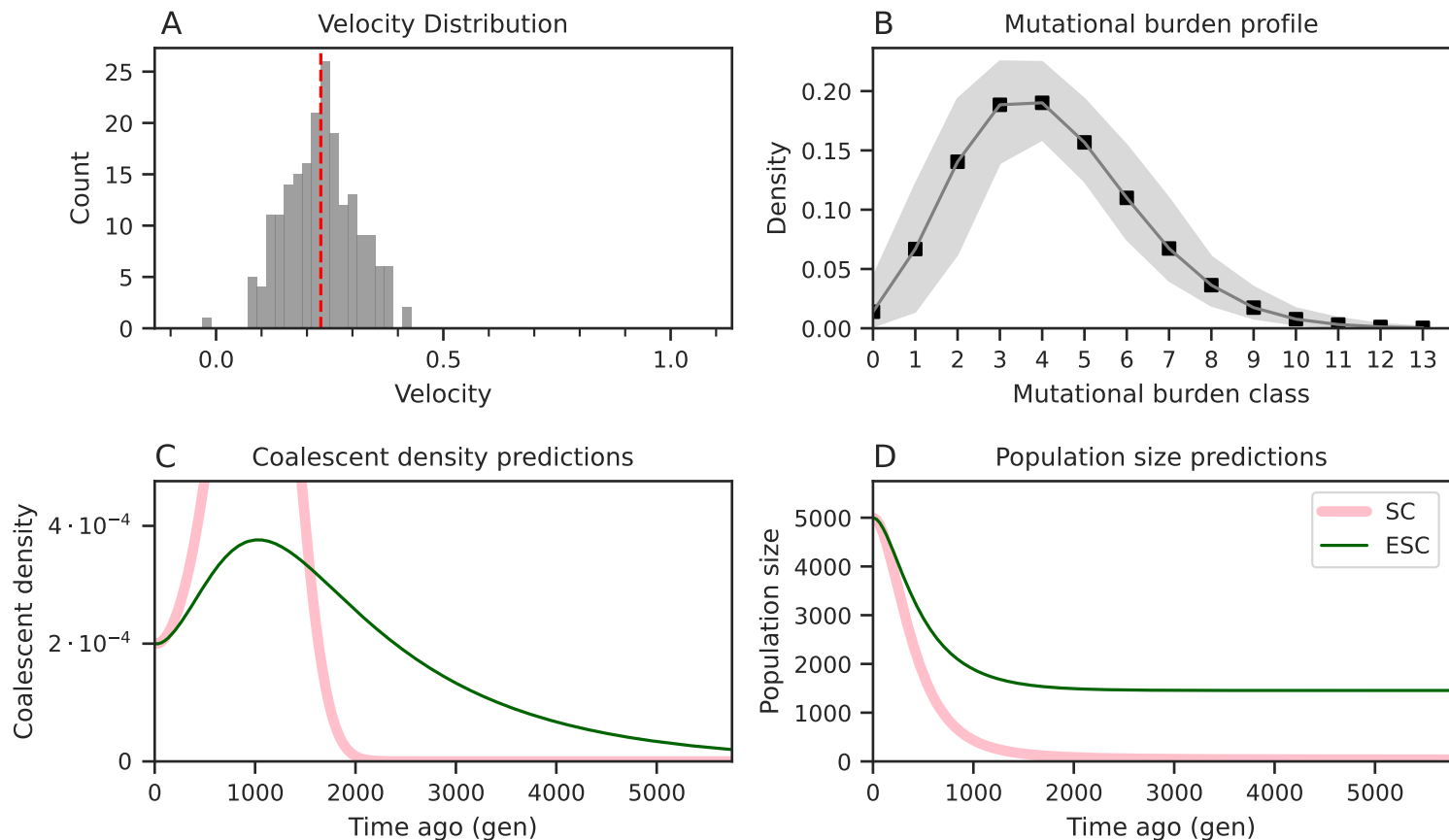
$N = 5e + 03$, $s = 0.00162$, $U_d = 0.006$, $\phi = 0.202$
Mean Velocity= 0.1659 ± 0.0632 , $n = 200$ simulations



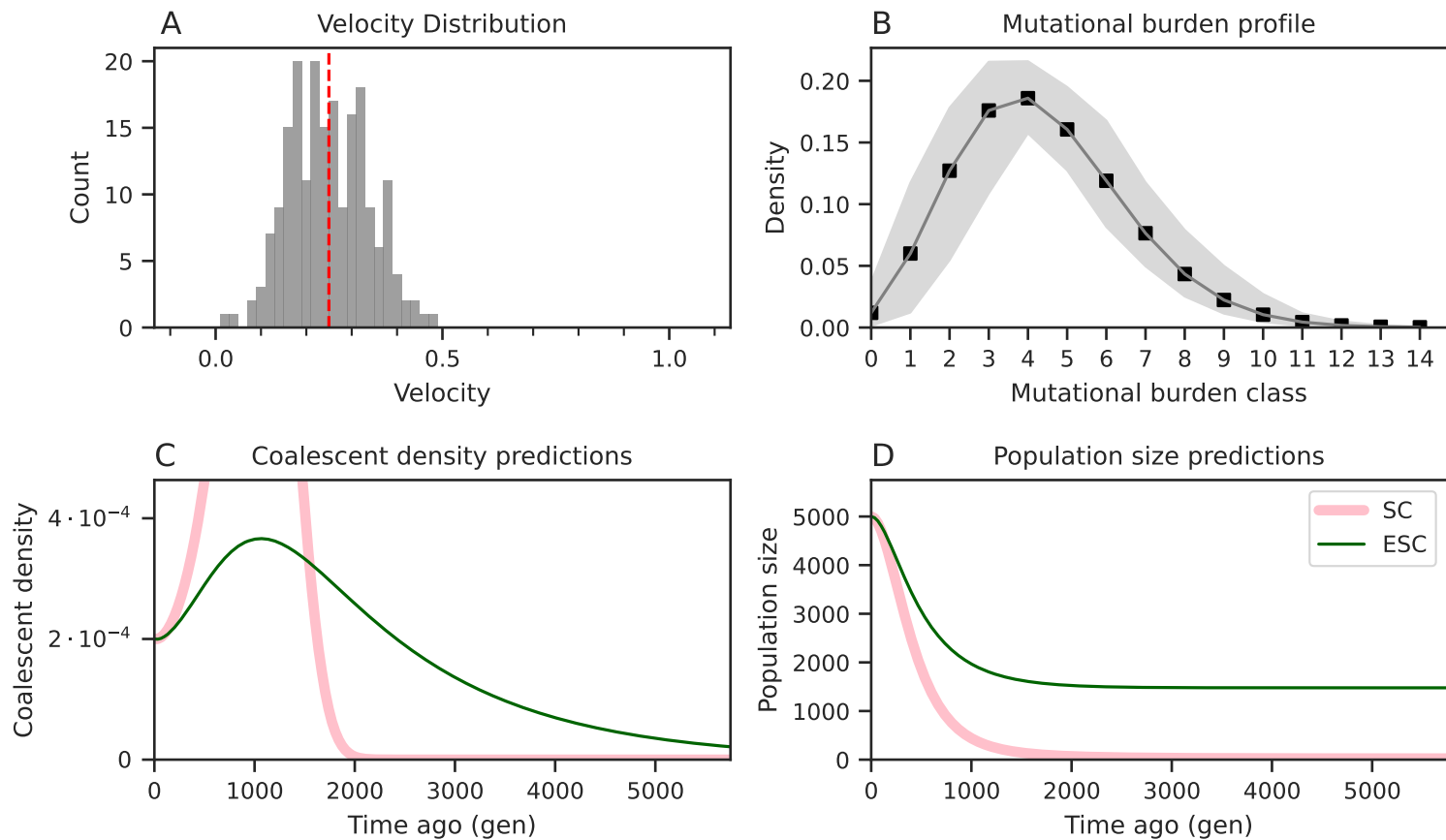
$N = 5e + 03$, $s = 0.00144$, $U_d = 0.006$, $\phi = 0.112$
Mean Velocity= 0.1905 ± 0.0664 , $n = 200$ simulations



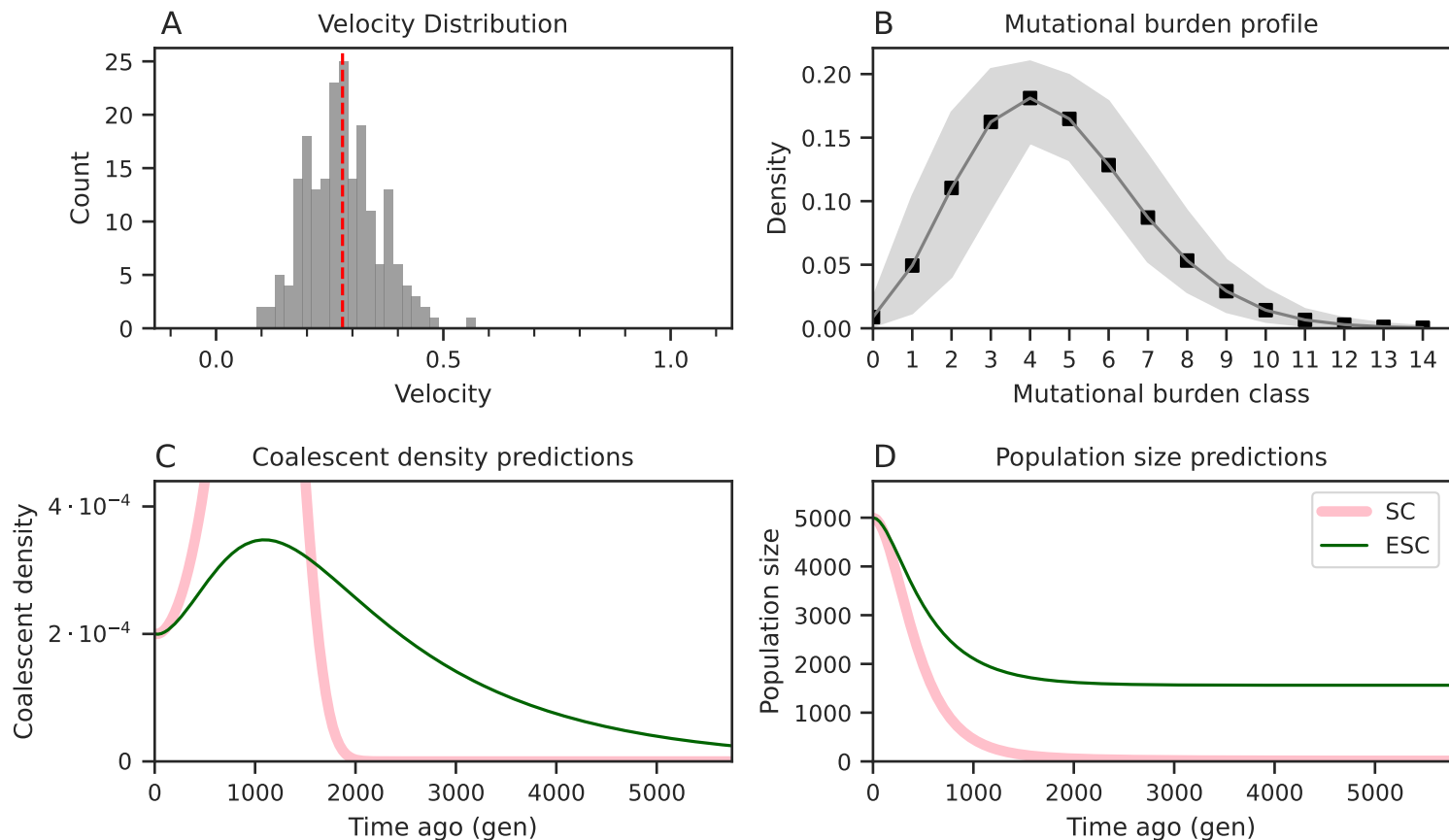
$N = 5e + 03$, $s = 0.00123$, $U_d = 0.006$, $\phi = 0.0464$
Mean Velocity= 0.2304 ± 0.0768 , $n = 200$ simulations



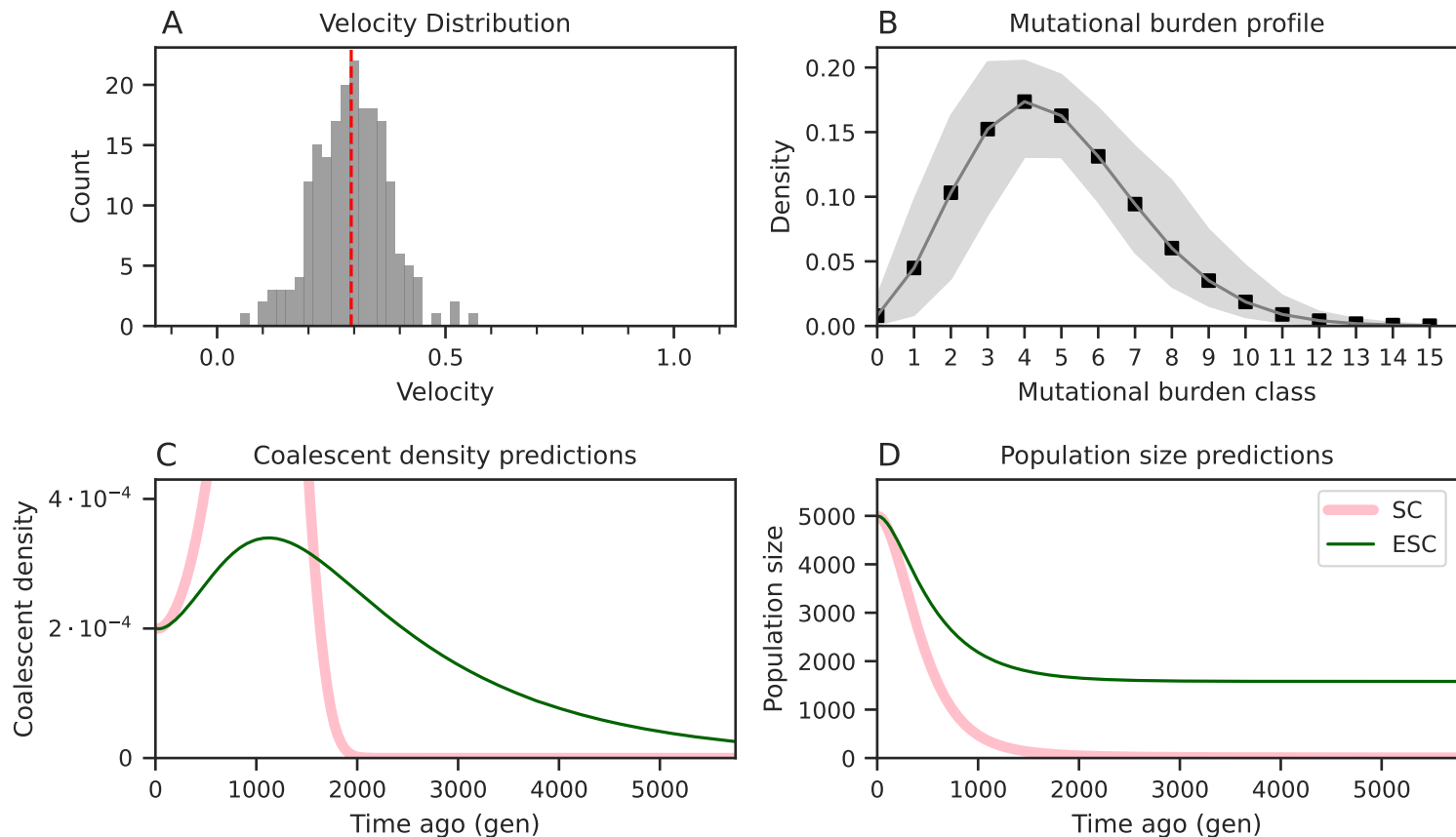
$N = 5e + 03$, $s = 0.00112$, $U_d = 0.006$, $\phi = 0.0264$
Mean Velocity= 0.2497 ± 0.0863 , $n = 200$ simulations



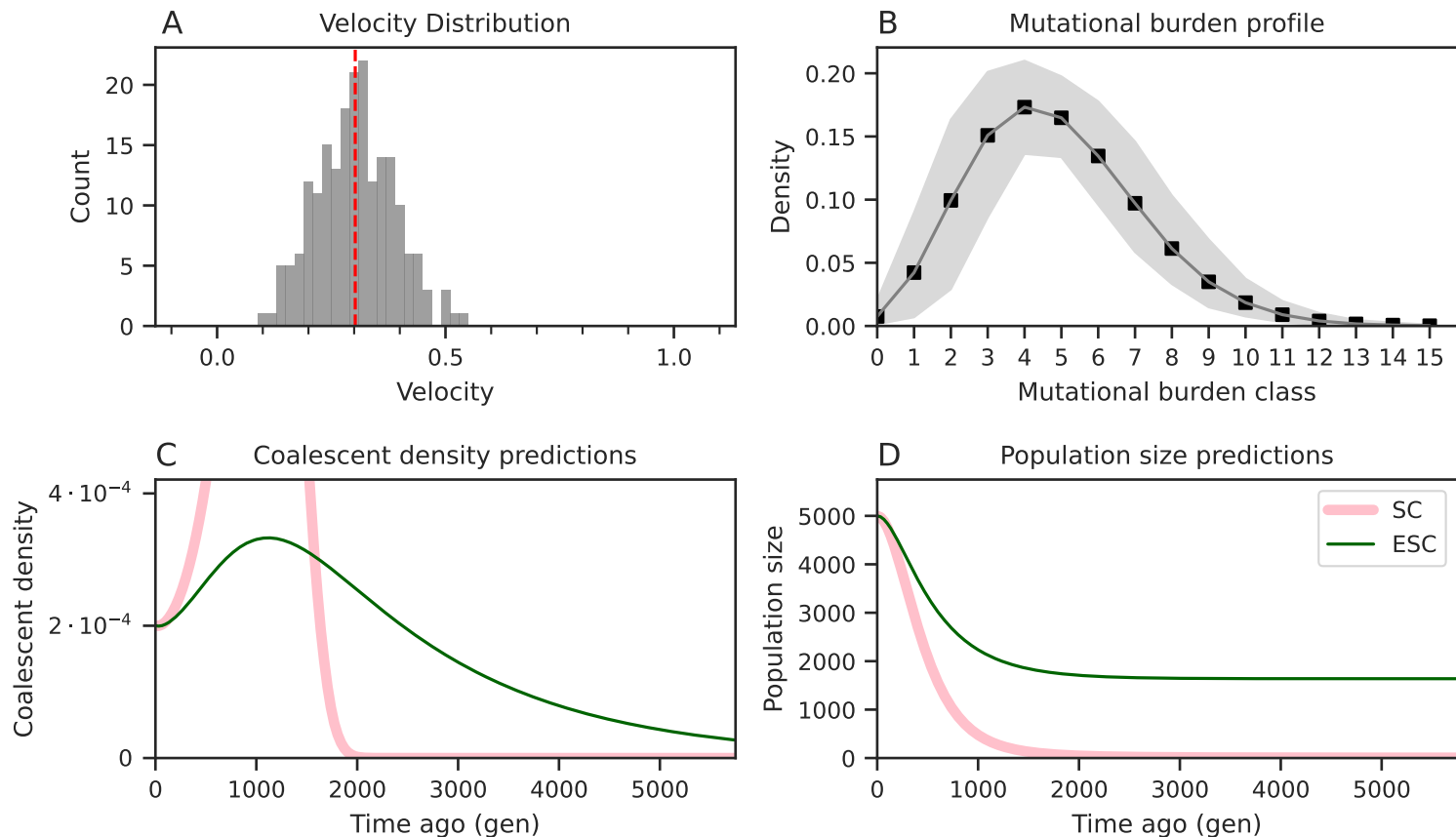
$N = 5e + 03$, $s = 0.001$, $U_d = 0.006$, $\phi = 0.0124$
Mean Velocity= 0.2782 ± 0.0796 , $n = 200$ simulations



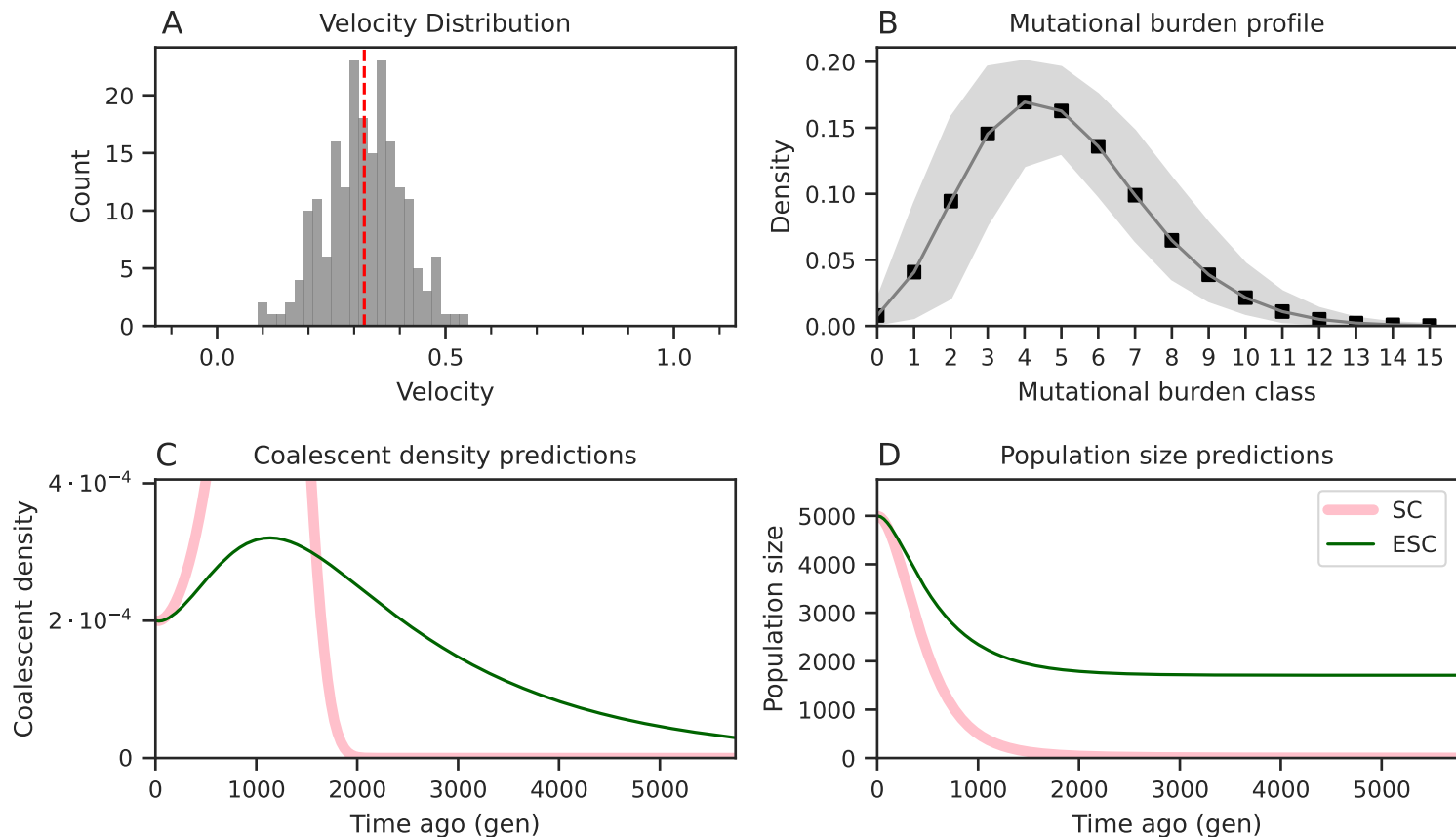
$N = 5e + 03$, $s = 9.2 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00677$
Mean Velocity = 0.2936 ± 0.0808 , $n = 200$ simulations



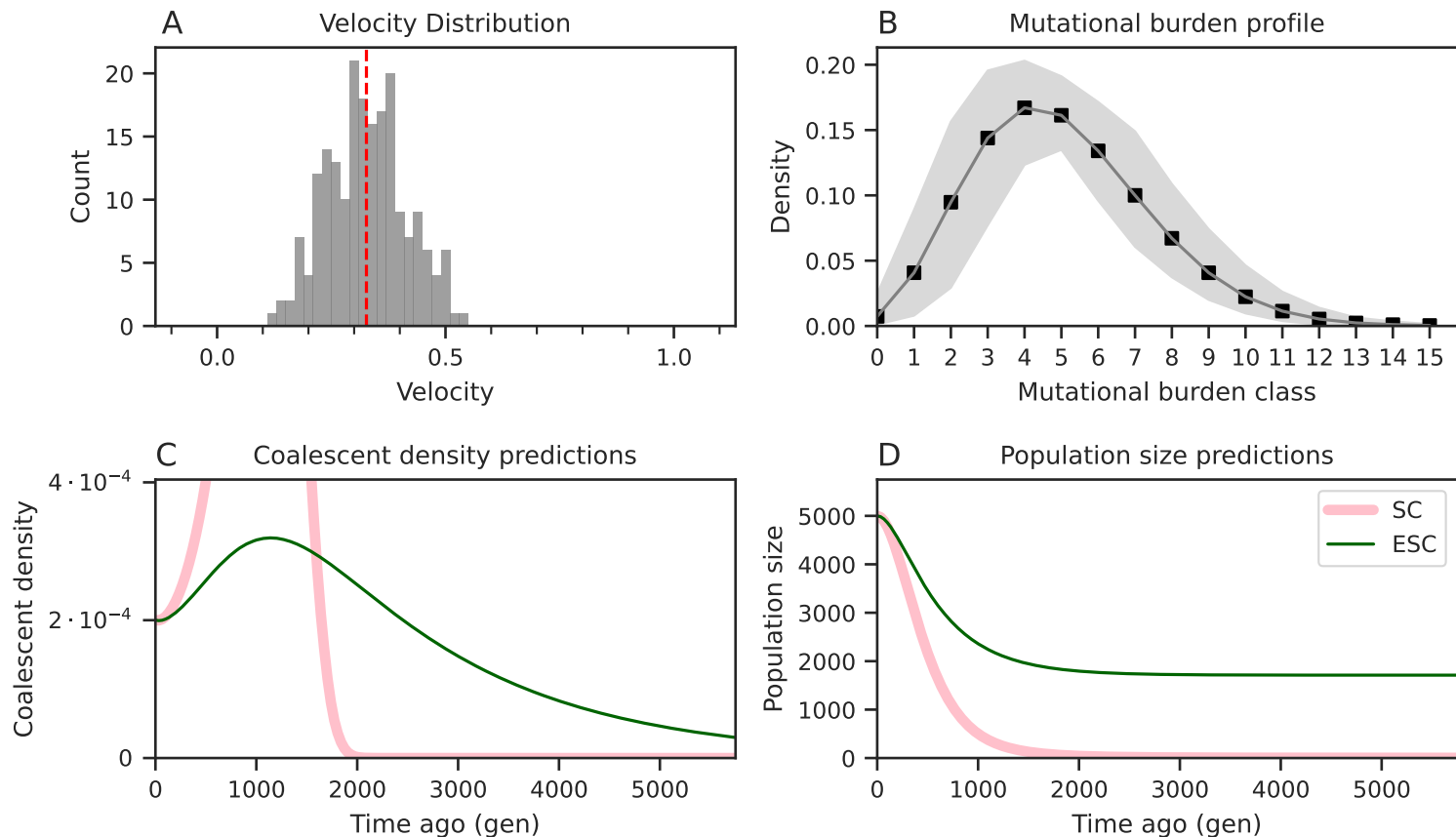
$N = 5e + 03$, $s = 9.01 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00578$
Mean Velocity= 0.3024 ± 0.0845 , $n = 200$ simulations



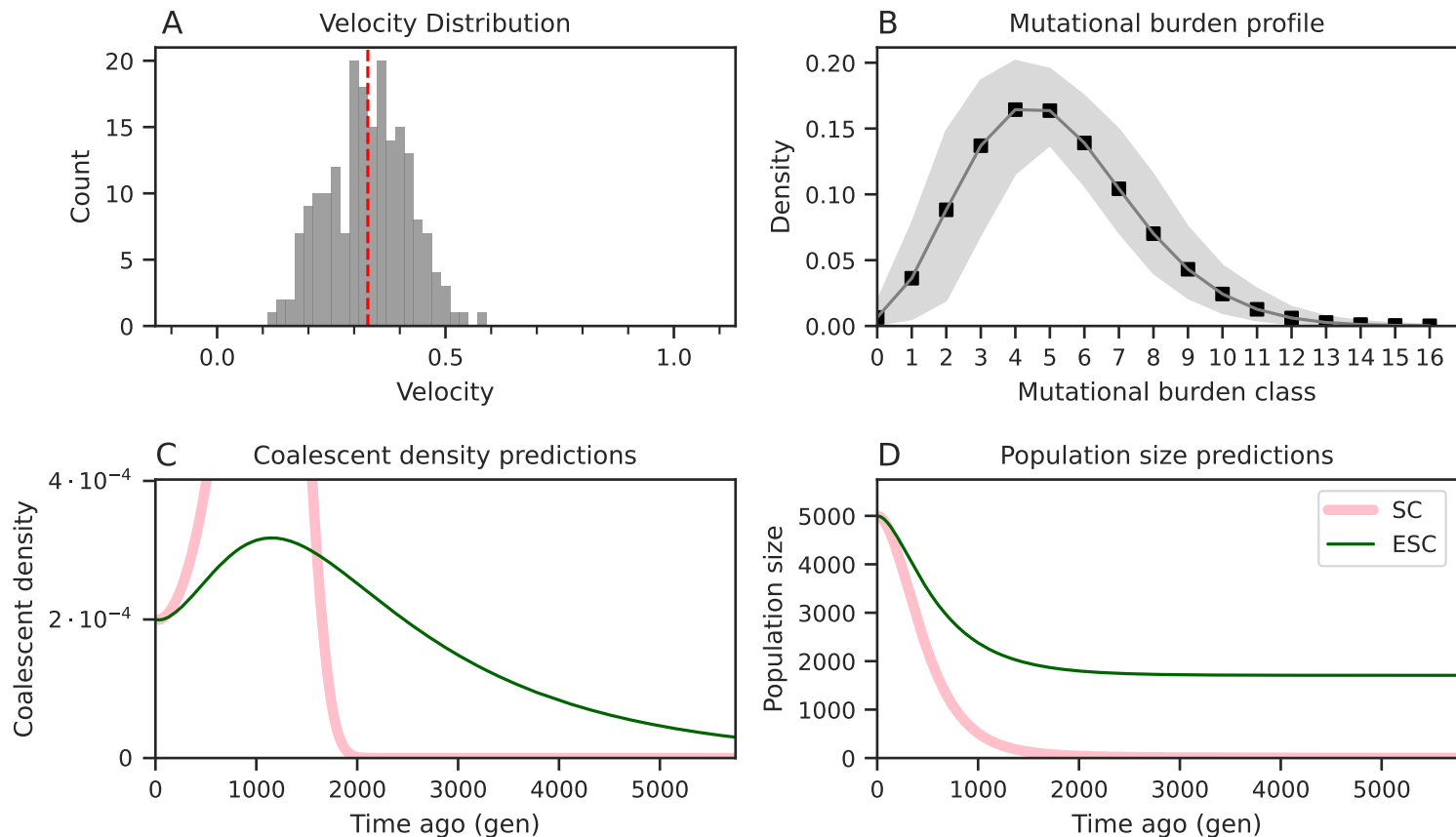
$N = 5e + 03$, $s = 8.4 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00332$
Mean Velocity= 0.3222 ± 0.0829 , $n = 200$ simulations



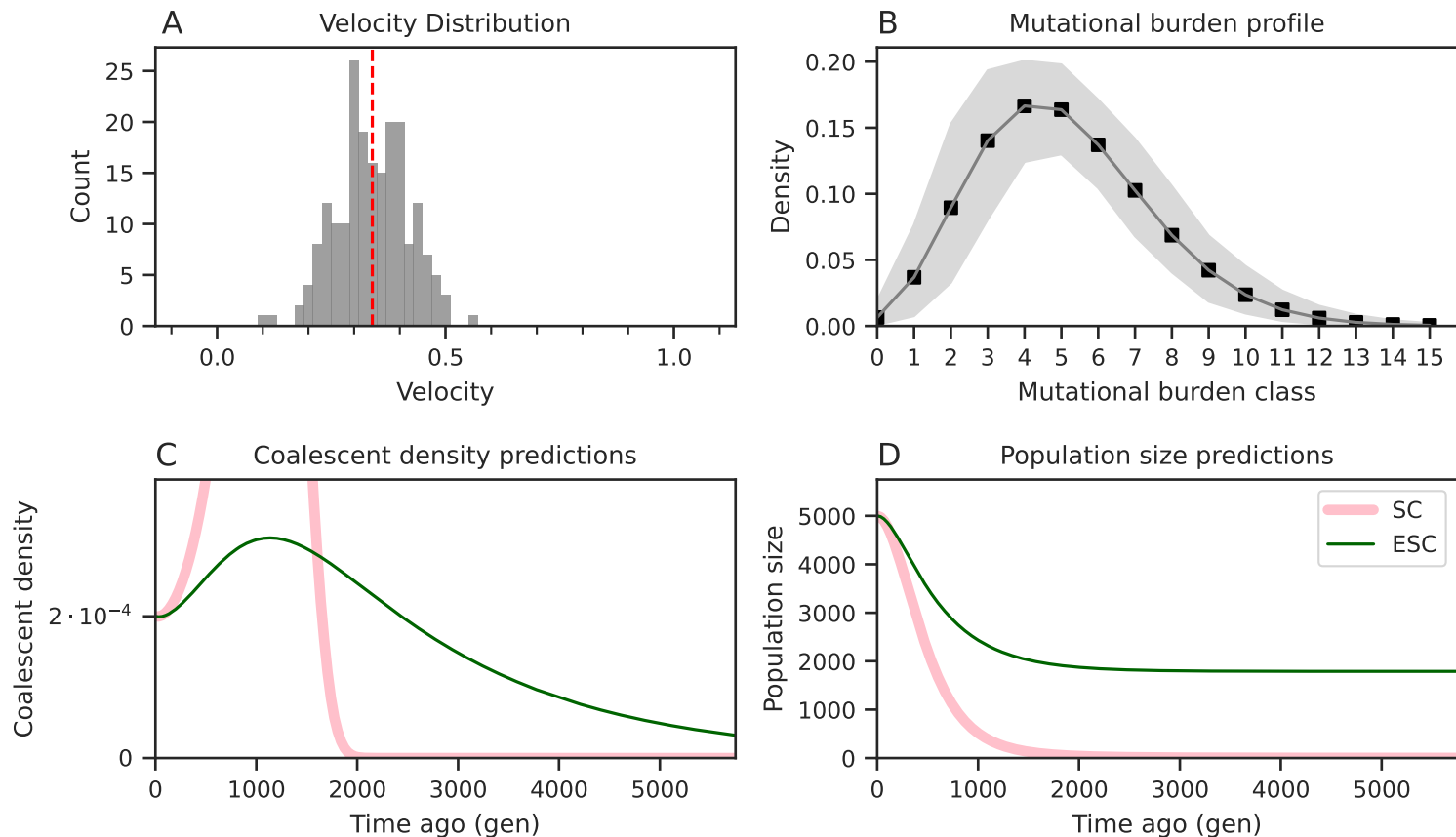
$N = 5e + 03$, $s = 8.32 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00307$
Mean Velocity= 0.3269 ± 0.0855 , $n = 200$ simulations



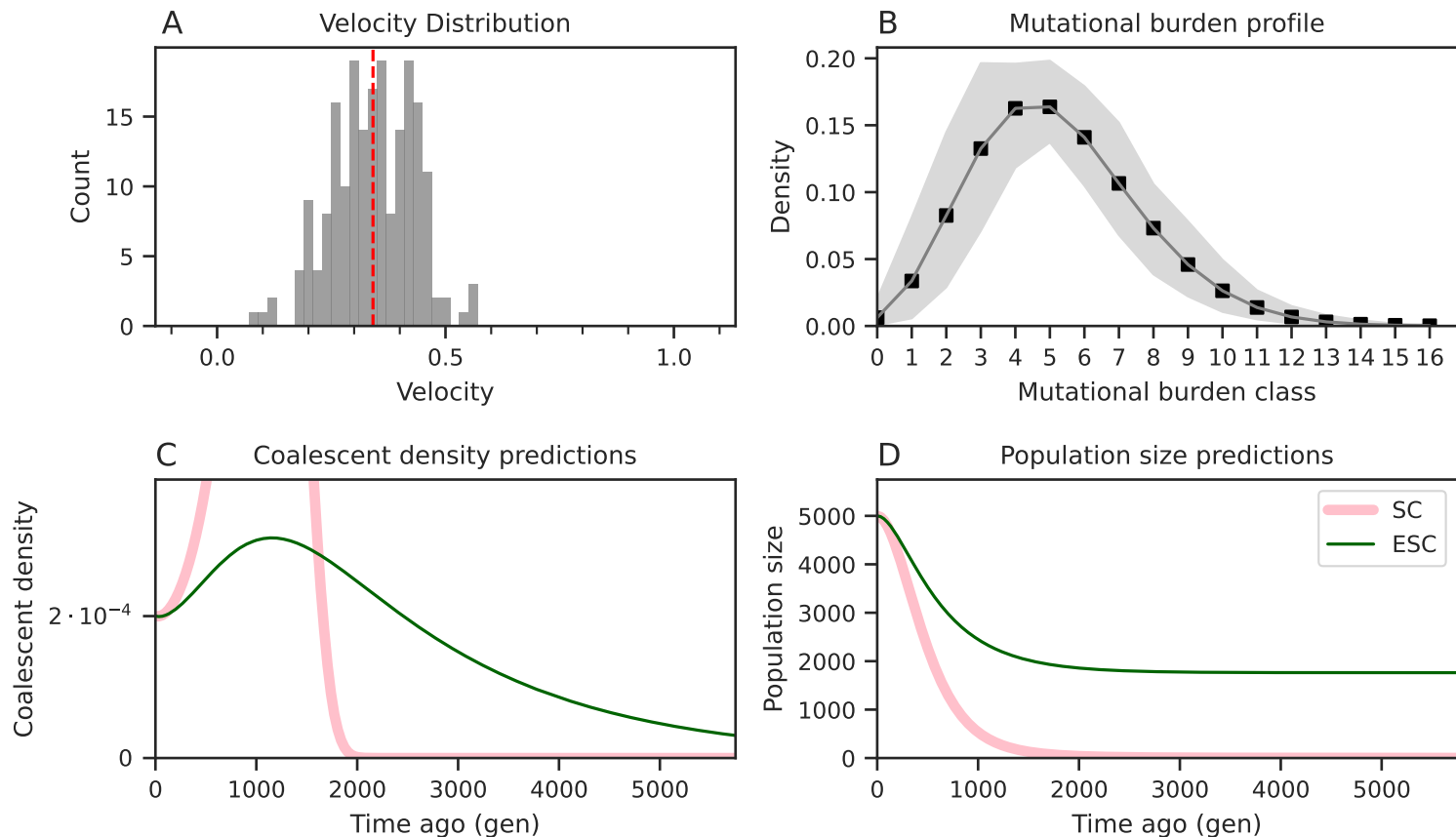
$N = 5e + 03$, $s = 8.02 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00226$
Mean Velocity= 0.3300 ± 0.0876 , $n = 200$ simulations



$N = 5e + 03$, $s = 8 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00221$
Mean Velocity= 0.3398 ± 0.0784 , $n = 200$ simulations

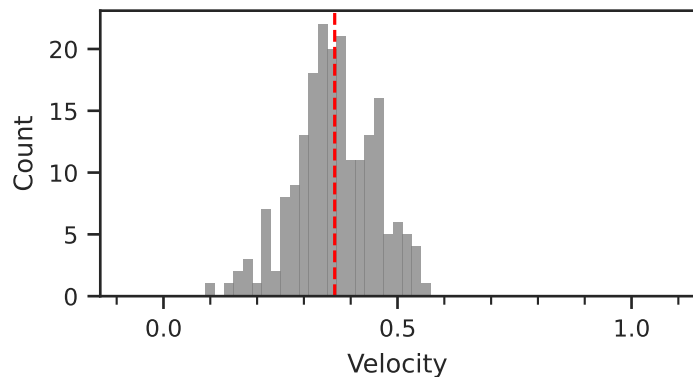


$N = 5e + 03$, $s = 7.6 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 0.00142$
Mean Velocity= 0.3413 ± 0.0896 , $n = 200$ simulations

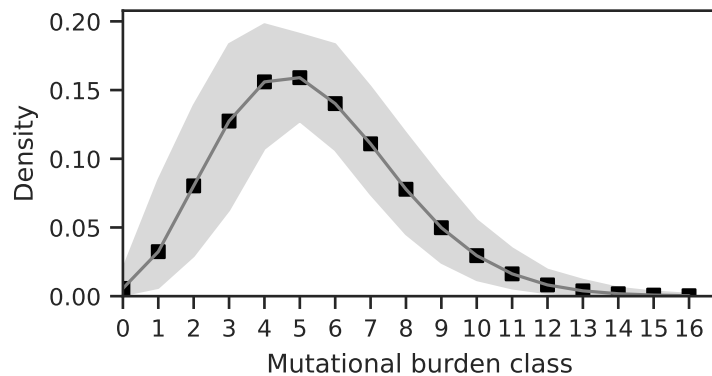


$N = 5e + 03$, $s = 7.03 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 6.91 \cdot 10^{-4}$
Mean Velocity= 0.3658 ± 0.0874 , $n = 200$ simulations

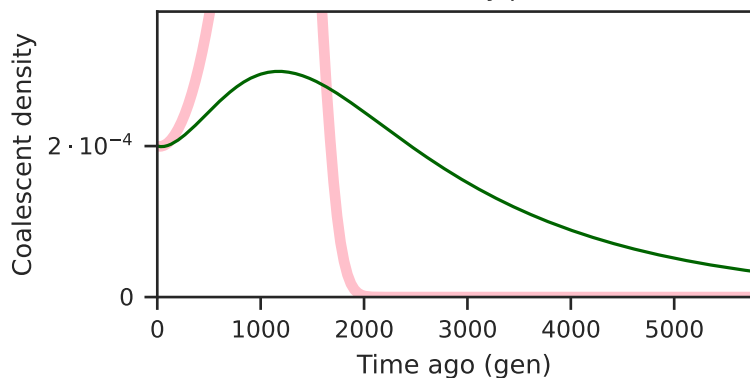
A Velocity Distribution



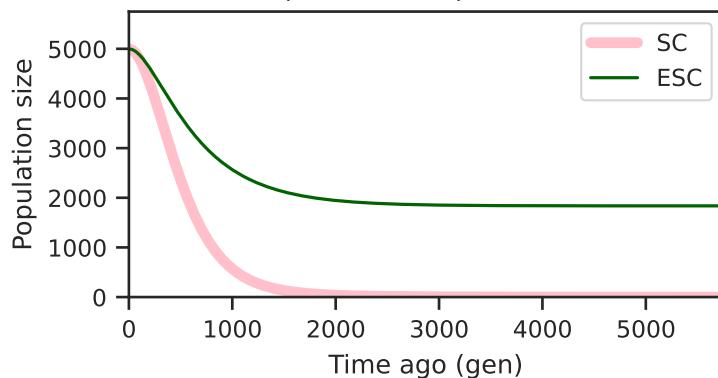
B Mutational burden profile



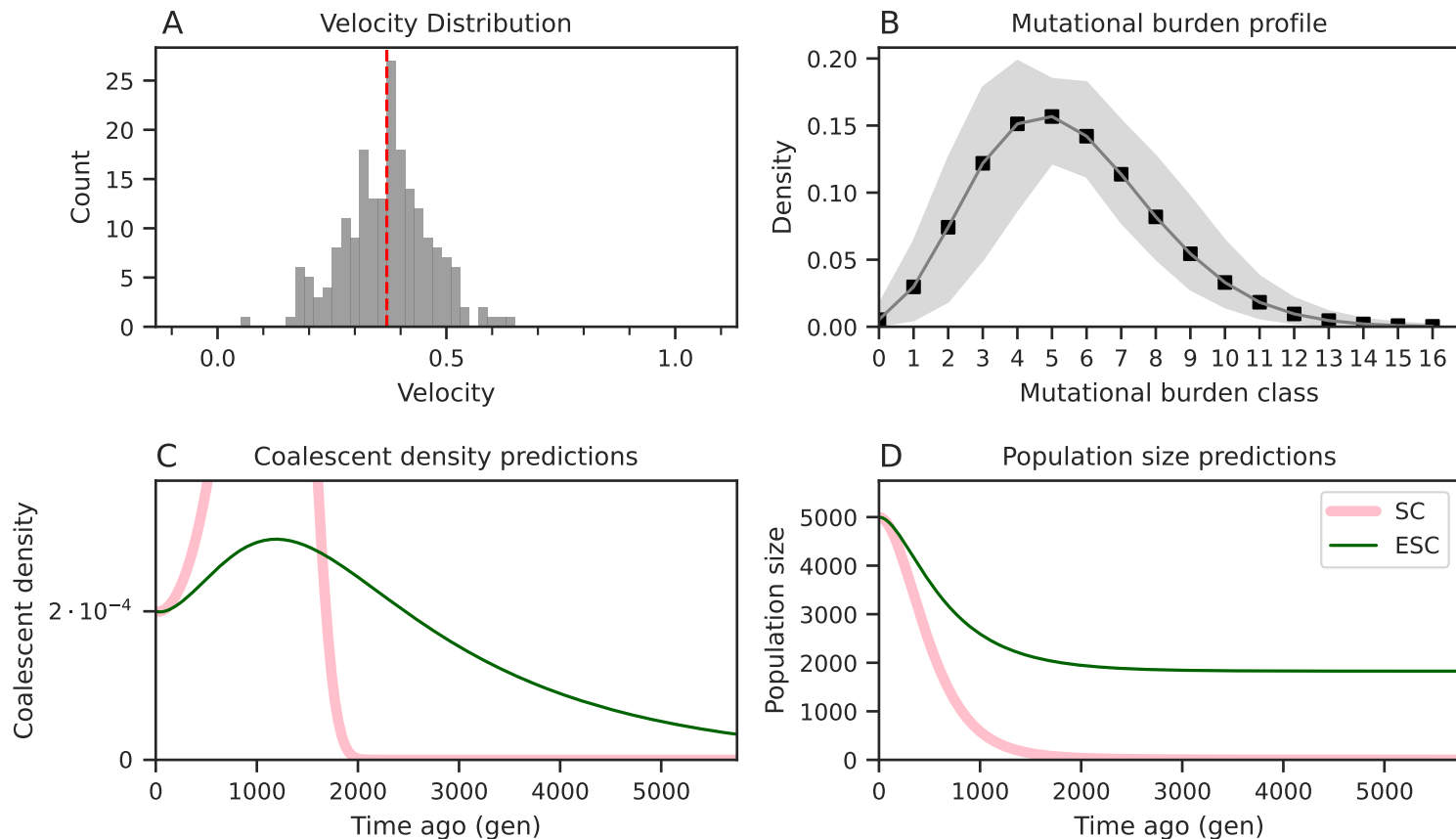
C Coalescent density predictions



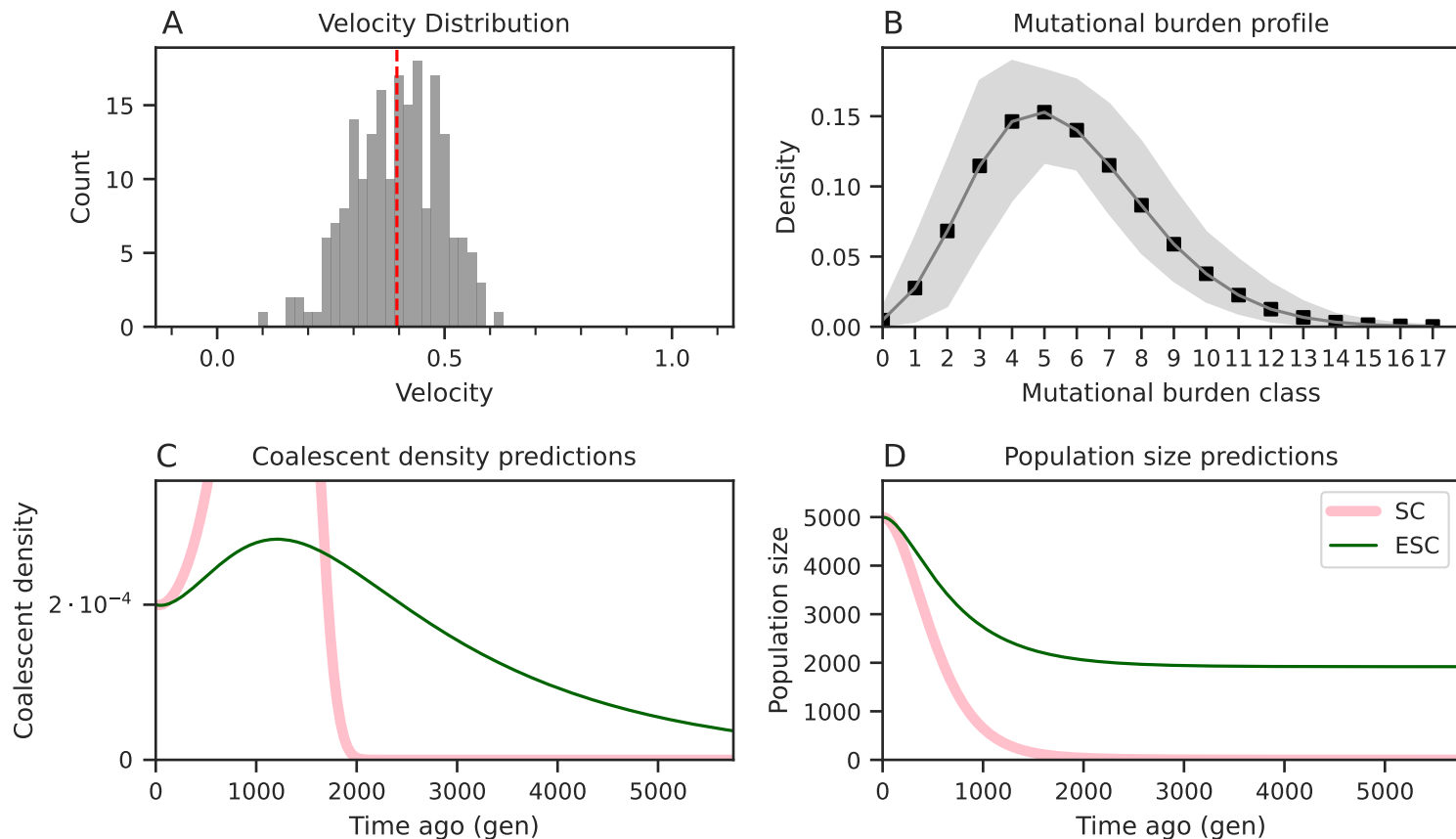
D Population size predictions



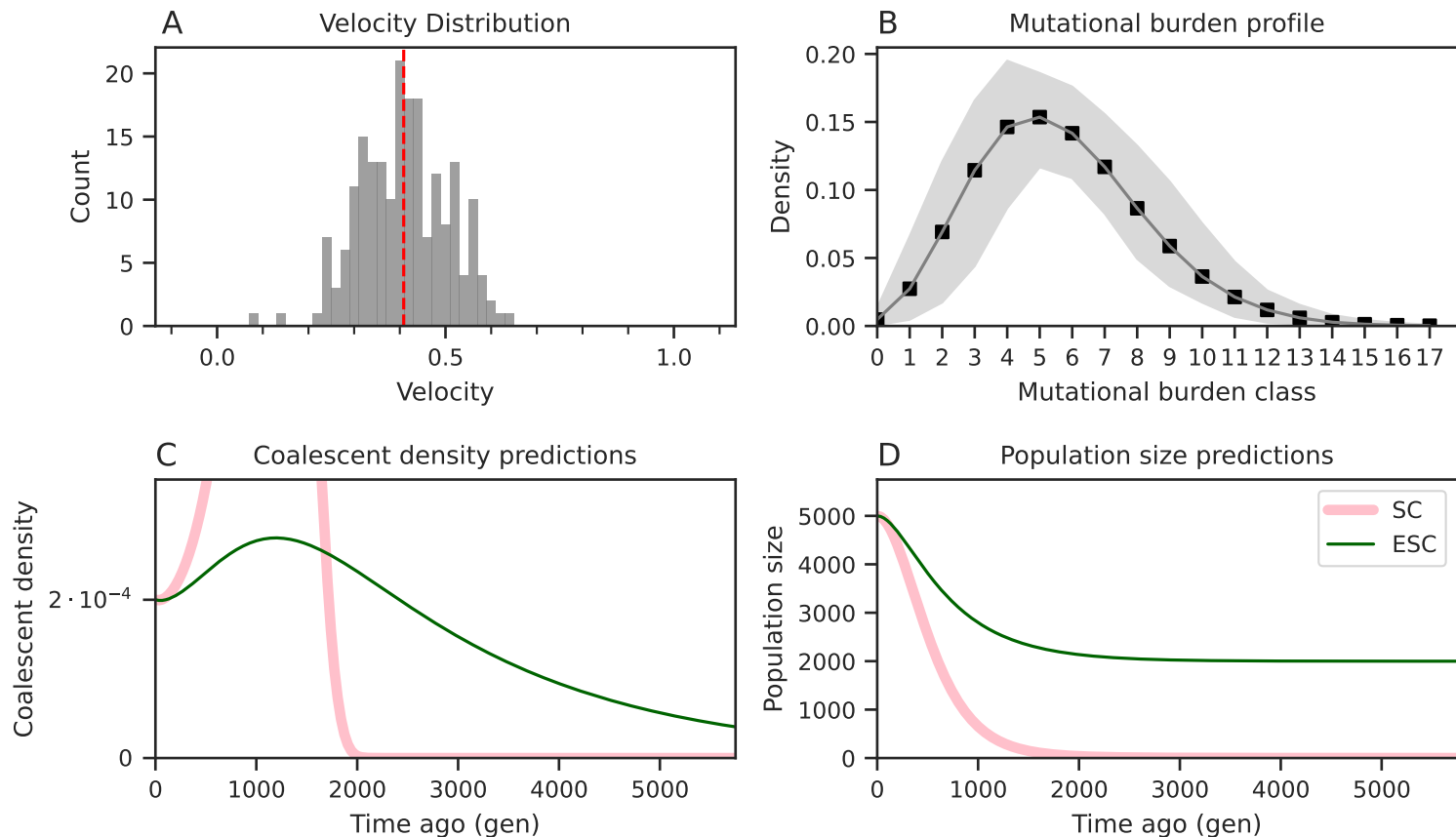
$N = 5e + 03$, $s = 6.8 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 5.01 \cdot 10^{-4}$
Mean Velocity= 0.3698 ± 0.0948 , $n = 200$ simulations



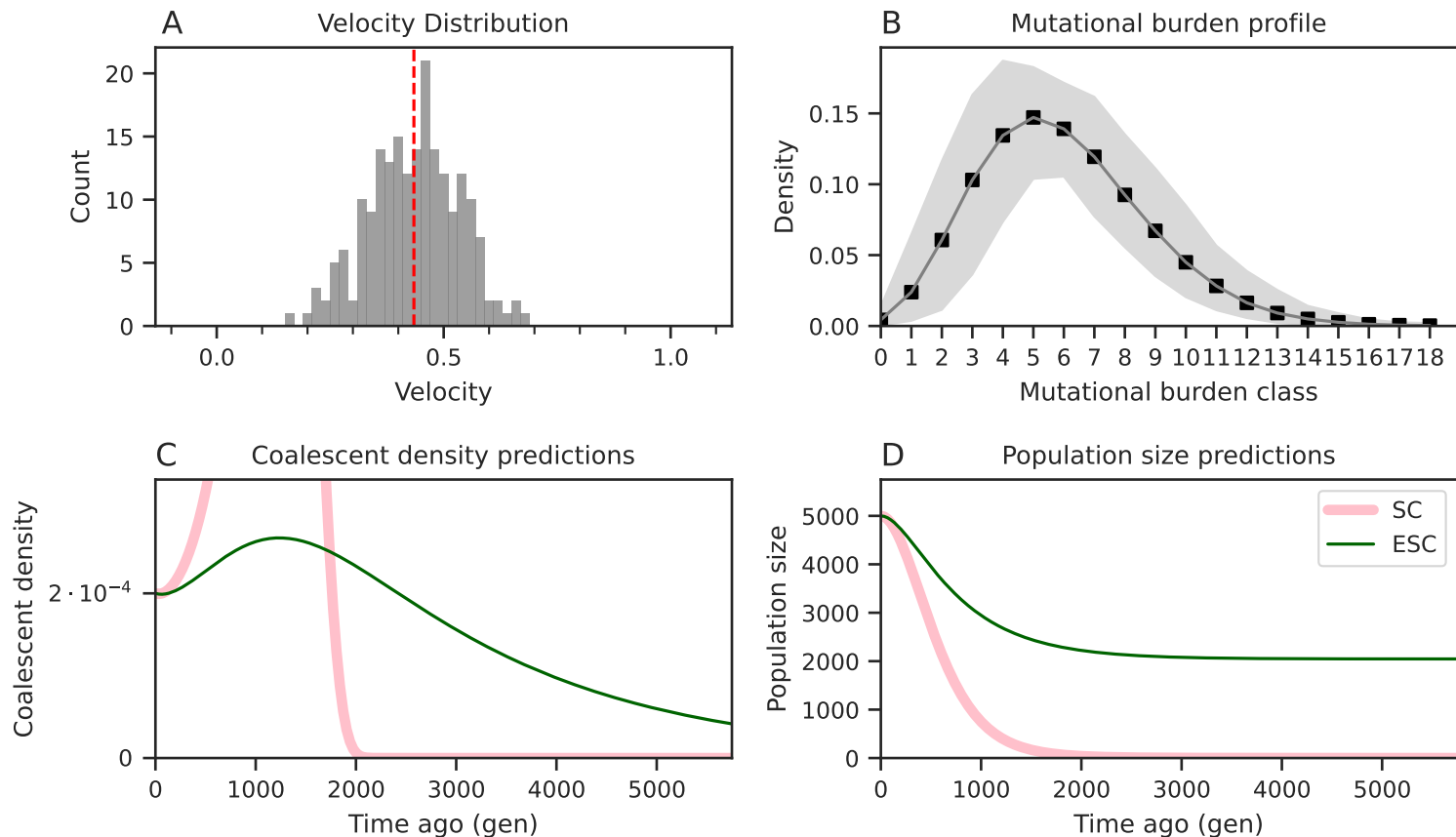
$N = 5e + 03$, $s = 6.04 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.46 \cdot 10^{-4}$
Mean Velocity = 0.3952 ± 0.0951 , $n = 200$ simulations



$N = 5e + 03$, $s = 6 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.36 \cdot 10^{-4}$
Mean Velocity = 0.4085 ± 0.0955 , $n = 200$ simulations

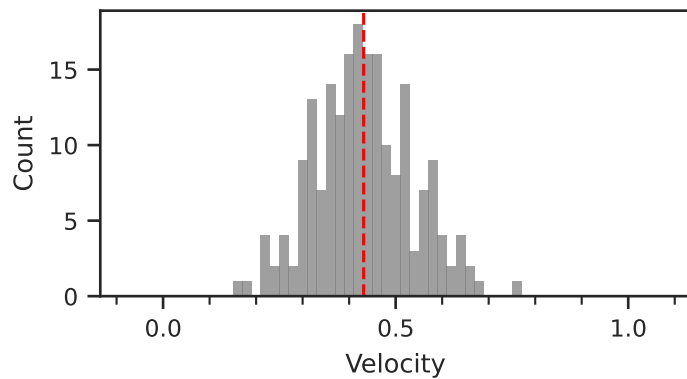


$N = 5e + 03$, $s = 5.2 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 2.53 \cdot 10^{-5}$
Mean Velocity = 0.4344 ± 0.0987 , $n = 200$ simulations

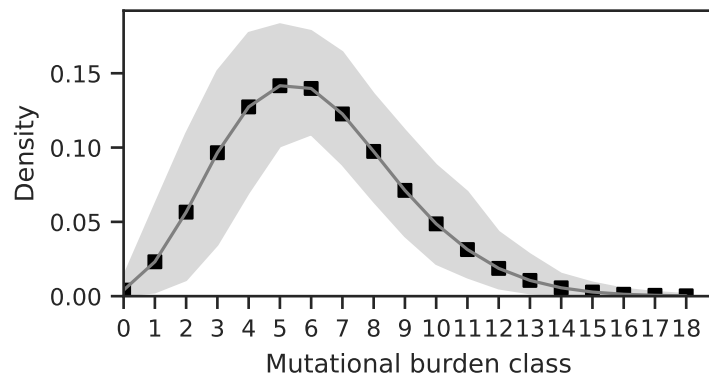


$N = 5e + 03$, $s = 5.05 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.75 \cdot 10^{-5}$
Mean Velocity = 0.4313 ± 0.1050 , $n = 200$ simulations

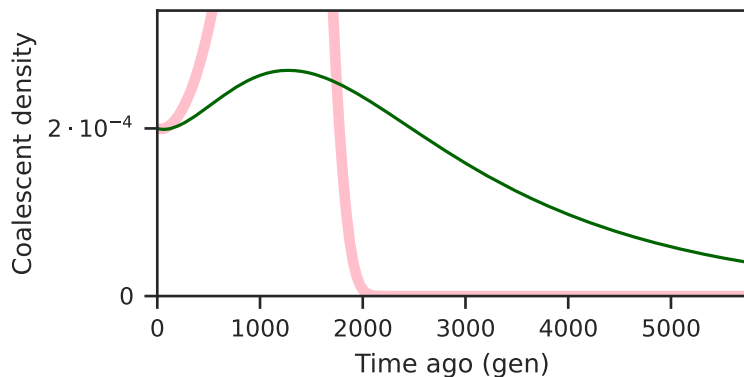
A Velocity Distribution



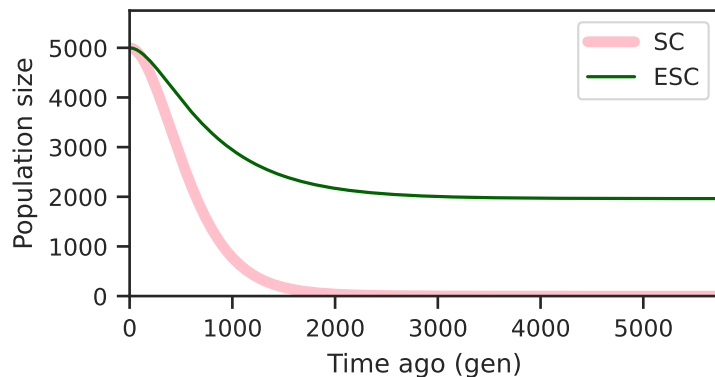
B Mutational burden profile



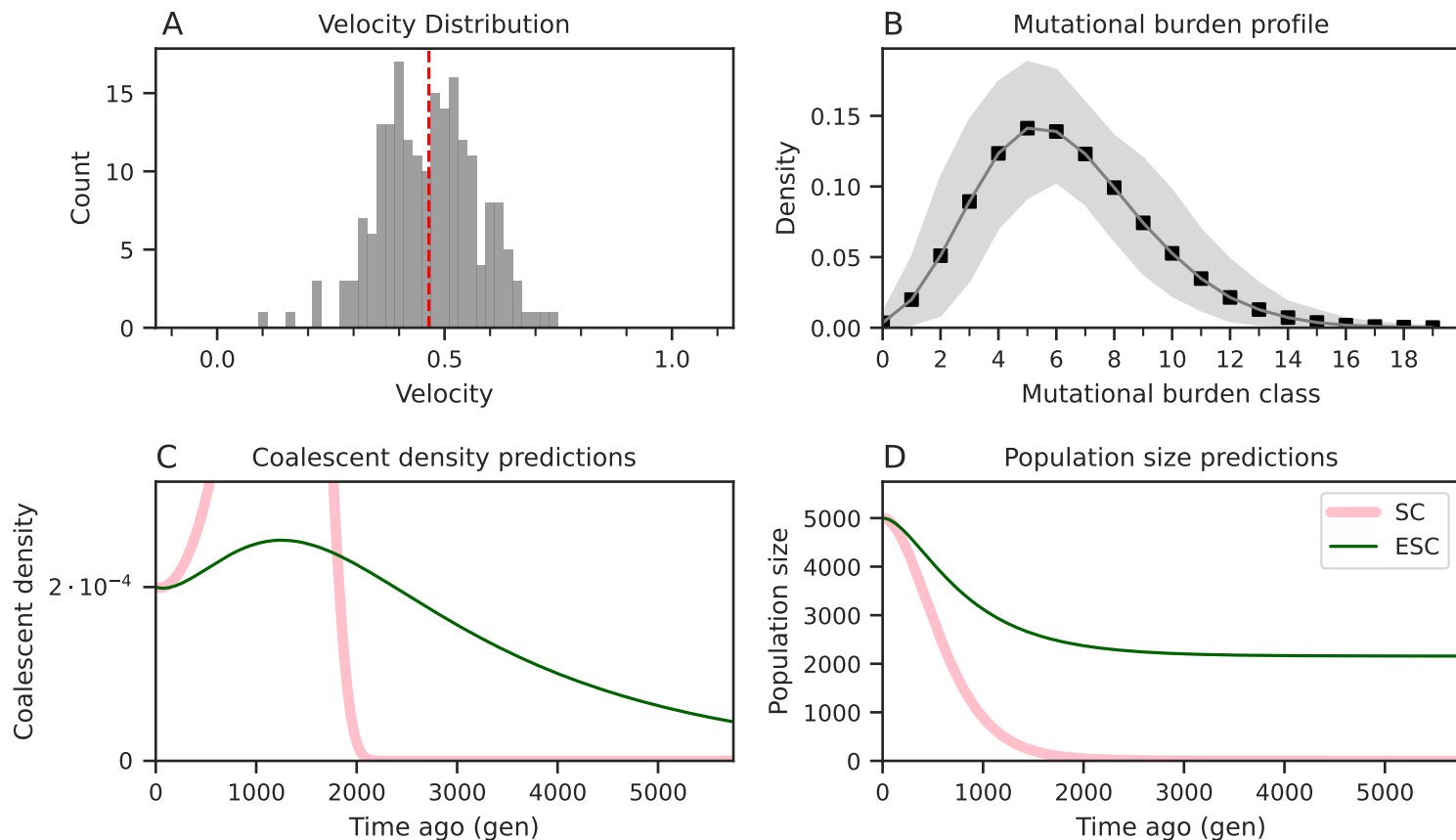
C Coalescent density predictions



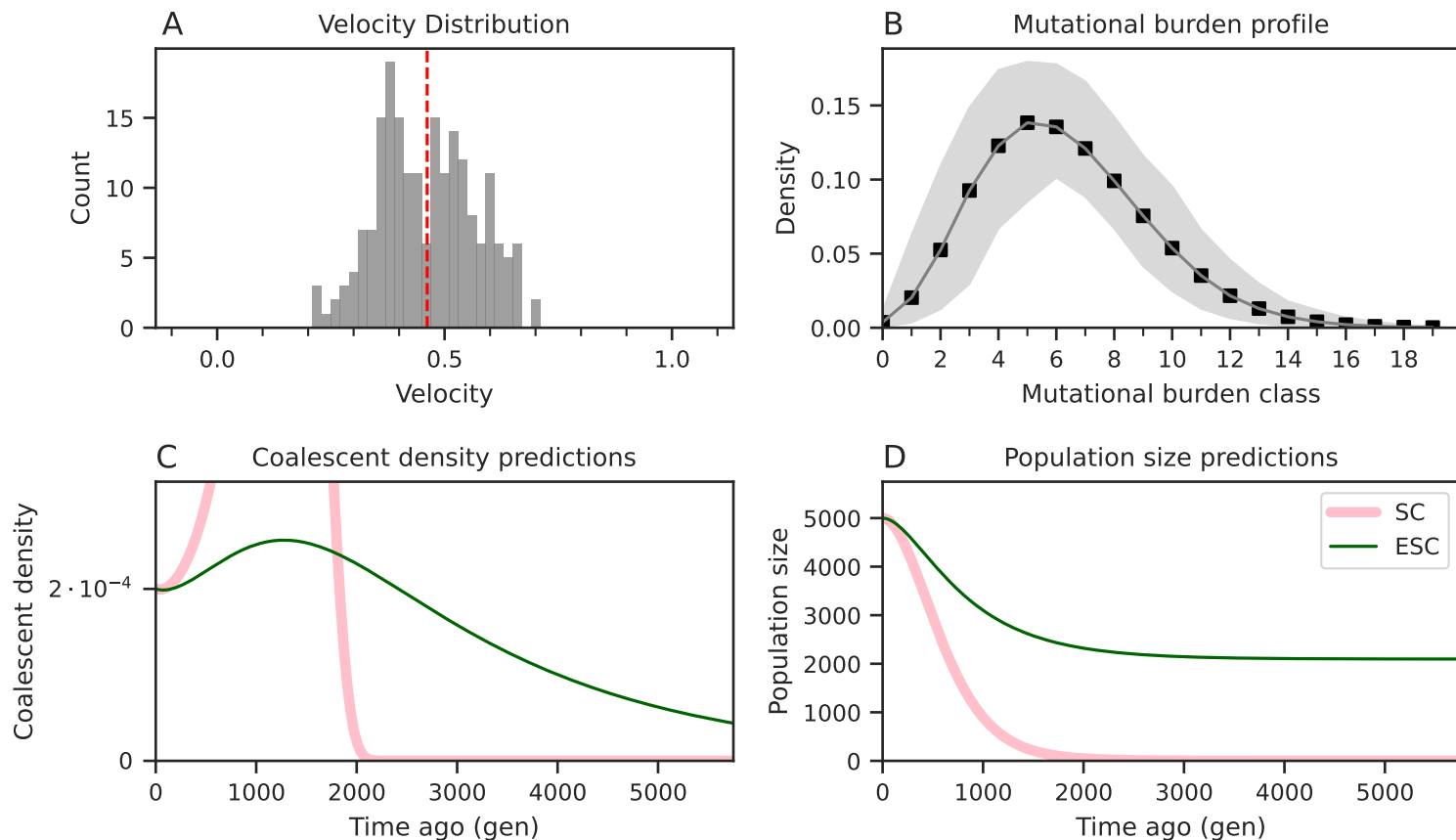
D Population size predictions



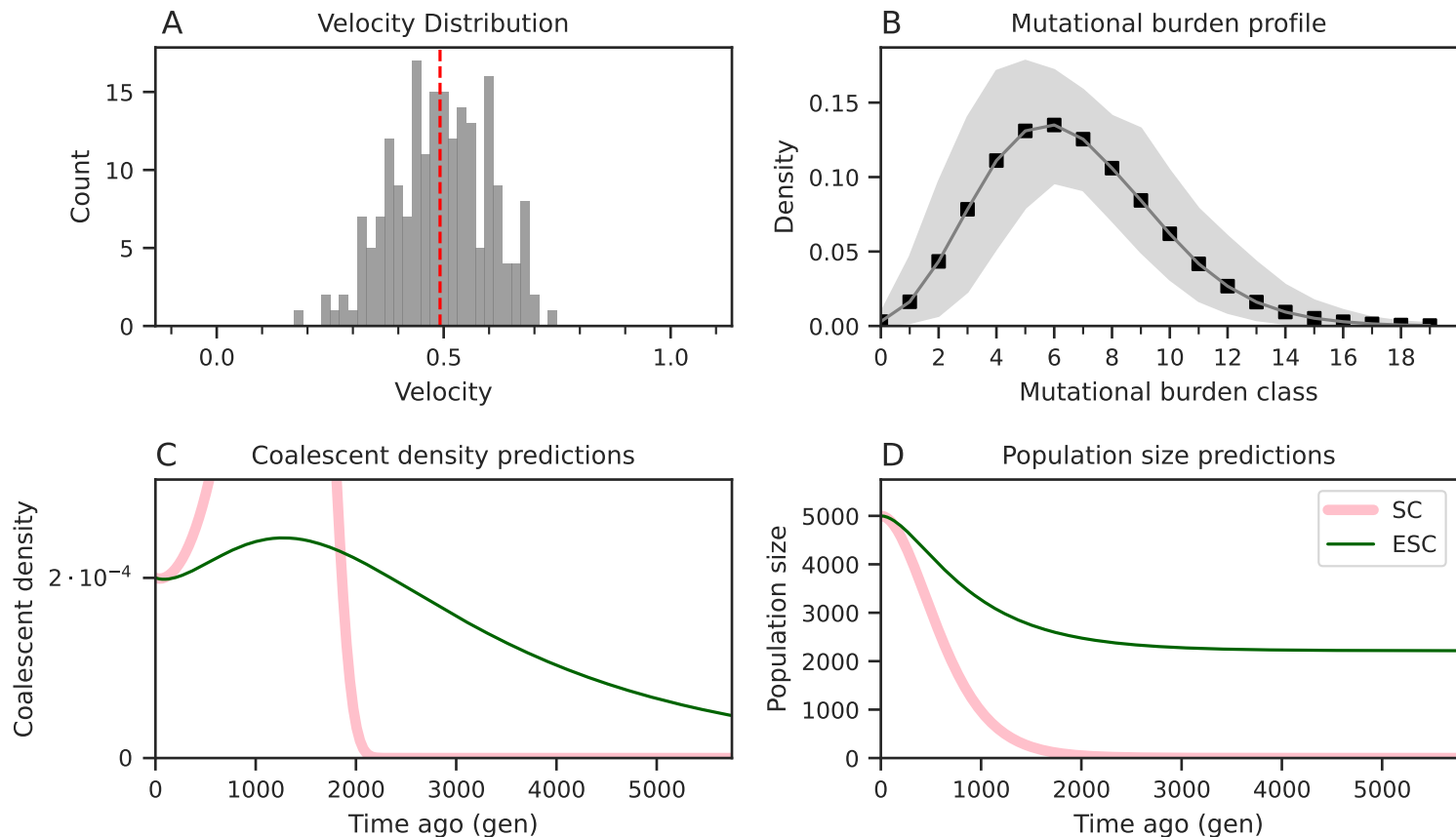
$N = 5e + 03$, $s = 4.4 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 2.63 \cdot 10^{-6}$
Mean Velocity = 0.4655 ± 0.1077 , $n = 200$ simulations



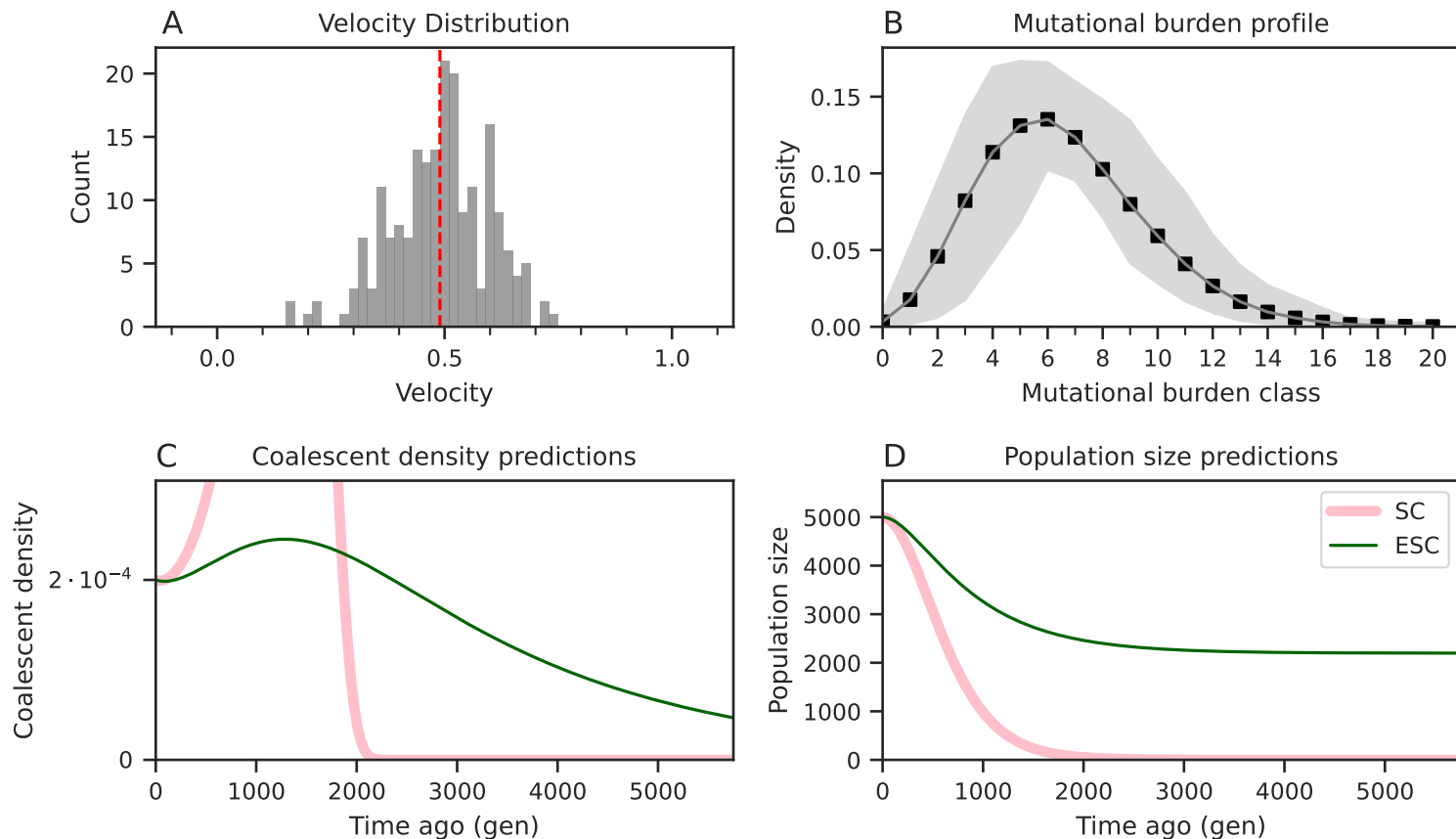
$N = 5e + 03$, $s = 4.36 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 2.3 \cdot 10^{-6}$
Mean Velocity= 0.4618 ± 0.1067 , $n = 200$ simulations



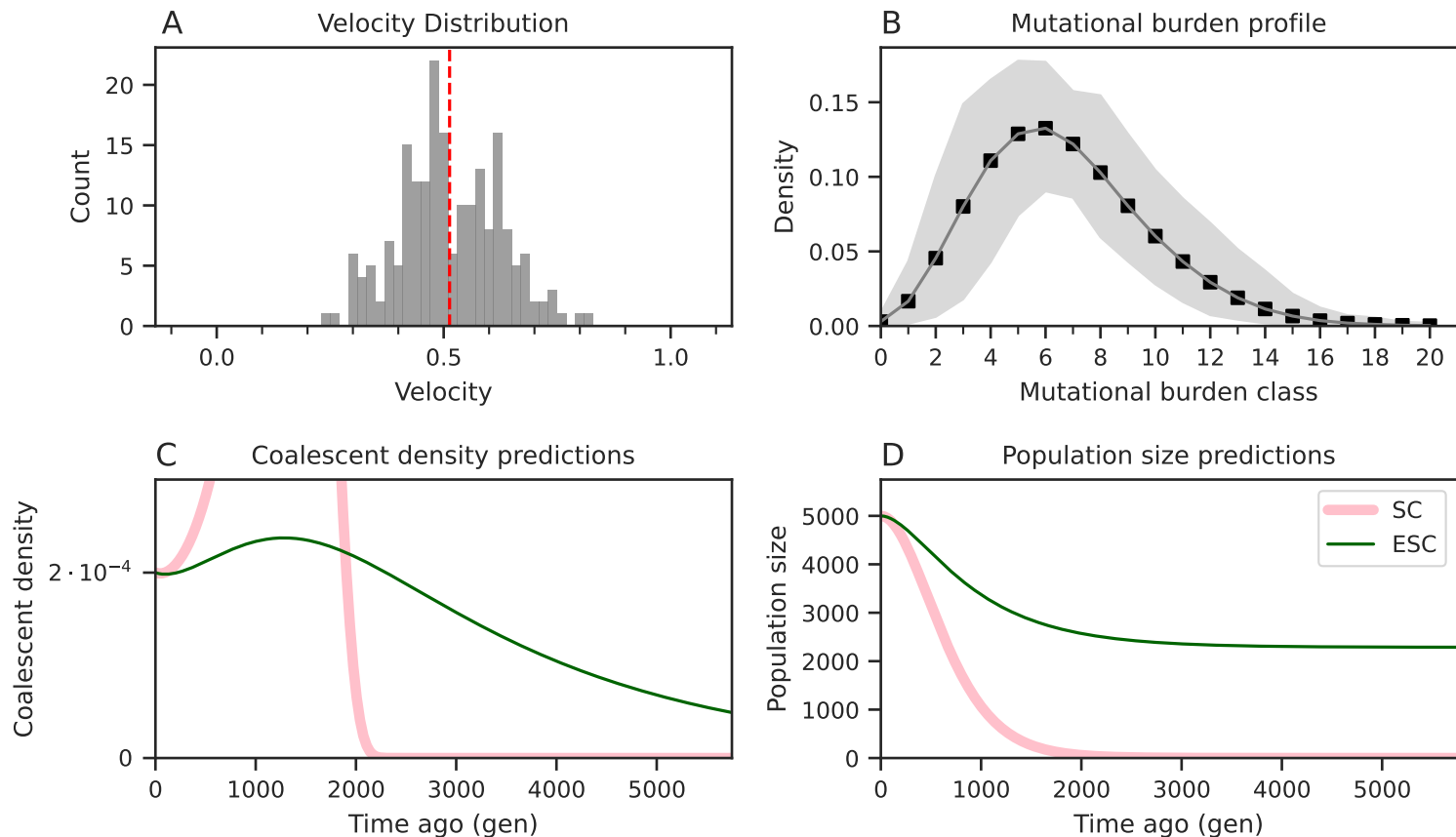
$N = 5e + 03$, $s = 4.06 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 7.75 \cdot 10^{-7}$
Mean Velocity = 0.4920 ± 0.1065 , $n = 200$ simulations



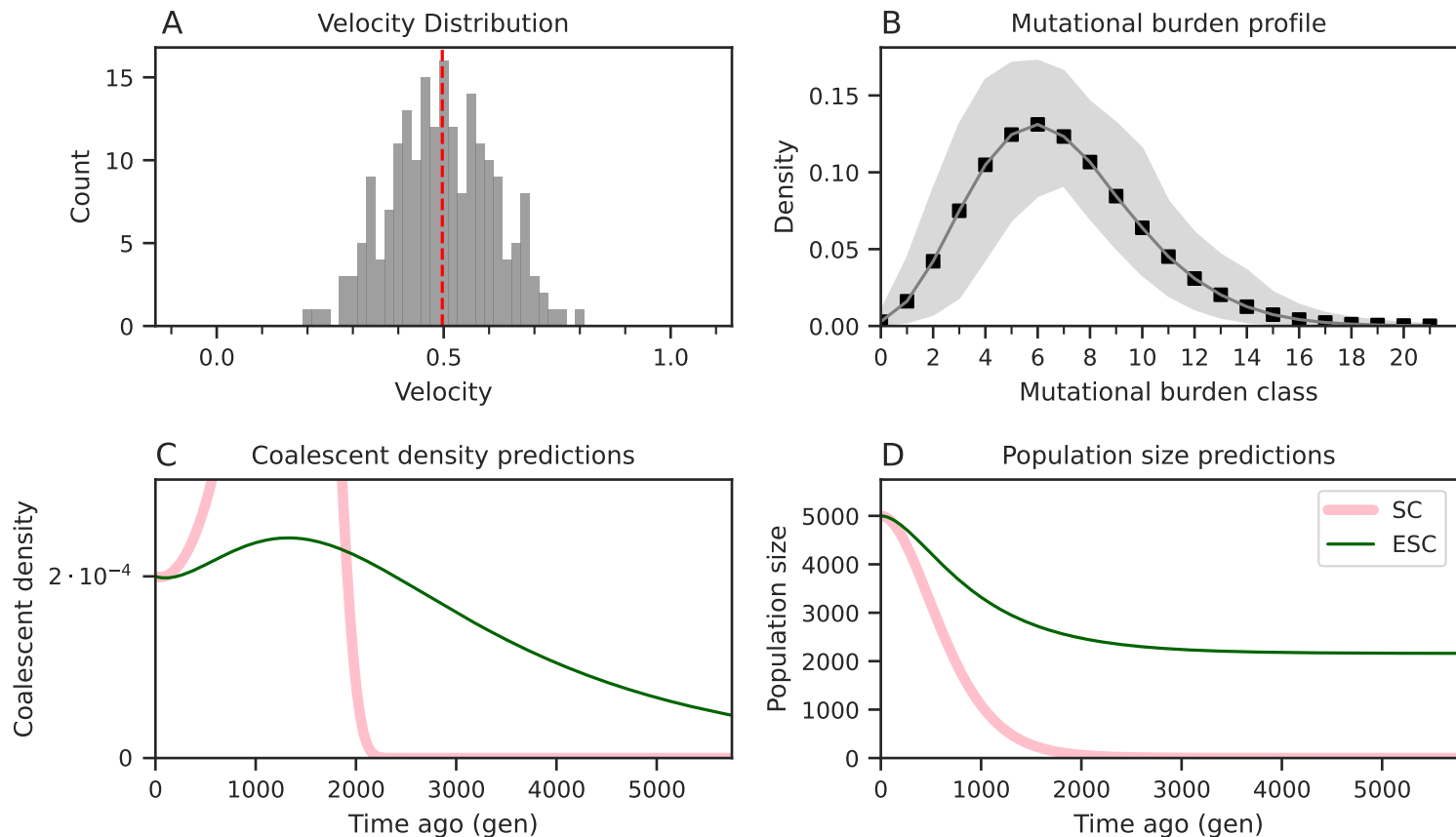
$N = 5e + 03$, $s = 4 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 6.12 \cdot 10^{-7}$
Mean Velocity= 0.4898 ± 0.1087 , $n = 200$ simulations



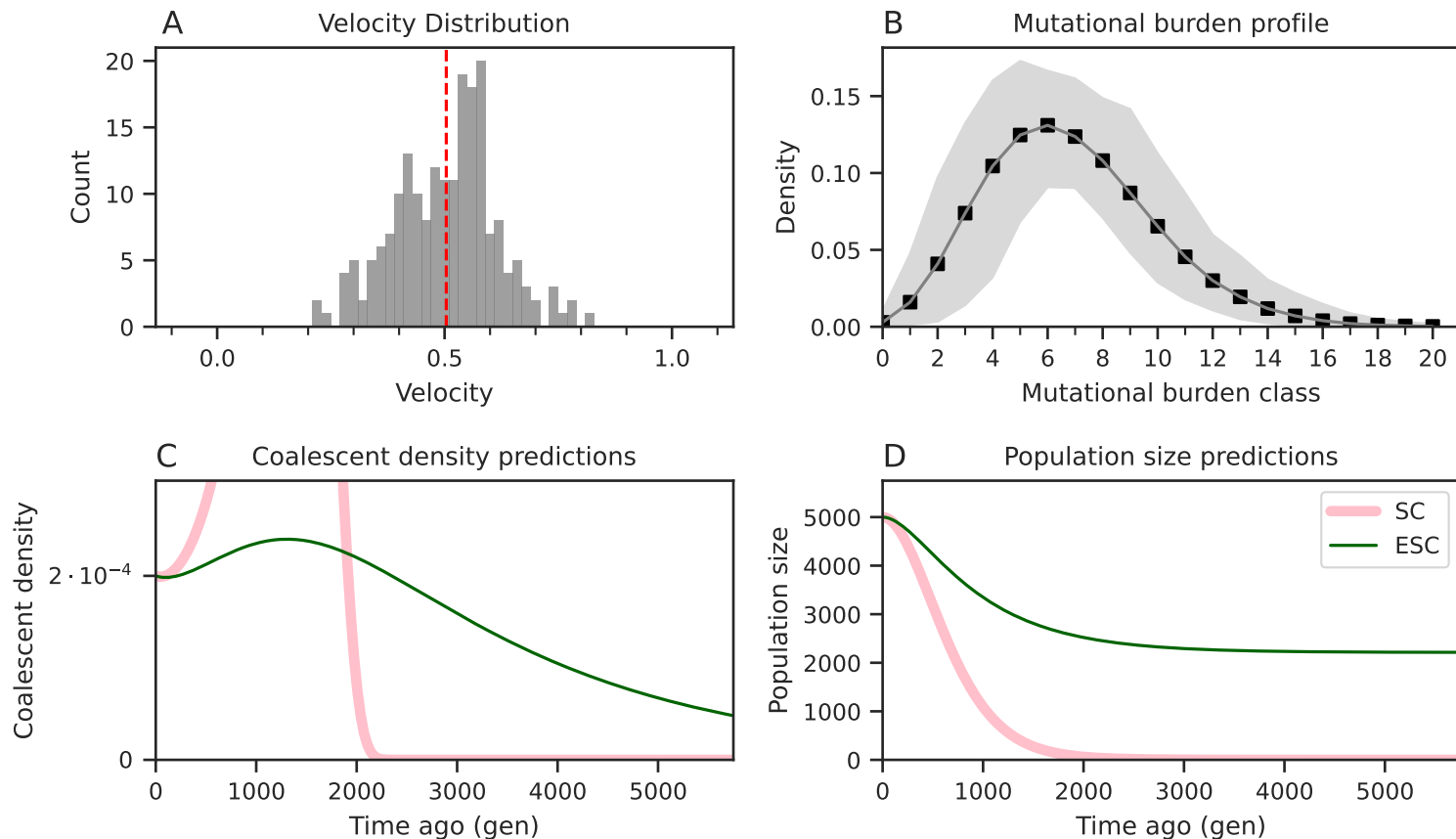
$N = 5e + 03$, $s = 3.68 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.53 \cdot 10^{-7}$
Mean Velocity = 0.5130 ± 0.1115 , $n = 200$ simulations



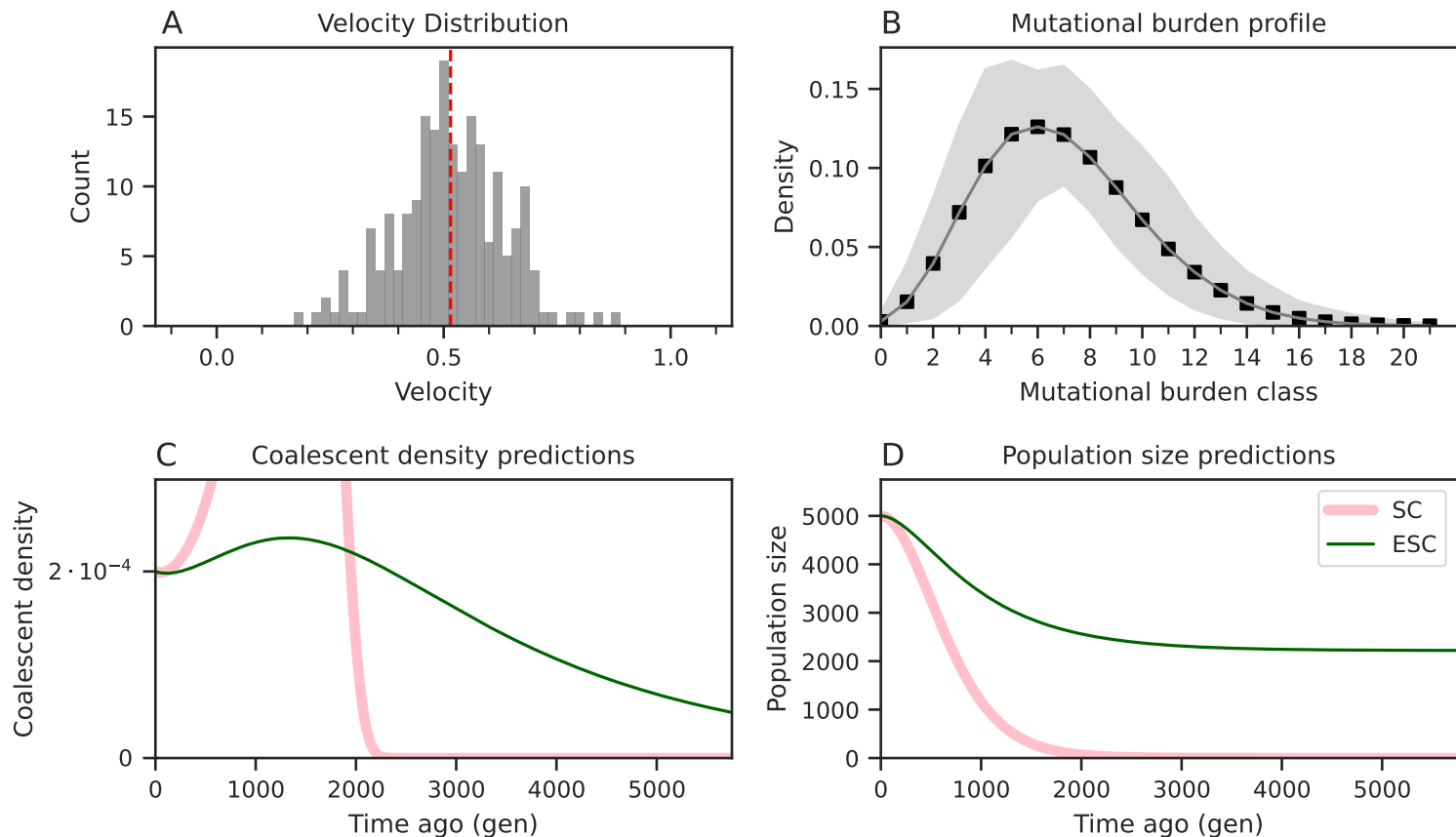
$N = 5e + 03$, $s = 3.6 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.06 \cdot 10^{-7}$
Mean Velocity = 0.4970 ± 0.1140 , $n = 200$ simulations



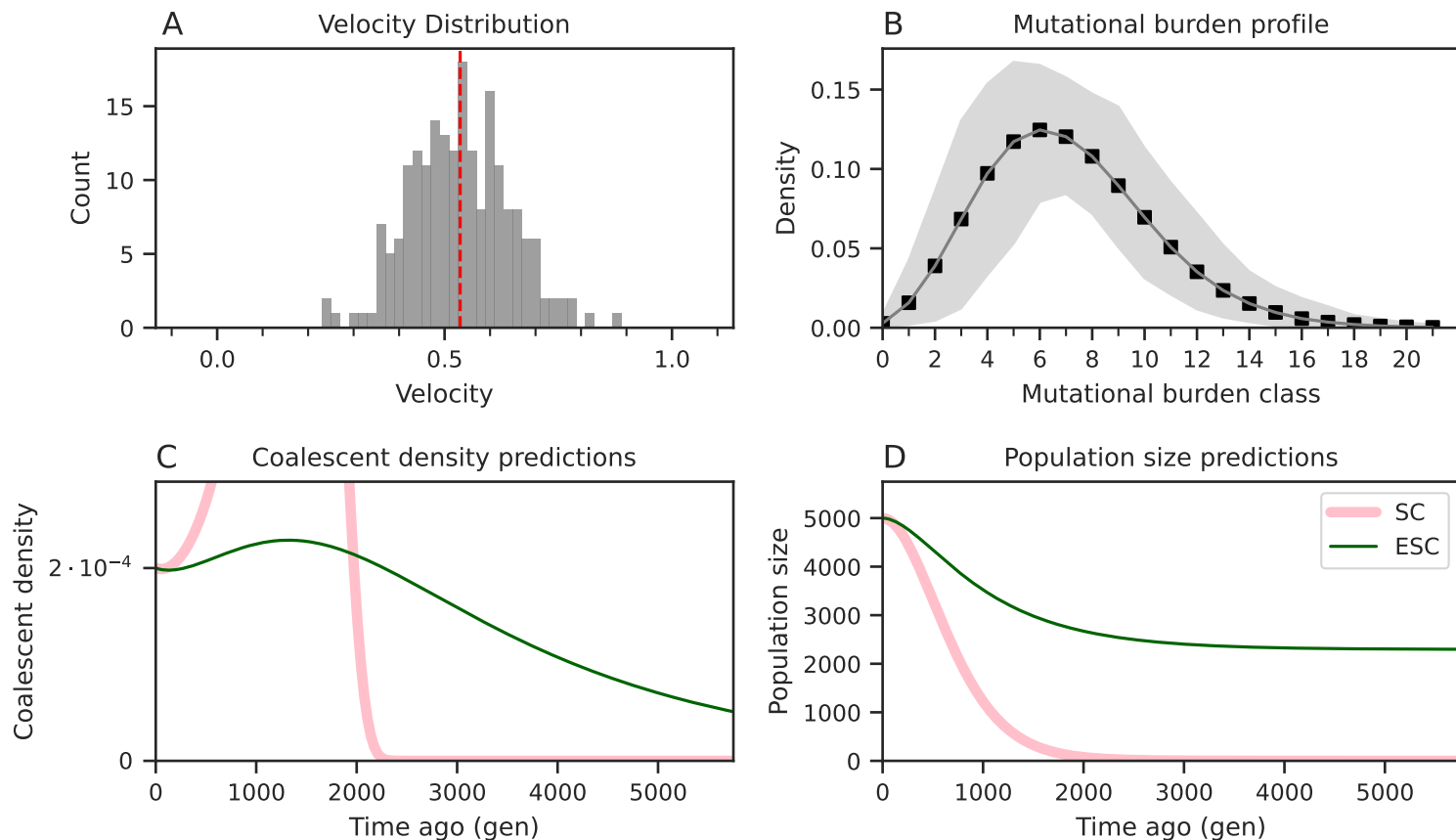
$N = 5e + 03$, $s = 3.6 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.04 \cdot 10^{-7}$
Mean Velocity = 0.5038 ± 0.1125 , $n = 200$ simulations



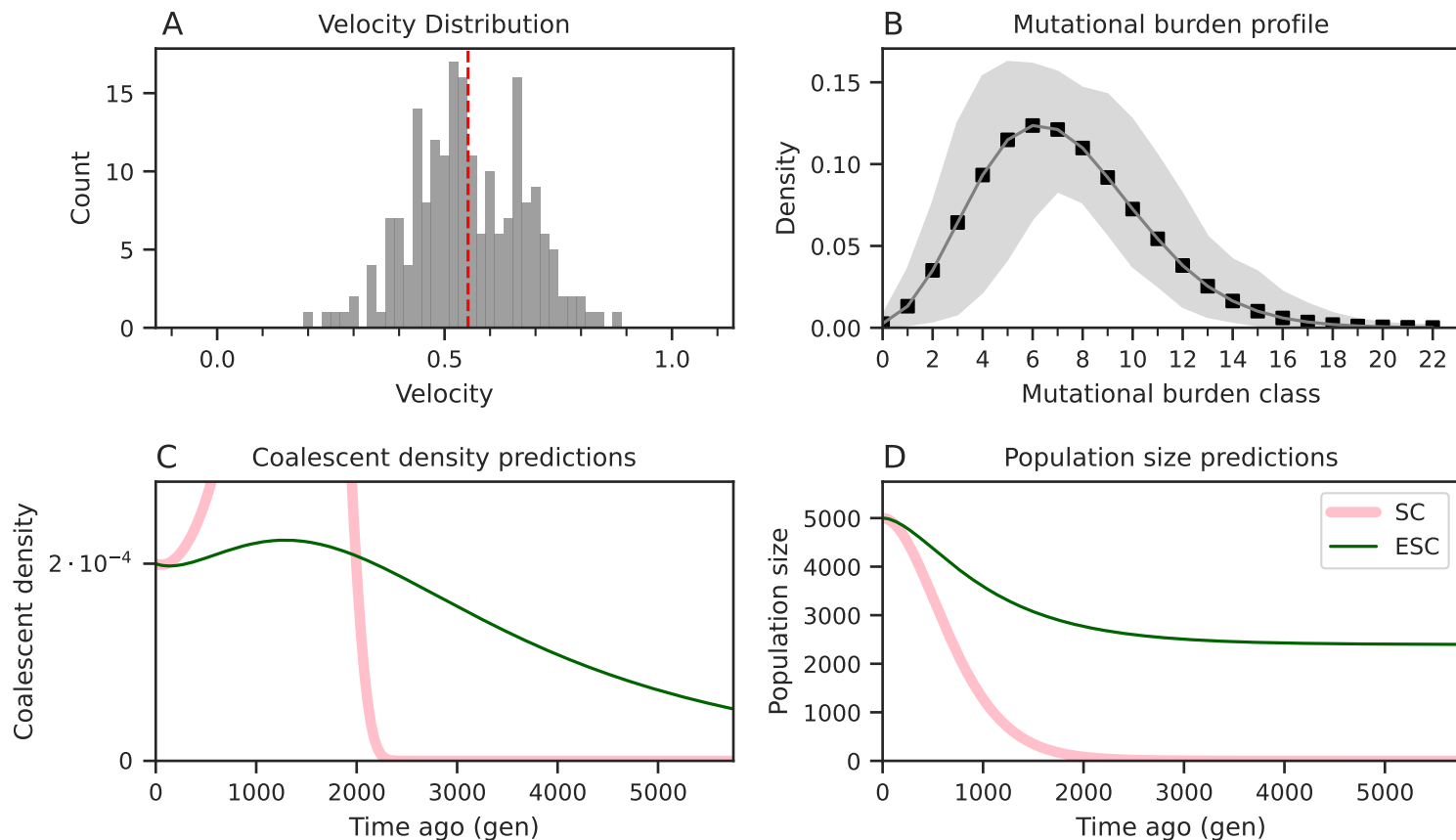
$N = 5e + 03$, $s = 3.36 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 2.95 \cdot 10^{-8}$
Mean Velocity = 0.5151 ± 0.1182 , $n = 200$ simulations



$N = 5e + 03$, $s = 3.21 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.21 \cdot 10^{-8}$
Mean Velocity= 0.5343 ± 0.1105 , $n = 200$ simulations

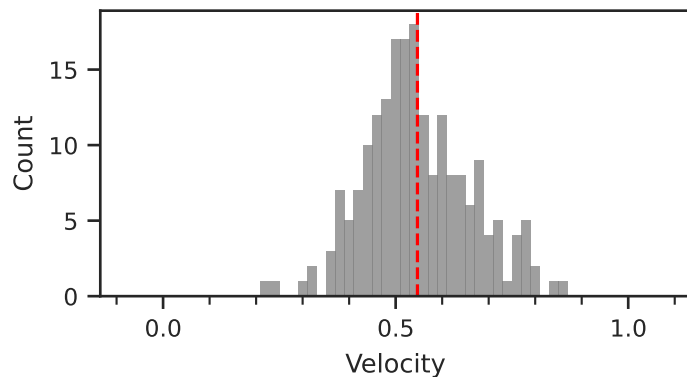


$N = 5e + 03$, $s = 3.07 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 4.99 \cdot 10^{-9}$
Mean Velocity= 0.5517 ± 0.1234 , $n = 200$ simulations

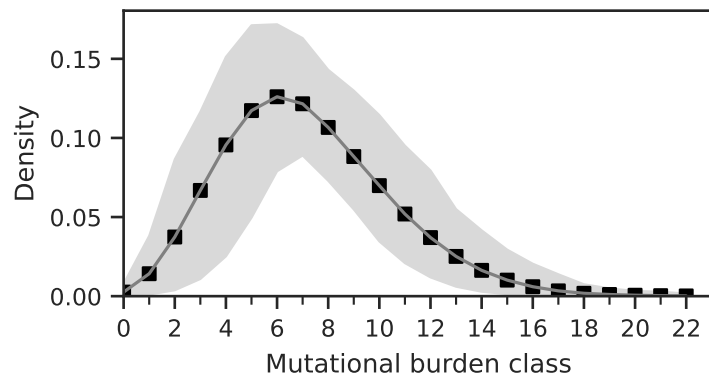


$N = 5e + 03$, $s = 3.04 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 4.08 \cdot 10^{-9}$
Mean Velocity = 0.5468 ± 0.1144 , $n = 200$ simulations

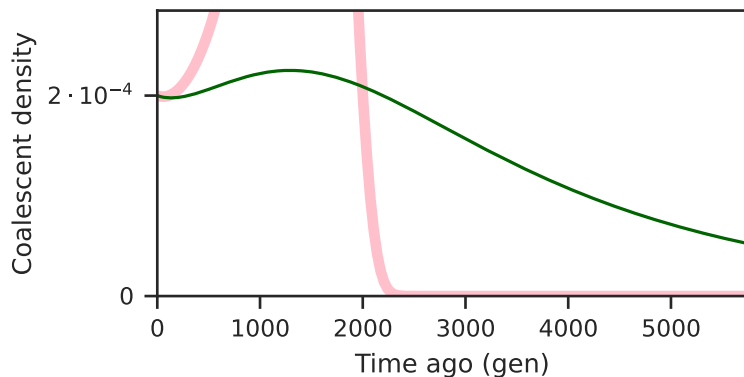
A Velocity Distribution



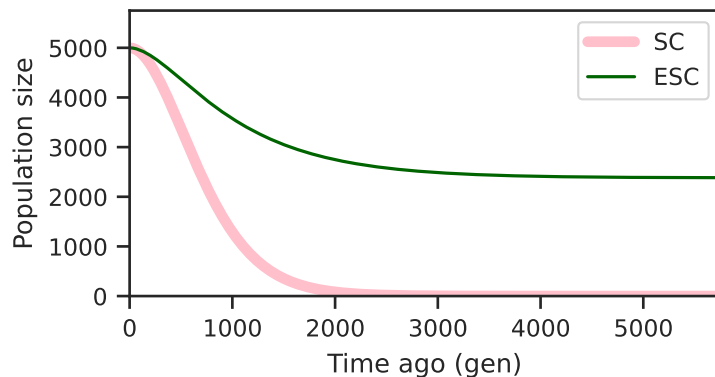
B Mutational burden profile



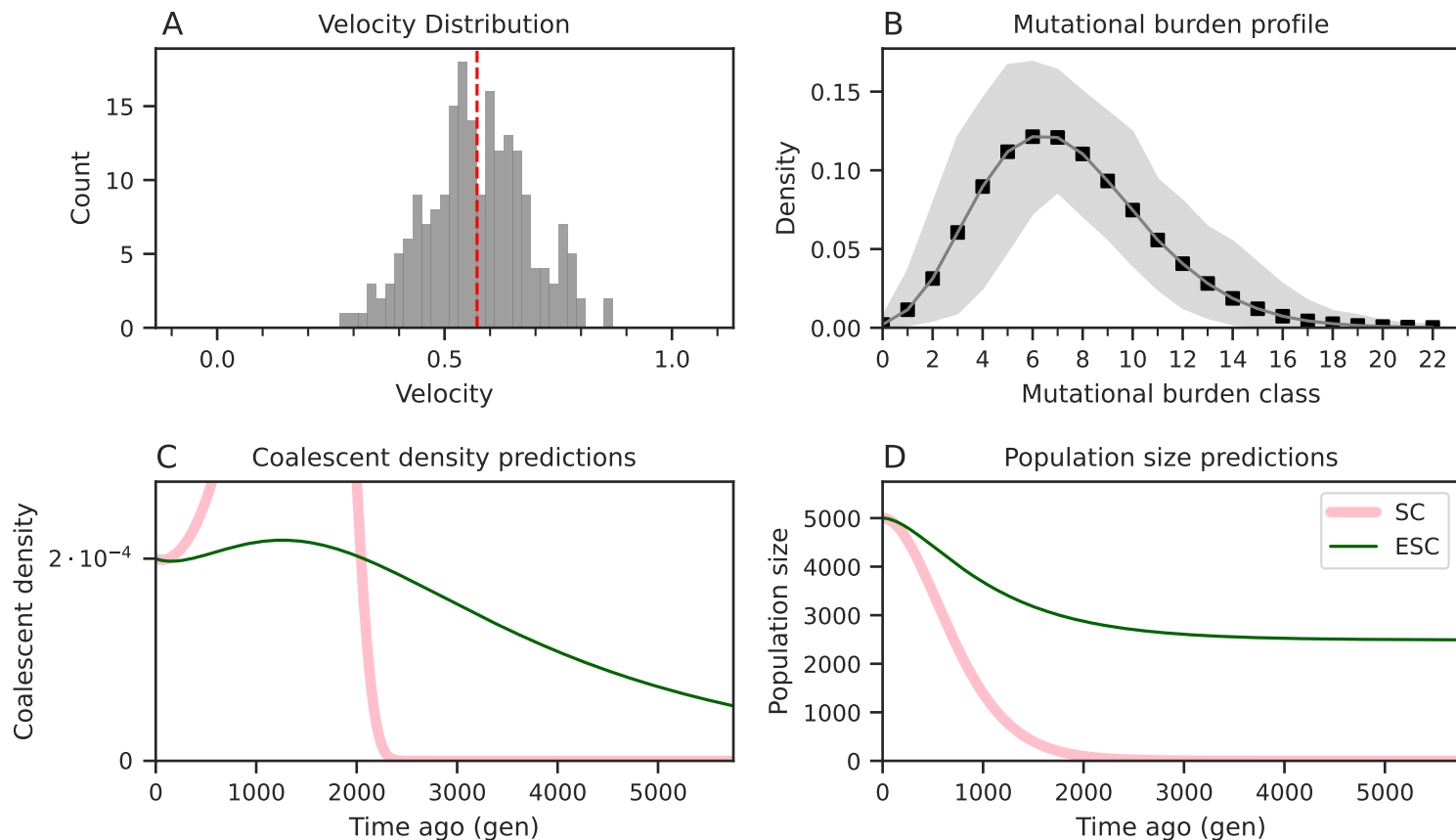
C Coalescent density predictions



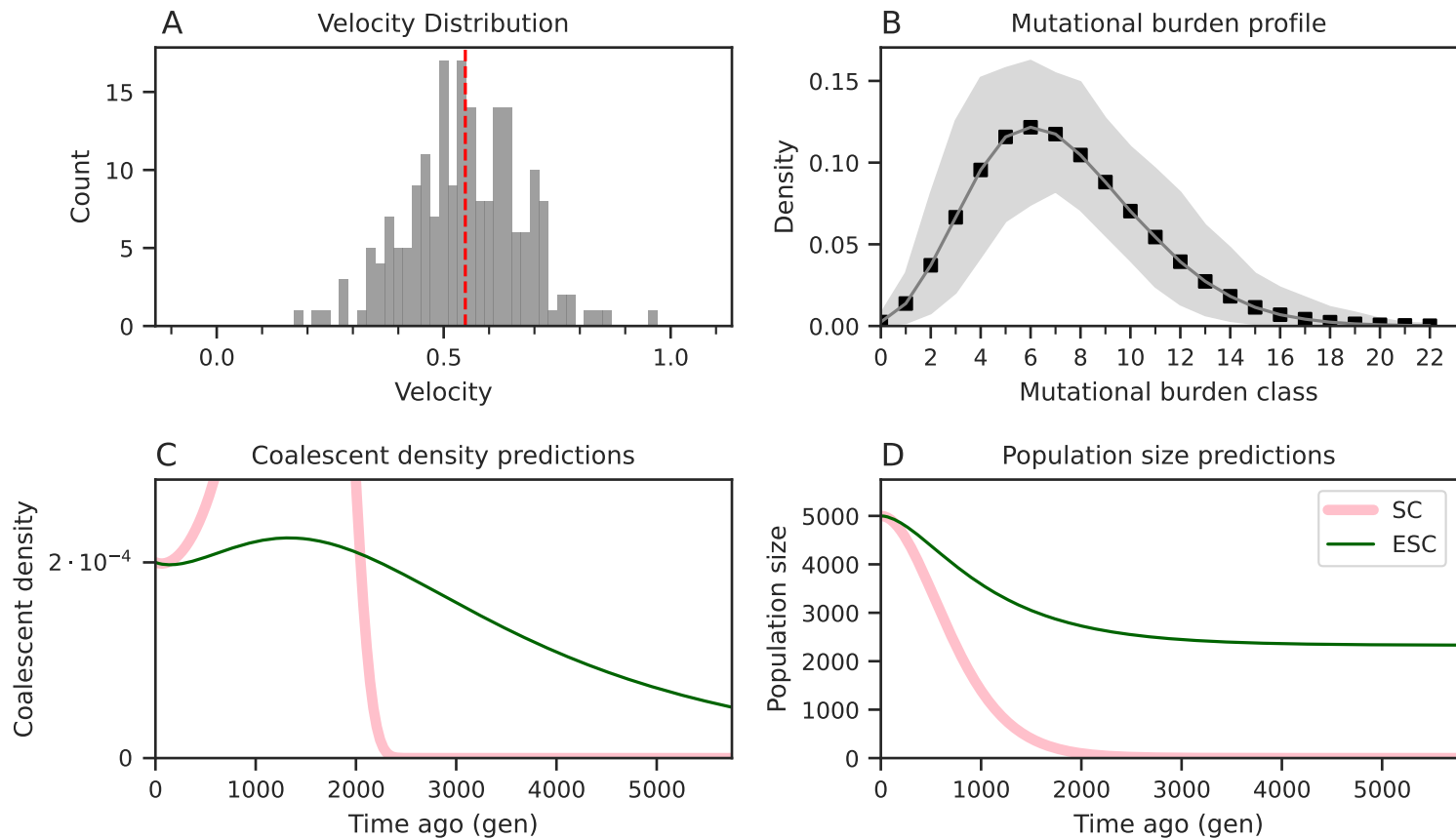
D Population size predictions



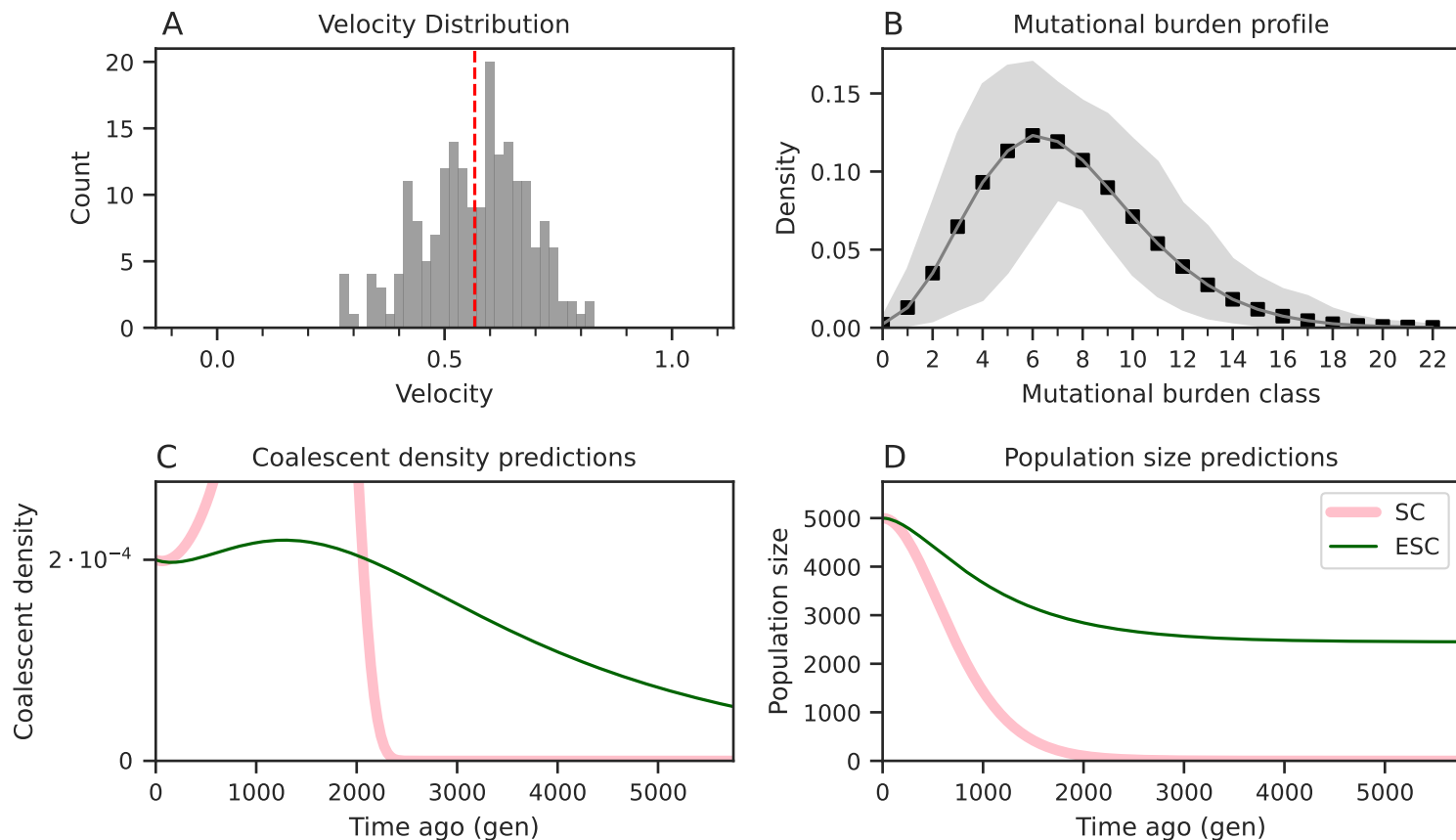
$N = 5e + 03$, $s = 2.81 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 7.61 \cdot 10^{-10}$
Mean Velocity= 0.5714 ± 0.1138 , $n = 200$ simulations



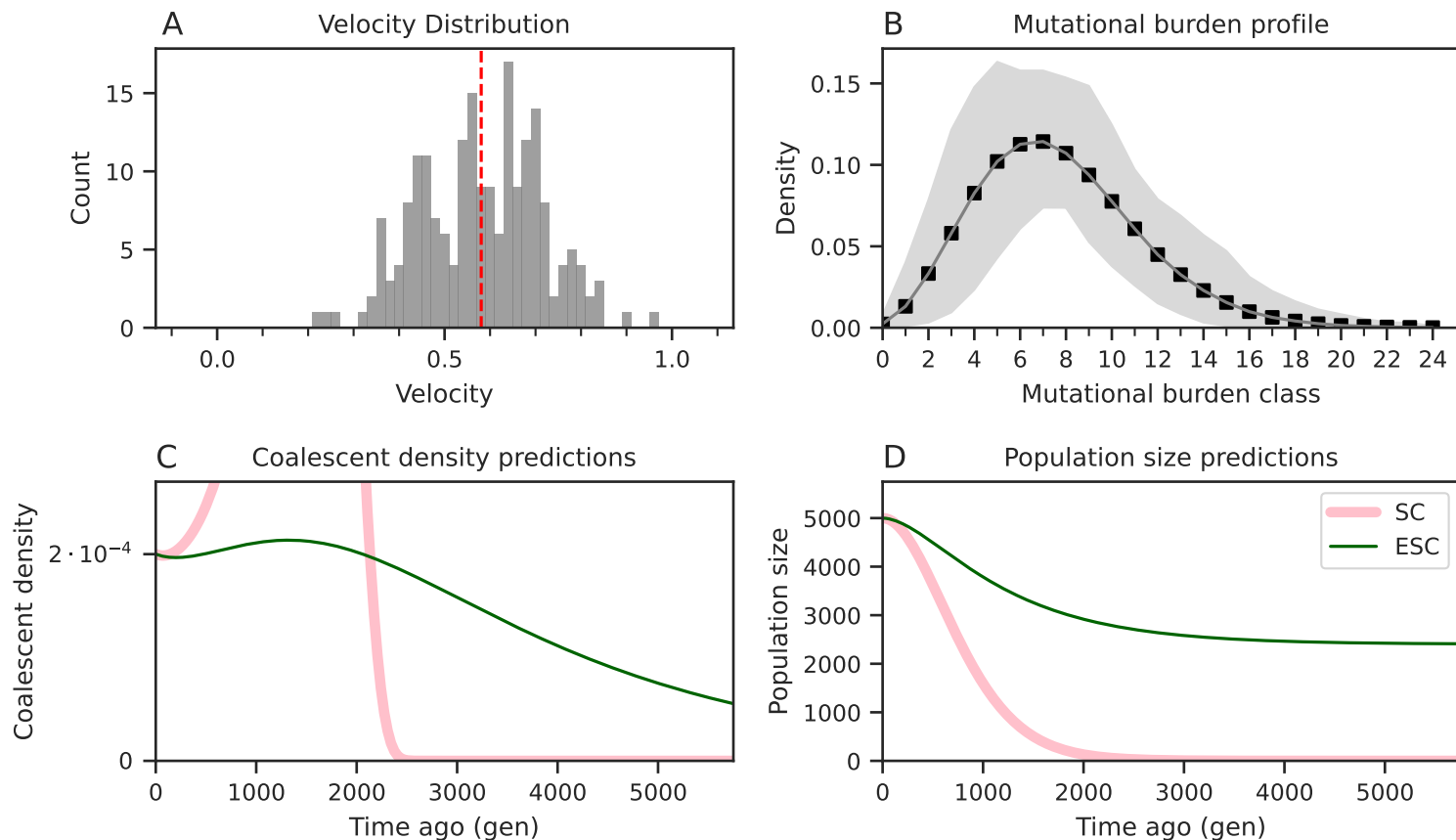
$N = 5e + 03$, $s = 2.8 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 6.92 \cdot 10^{-10}$
Mean Velocity = 0.5477 ± 0.1251 , $n = 200$ simulations



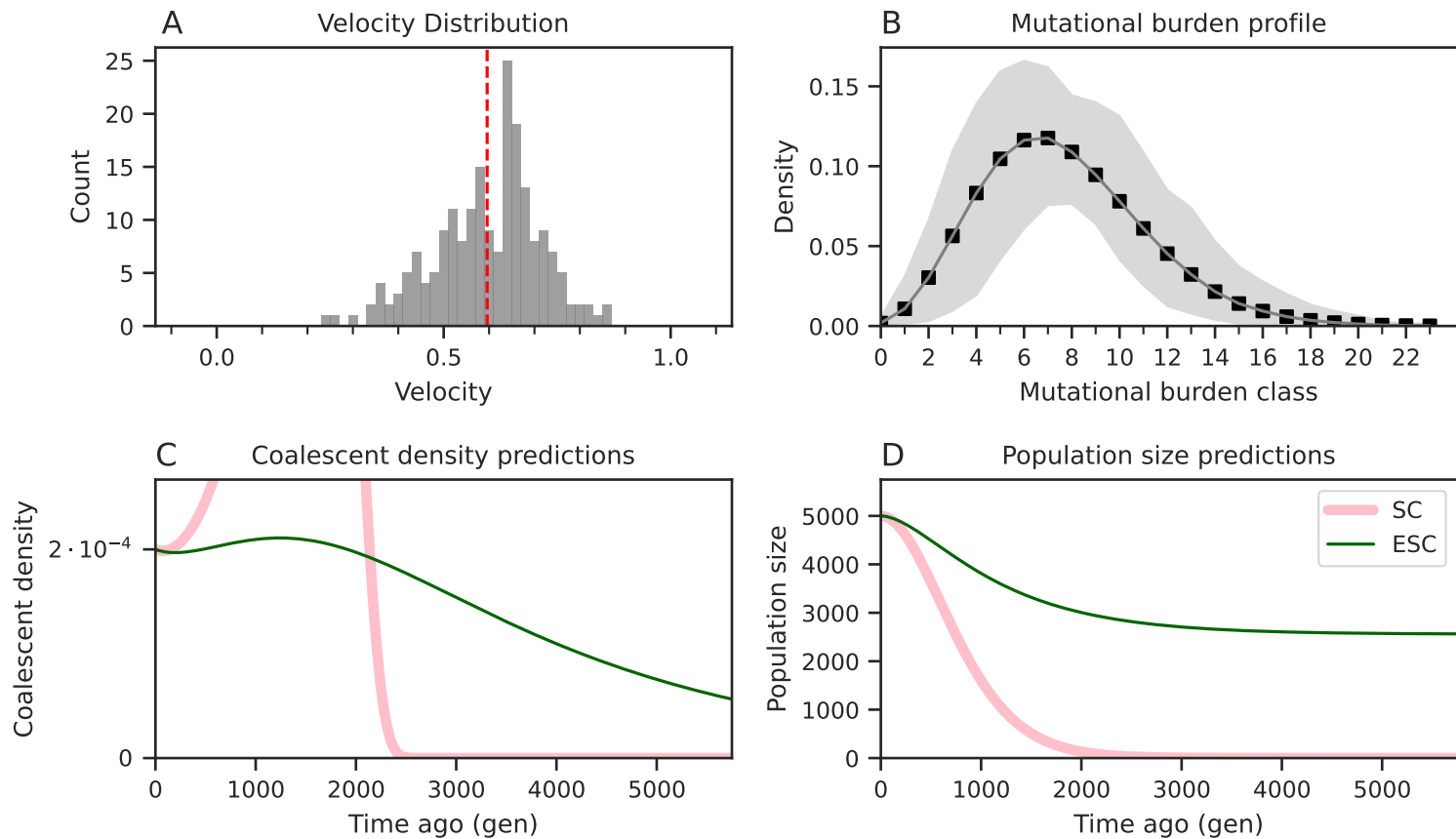
$N = 5e + 03$, $s = 2.72 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 3.58 \cdot 10^{-10}$
Mean Velocity= 0.5663 ± 0.1149 , $n = 200$ simulations



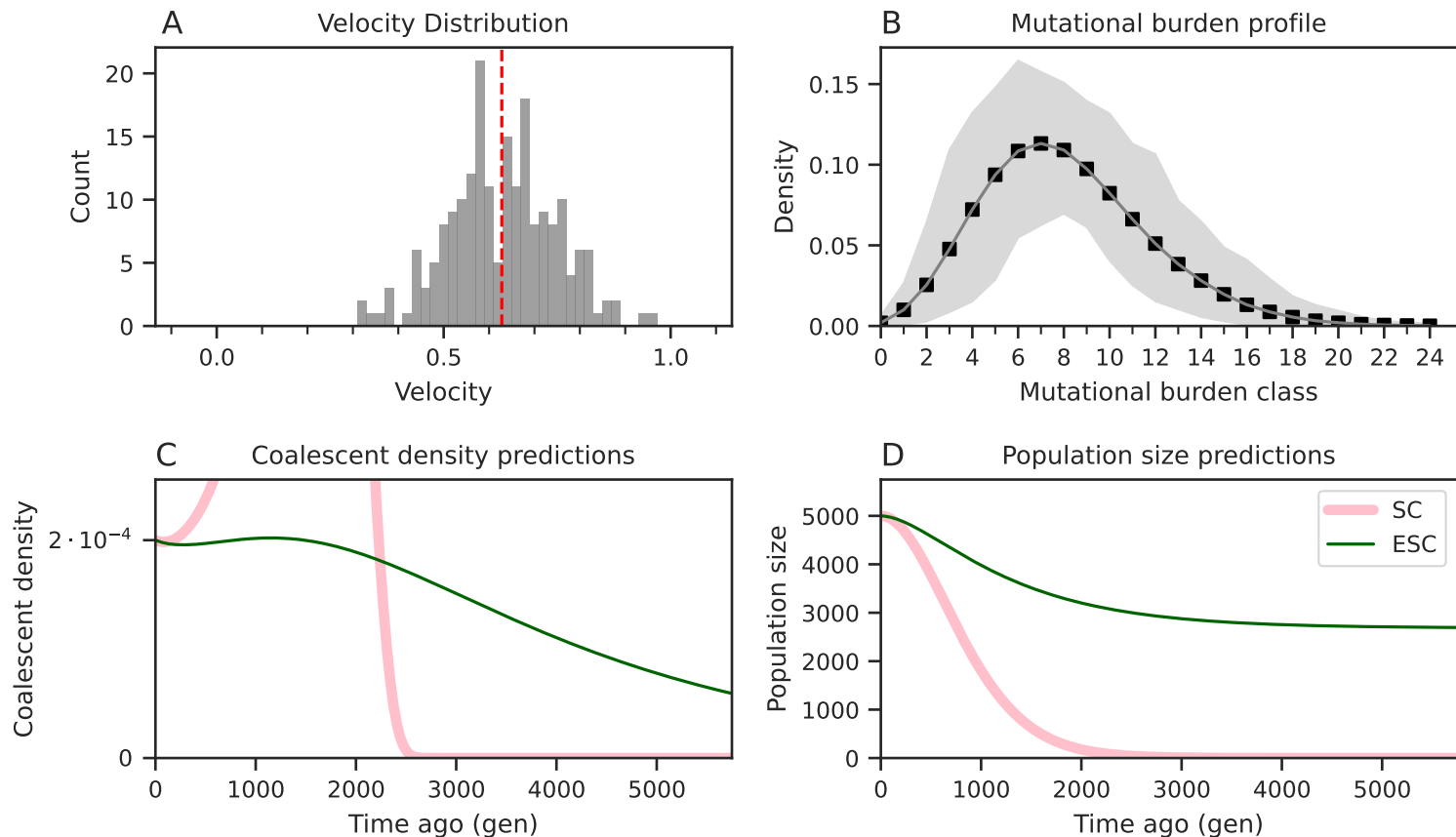
$N = 5e + 03$, $s = 2.42 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.98 \cdot 10^{-11}$
Mean Velocity= 0.5805 ± 0.1341 , $n = 200$ simulations



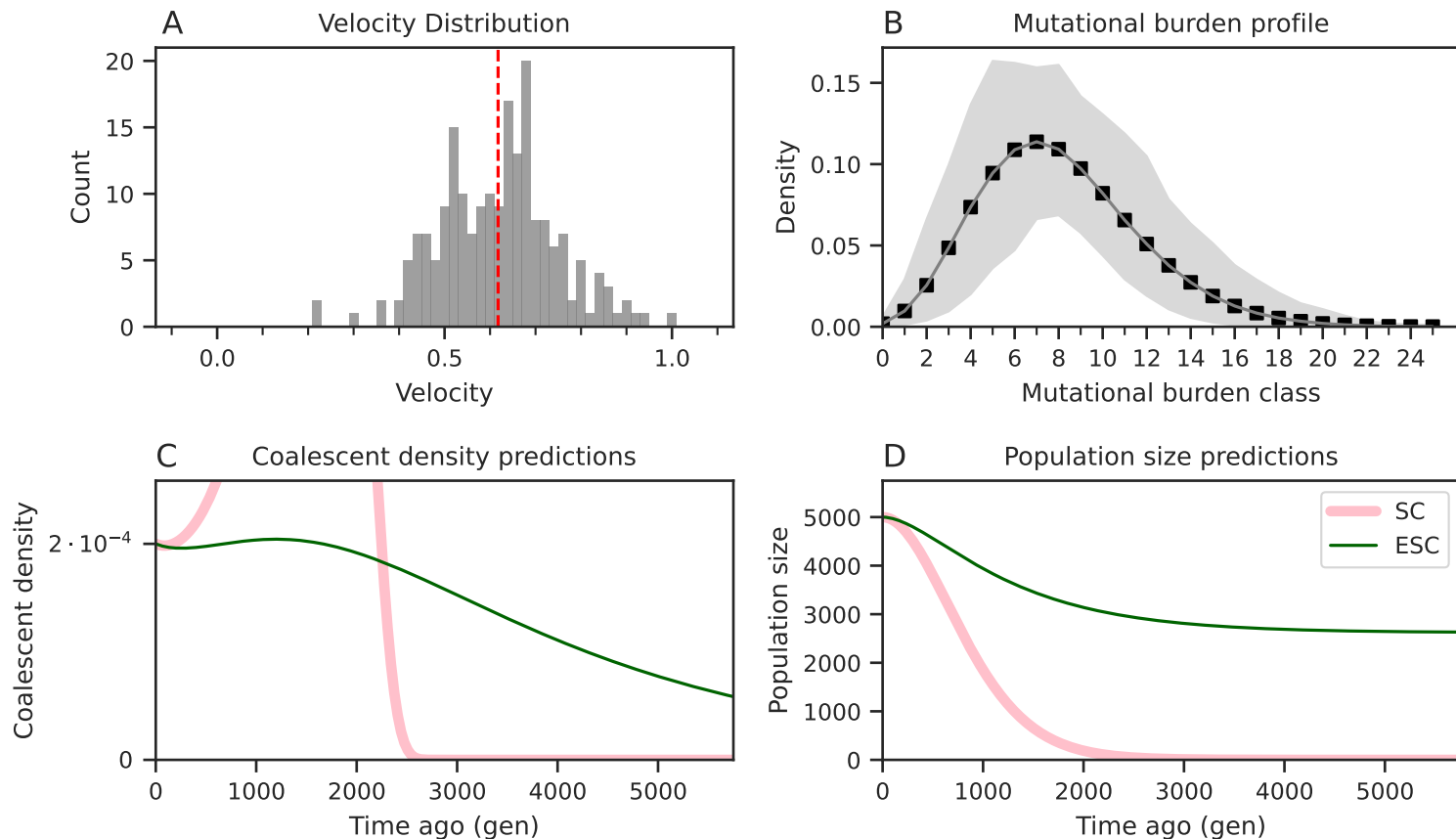
$N = 5e + 03$, $s = 2.4 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.67 \cdot 10^{-11}$
Mean Velocity = 0.5957 ± 0.1163 , $n = 200$ simulations



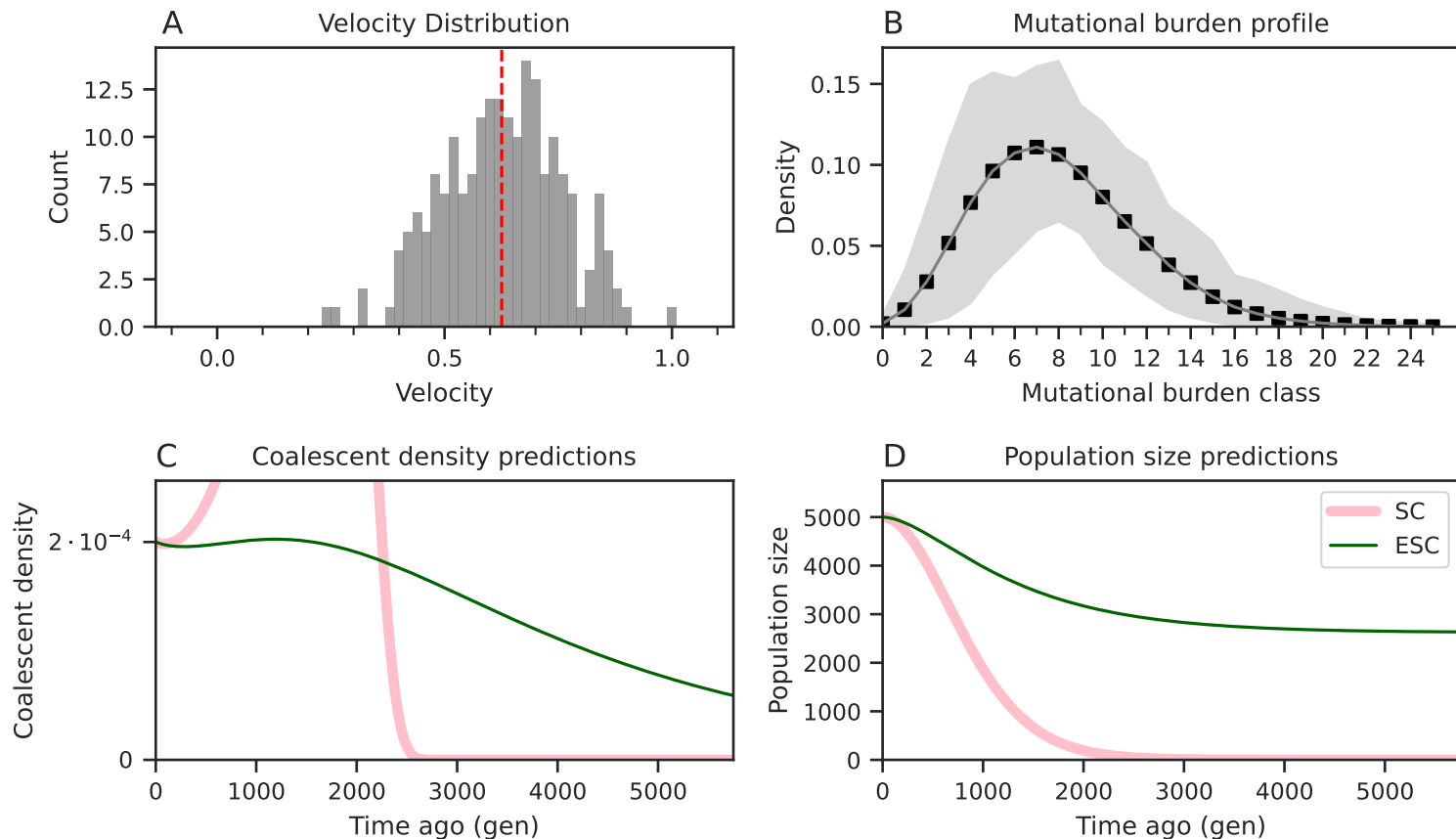
$N = 5e + 03$, $s = 2.08 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 3.09 \cdot 10^{-13}$
Mean Velocity= 0.6281 ± 0.1182 , $n = 200$ simulations



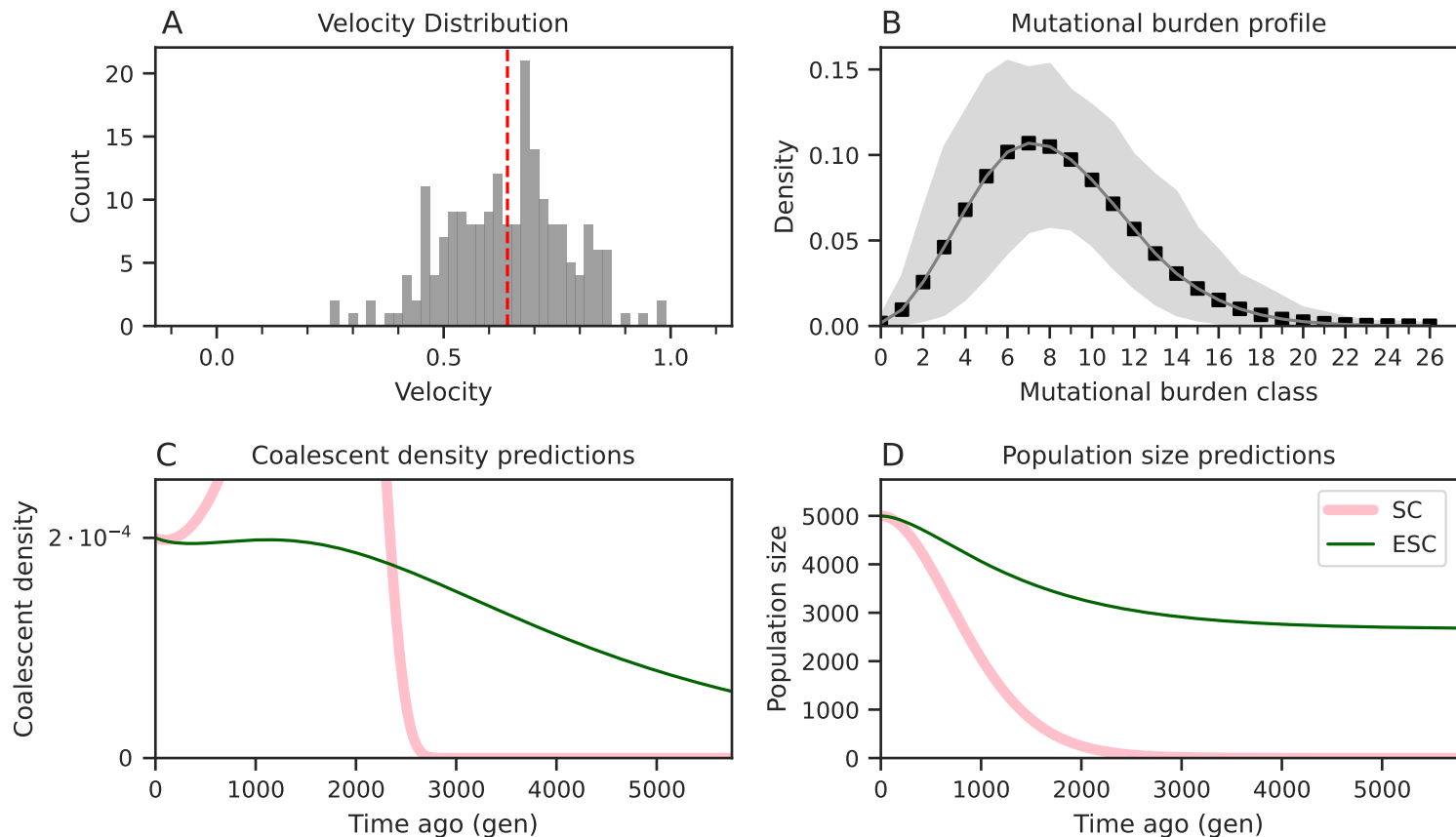
$N = 5e + 03$, $s = 2.02 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.27 \cdot 10^{-13}$
Mean Velocity= 0.6177 ± 0.1298 , $n = 200$ simulations



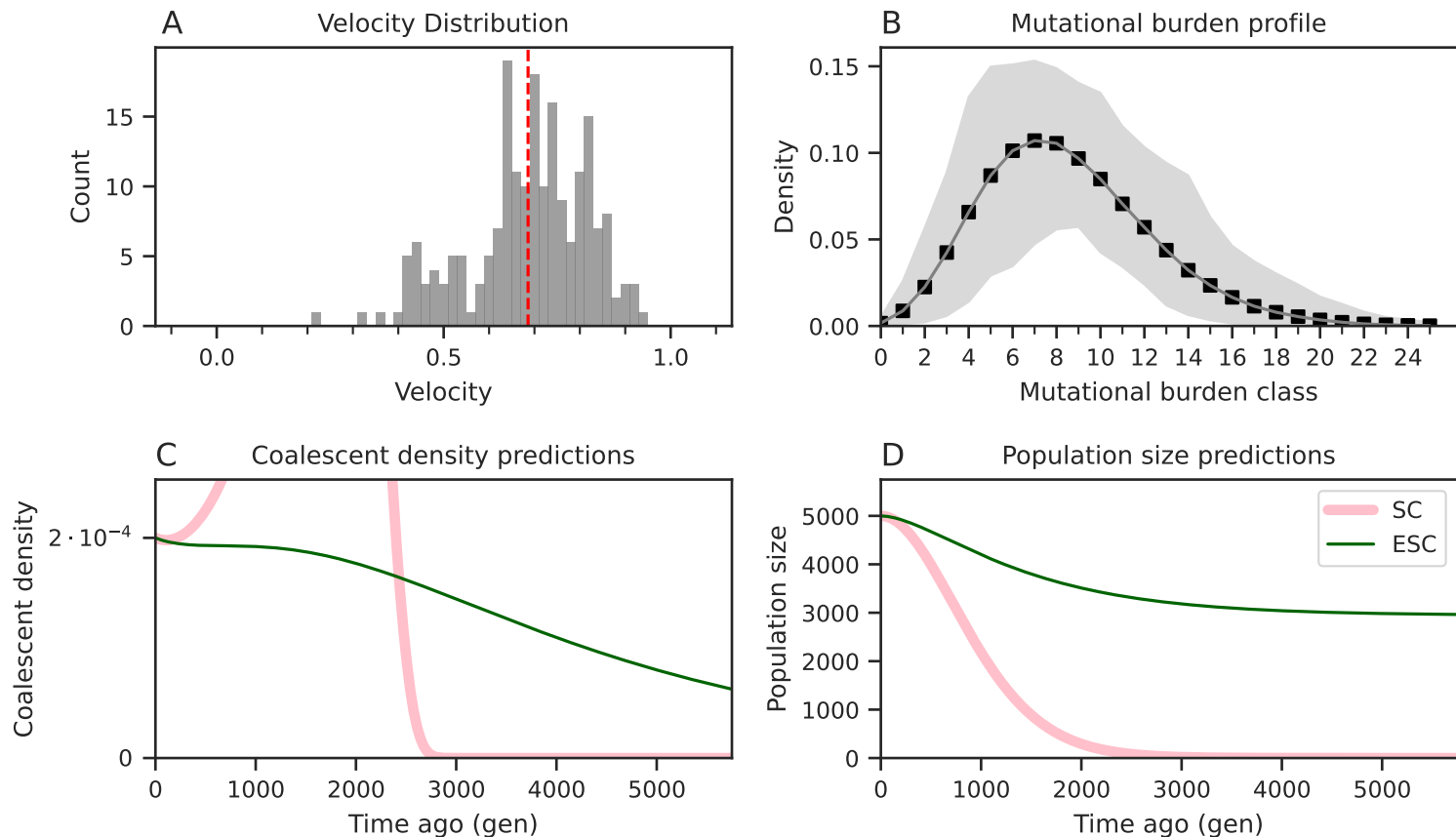
$N = 5e + 03$, $s = 2 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 9.36 \cdot 10^{-14}$
Mean Velocity = 0.6256 ± 0.1318 , $n = 200$ simulations



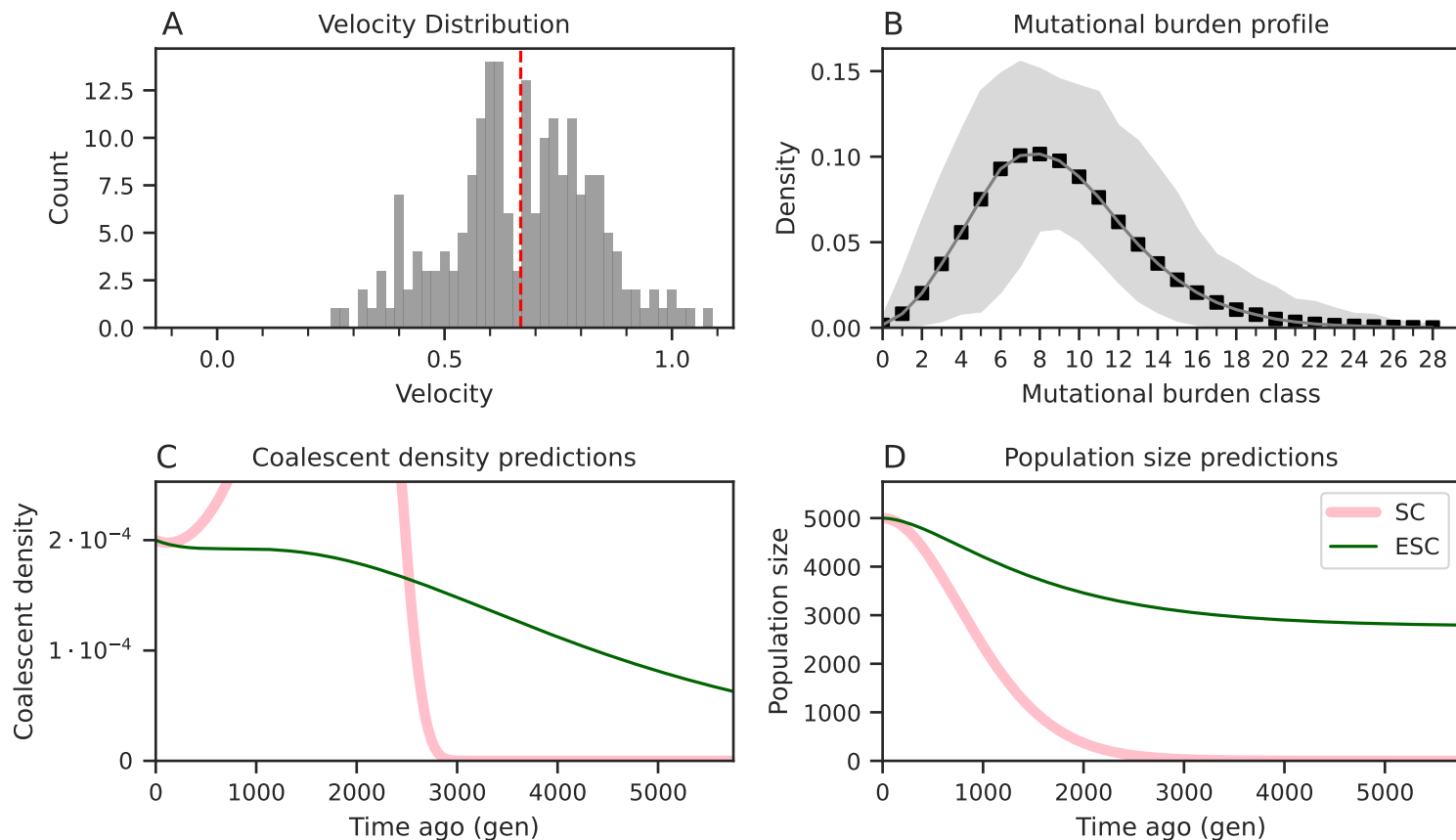
$N = 5e + 03$, $s = 1.76 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 1.38 \cdot 10^{-15}$
Mean Velocity= 0.6405 ± 0.1345 , $n = 200$ simulations



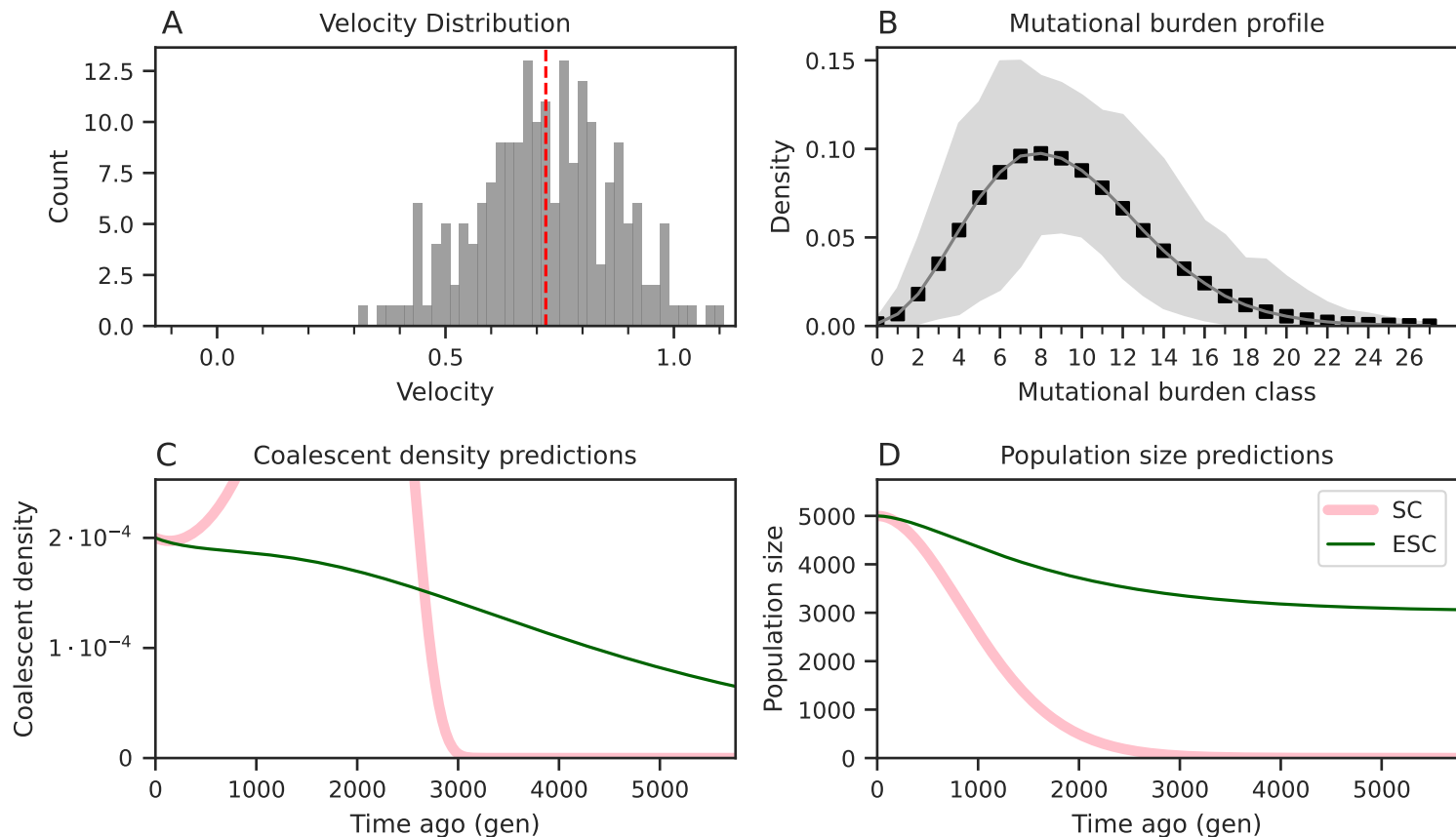
$N = 5e + 03$, $s = 1.62 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 7.31 \cdot 10^{-17}$
Mean Velocity = 0.6859 ± 0.1328 , $n = 200$ simulations



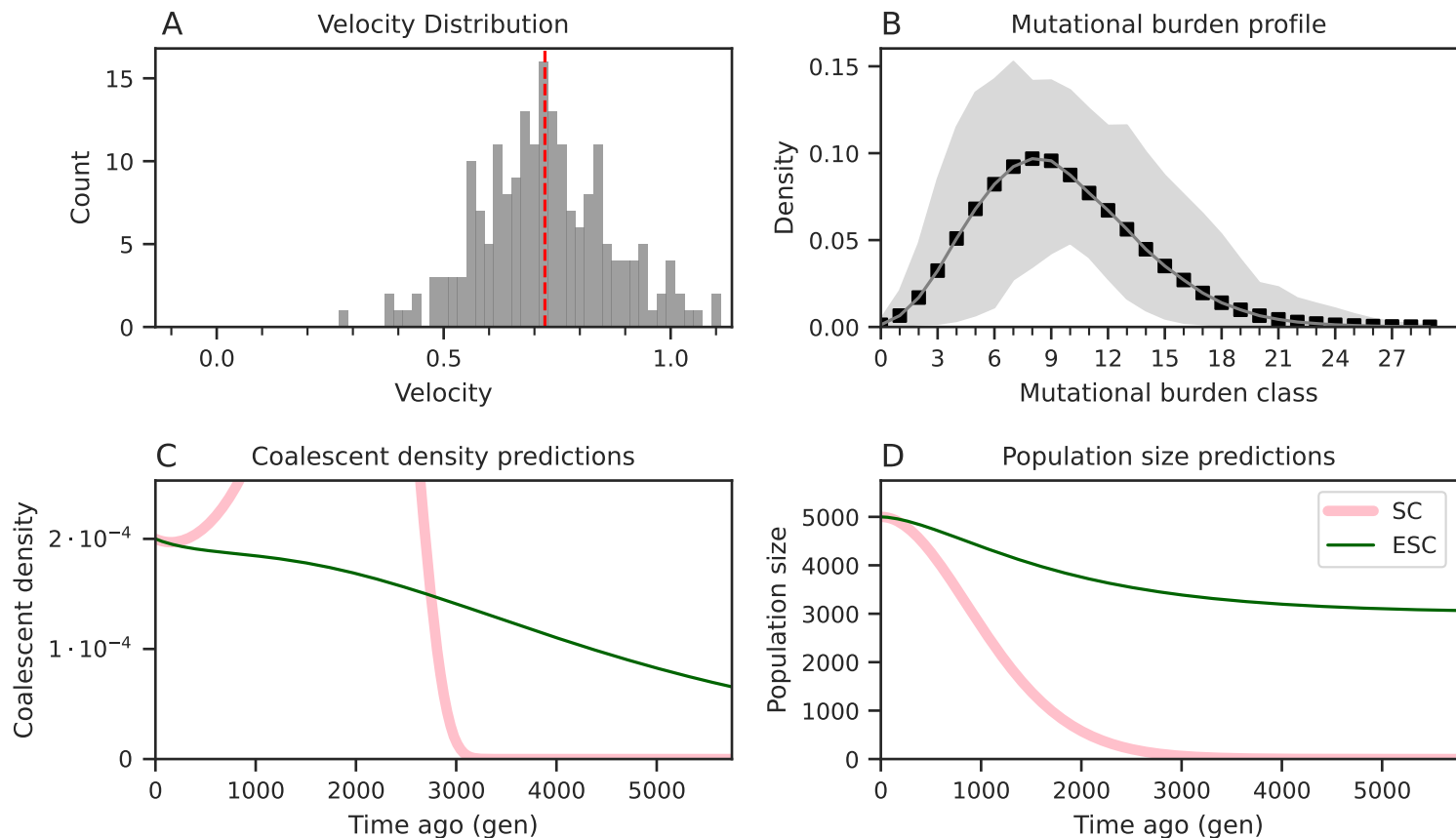
$N = 5e + 03$, $s = 1.44 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 5.78 \cdot 10^{-19}$
Mean Velocity= 0.6674 ± 0.1584 , $n = 200$ simulations



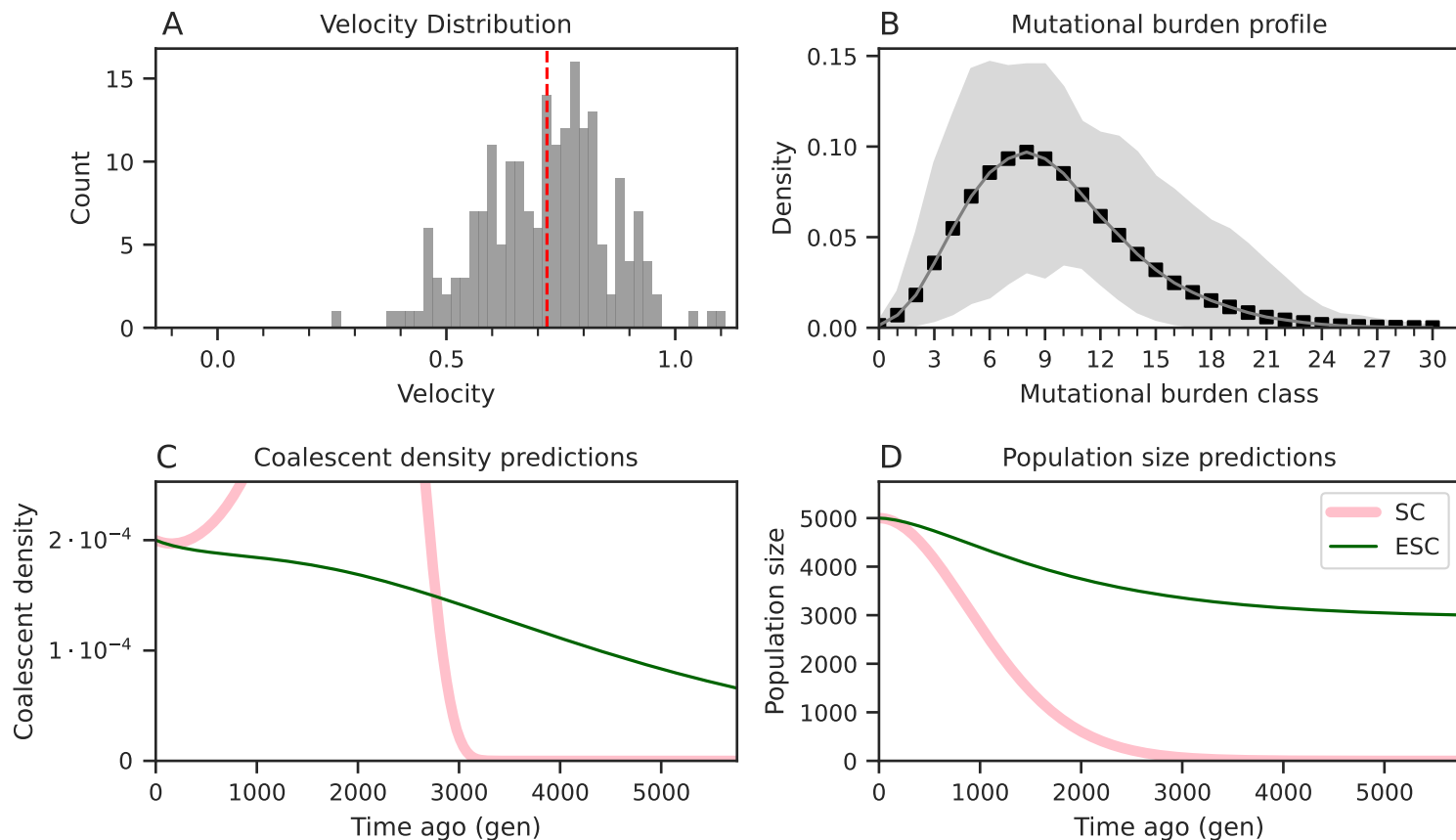
$N = 5e + 03$, $s = 1.23 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 3.7 \cdot 10^{-22}$
Mean Velocity = 0.7196 ± 0.1534 , $n = 200$ simulations



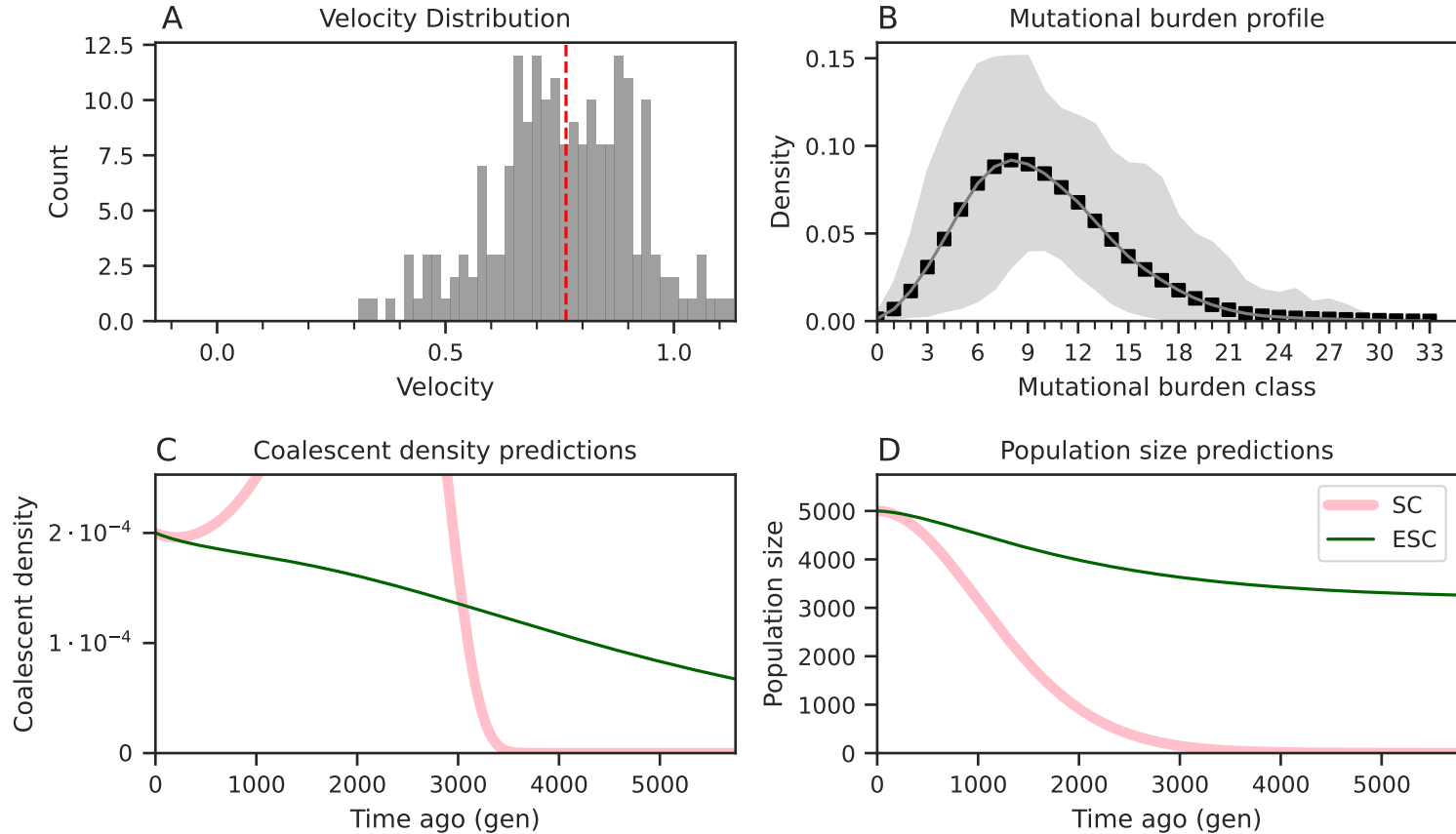
$N = 5e + 03$, $s = 1.12 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 3.04 \cdot 10^{-24}$
Mean Velocity= 0.7231 ± 0.1454 , $n = 200$ simulations



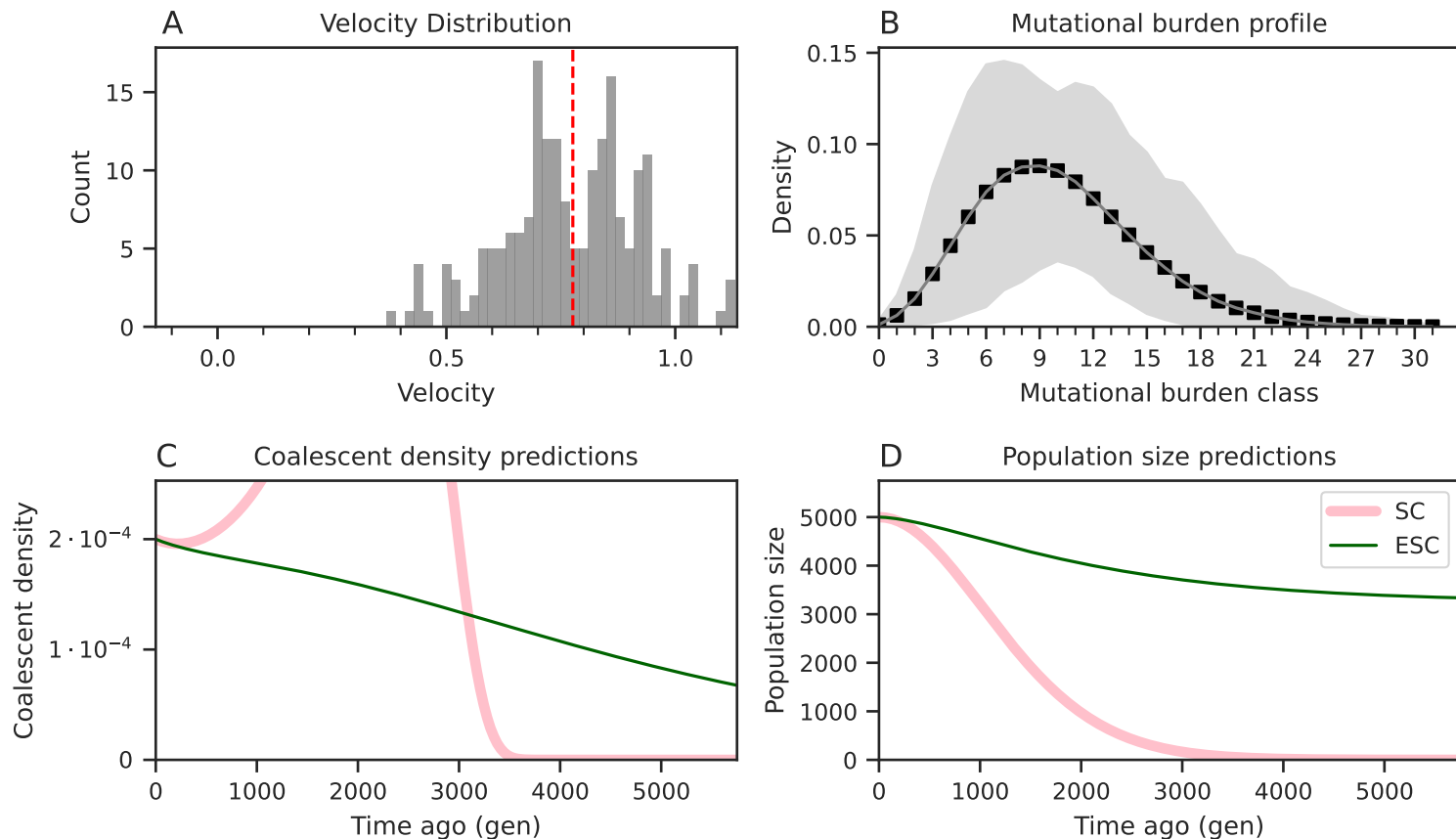
$N = 5e + 03$, $s = 1.09 \cdot 10^{-4}$, $U_d = 0.006$, $\phi = 6.77 \cdot 10^{-25}$
Mean Velocity= 0.7198 ± 0.1426 , $n = 200$ simulations



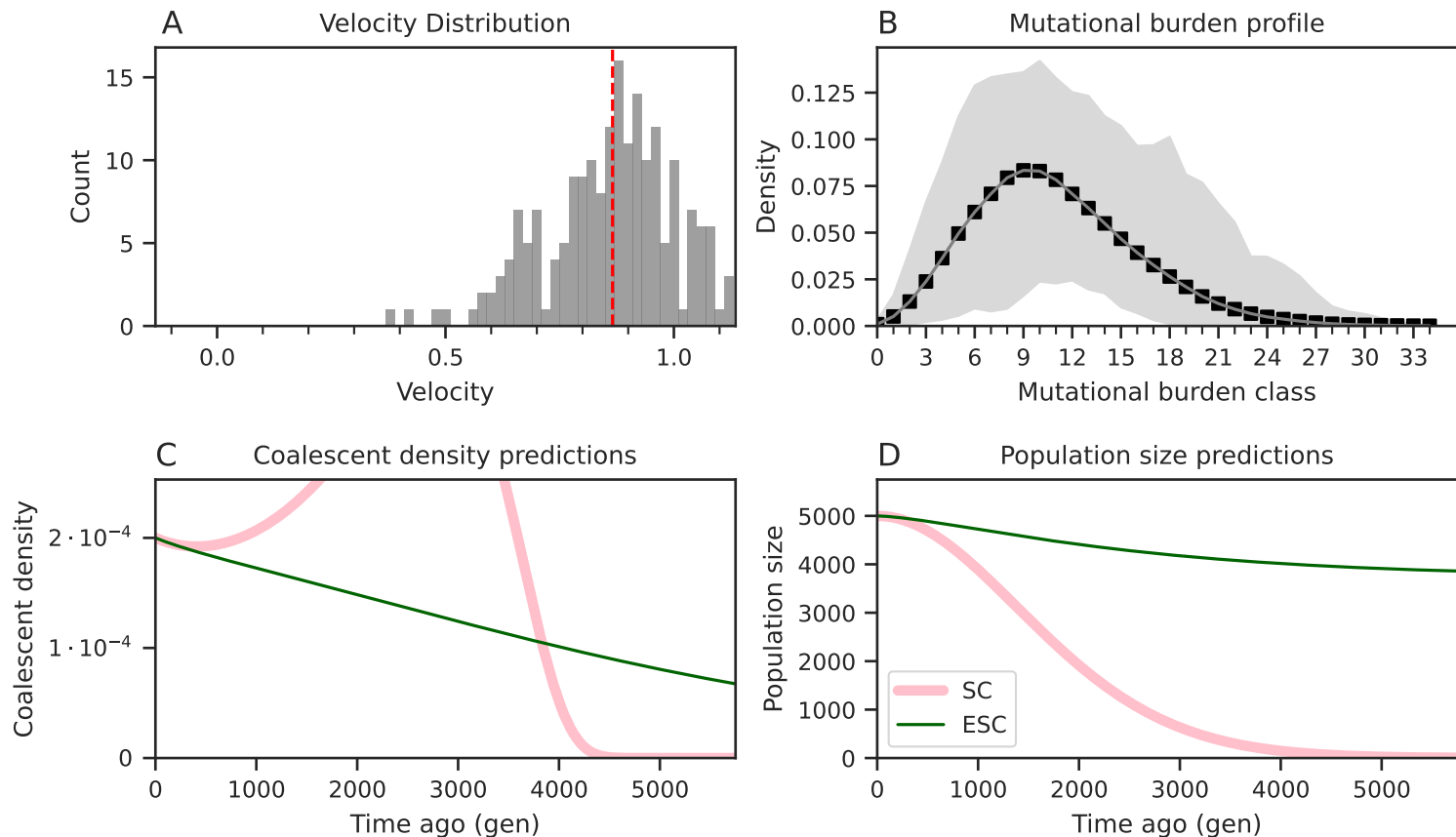
$N = 5e + 03$, $s = 8.32 \cdot 10^{-5}$, $U_d = 0.006$, $\phi = 1.99 \cdot 10^{-32}$
Mean Velocity = 0.7638 ± 0.1626 , $n = 200$ simulations



$N = 5e + 03$, $s = 8 \cdot 10^{-5}$, $U_d = 0.006$, $\phi = 1.07 \cdot 10^{-33}$
Mean Velocity= 0.7762 ± 0.1527 , $n = 200$ simulations

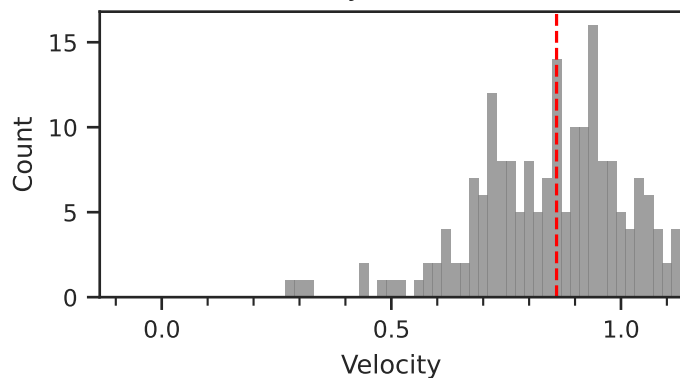


$N = 5e + 03$, $s = 4.36 \cdot 10^{-5}$, $U_d = 0.006$, $\phi = 3.74 \cdot 10^{-61}$
Mean Velocity= 0.8657 ± 0.1502 , $n = 200$ simulations

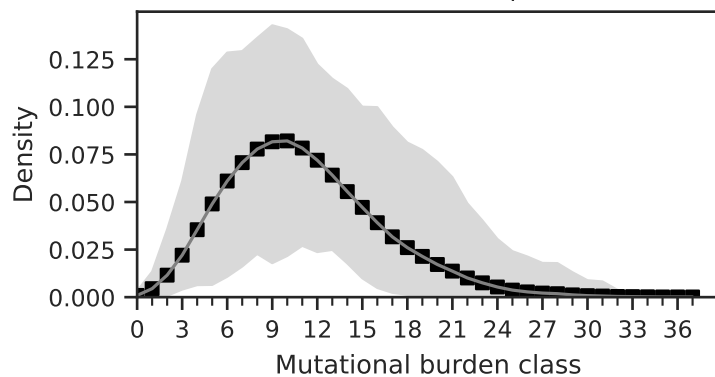


$N = 5e + 03$, $s = 4 \cdot 10^{-5}$, $U_d = 0.006$, $\phi = 1.44 \cdot 10^{-66}$
Mean Velocity= 0.8598 ± 0.1741 , $n = 200$ simulations

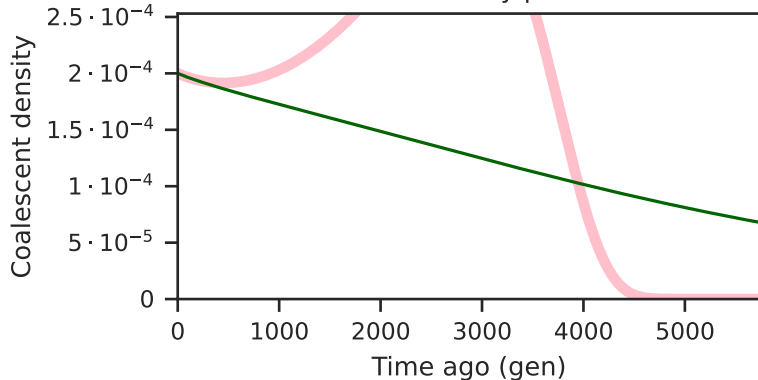
A Velocity Distribution



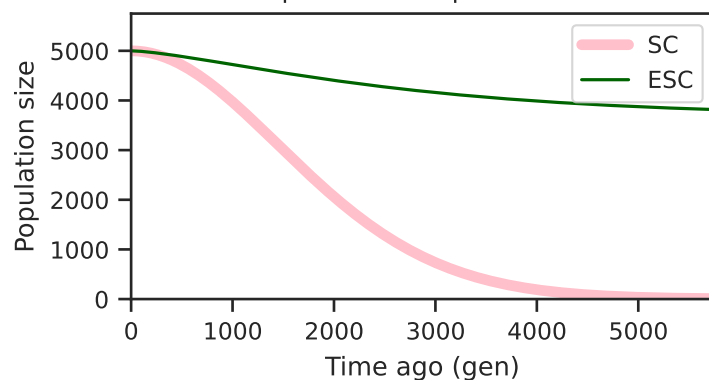
B Mutational burden profile



C Coalescent density predictions

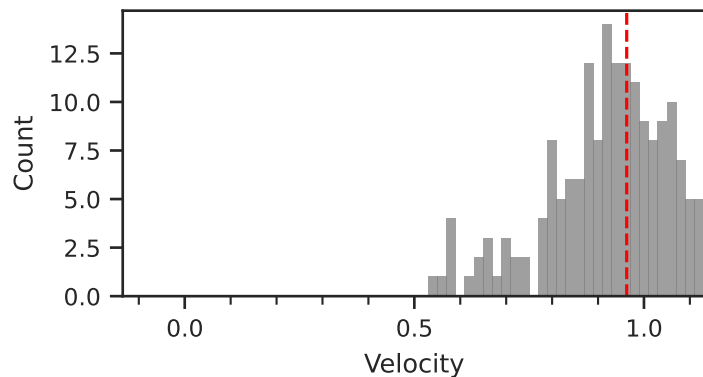


D Population size predictions

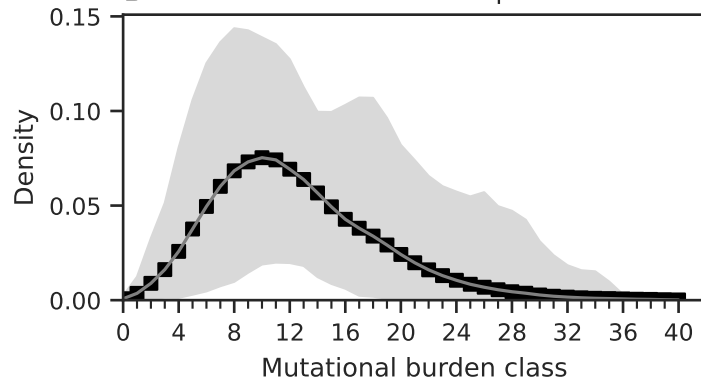


$N = 5e + 03$, $s = 1 \cdot 10^{-5}$, $U_d = 0.006$, $\phi = 1.33 \cdot 10^{-262}$
Mean Velocity= 0.9624 ± 0.1660 , $n = 200$ simulations

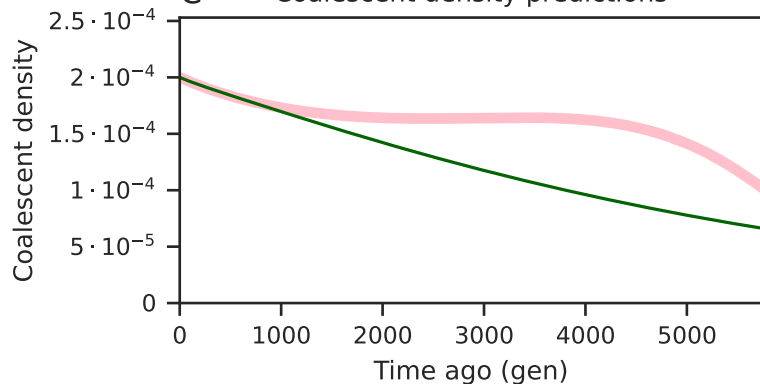
A Velocity Distribution



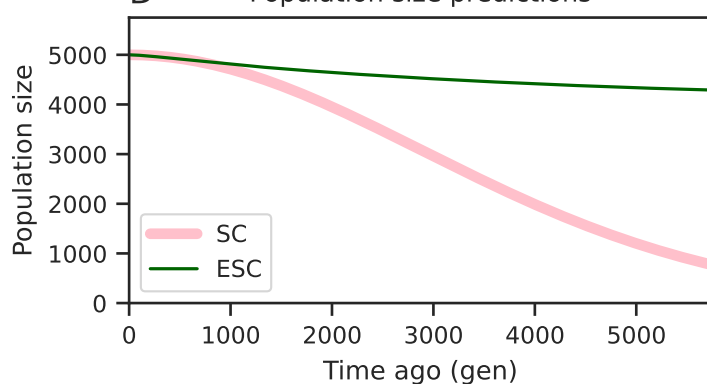
B Mutational burden profile



C Coalescent density predictions

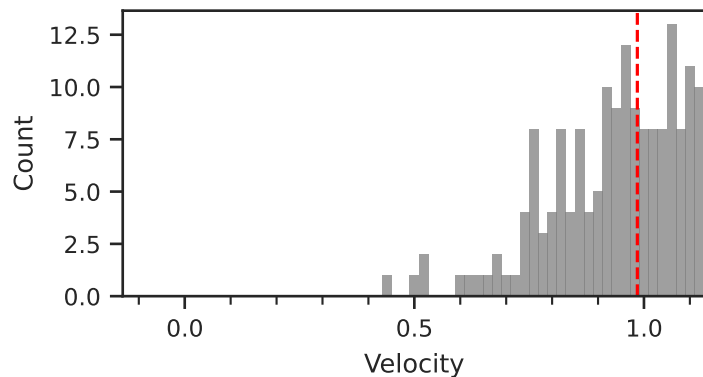


D Population size predictions

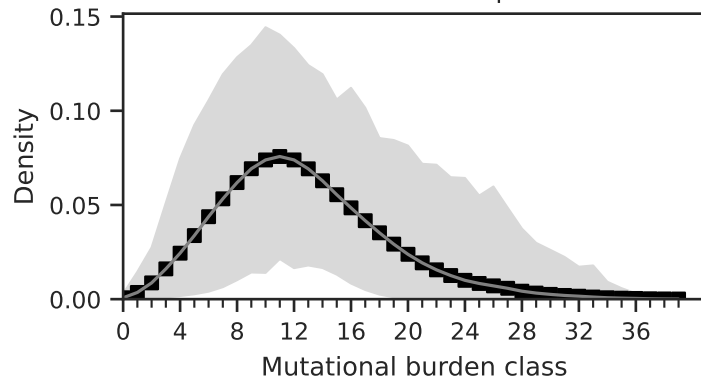


$N = 5e + 03$, $s = 4 \cdot 10^{-6}$, $U_d = 0.006$, $\phi = 0$
Mean Velocity = 0.9856 ± 0.1803 , $n = 200$ simulations

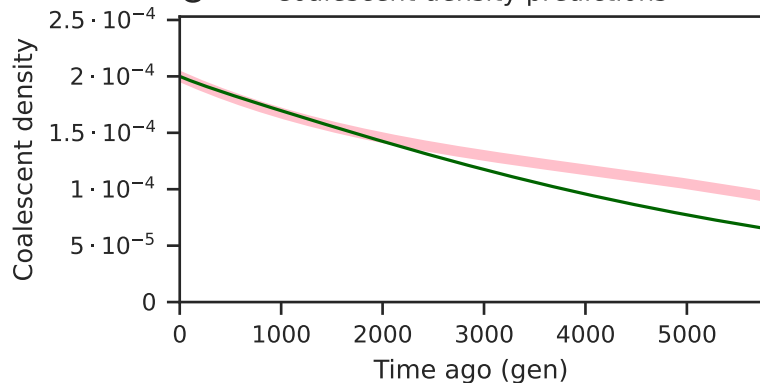
A Velocity Distribution



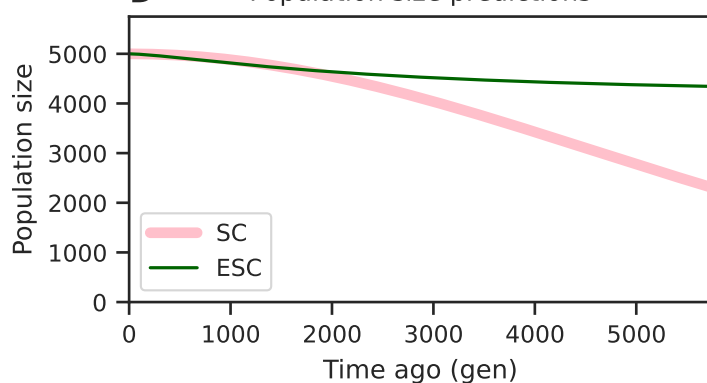
B Mutational burden profile



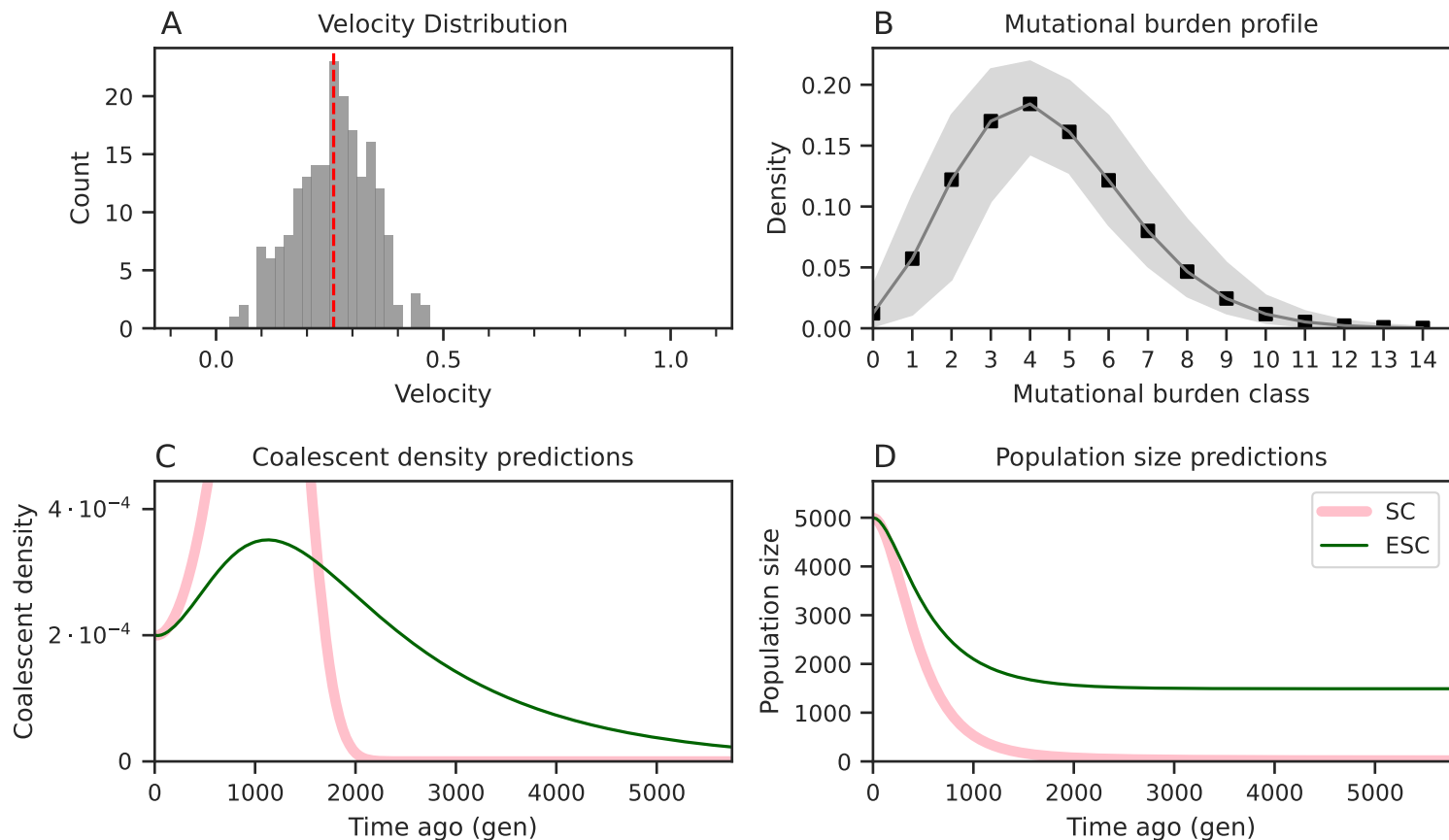
C Coalescent density predictions



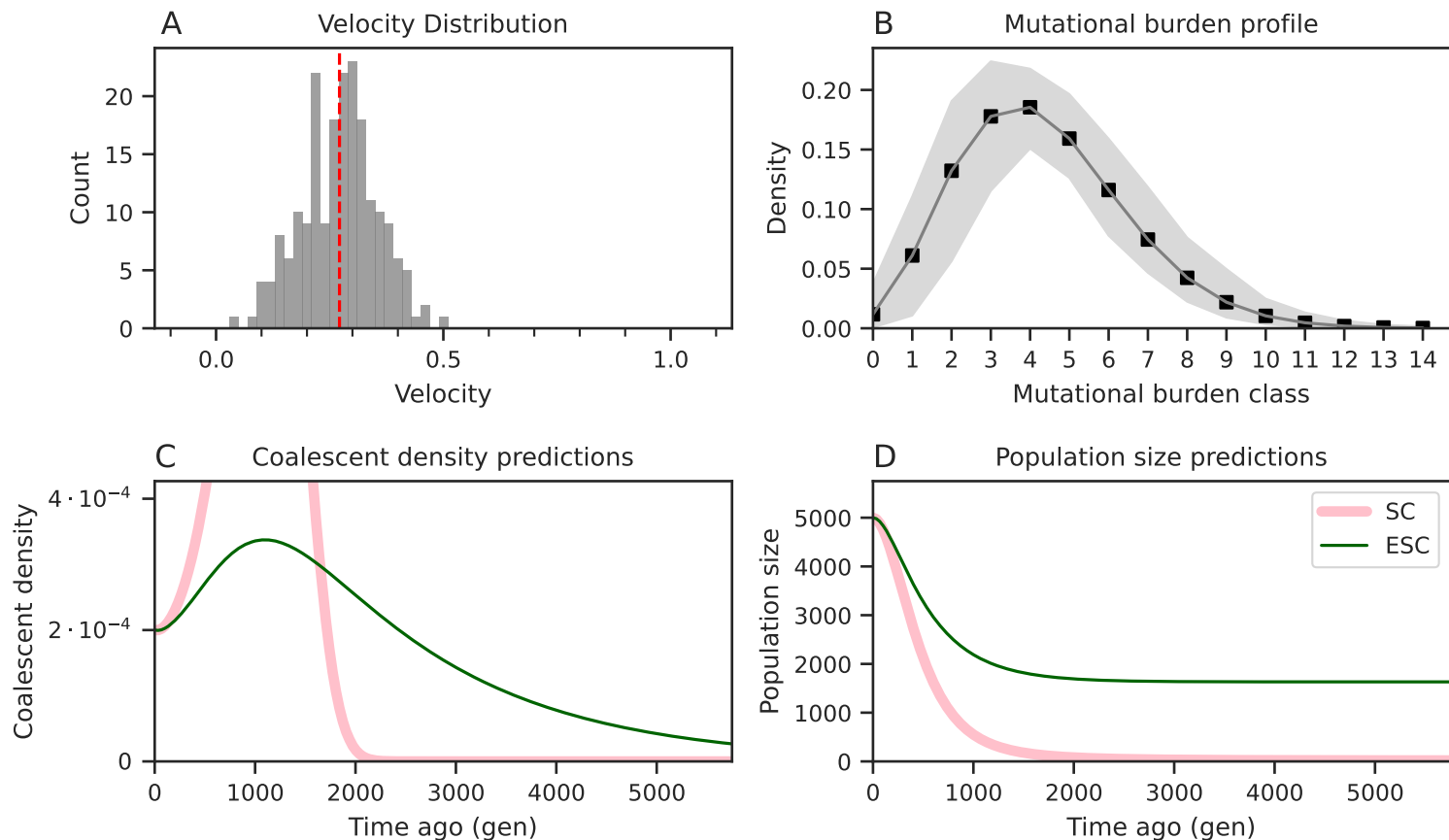
D Population size predictions



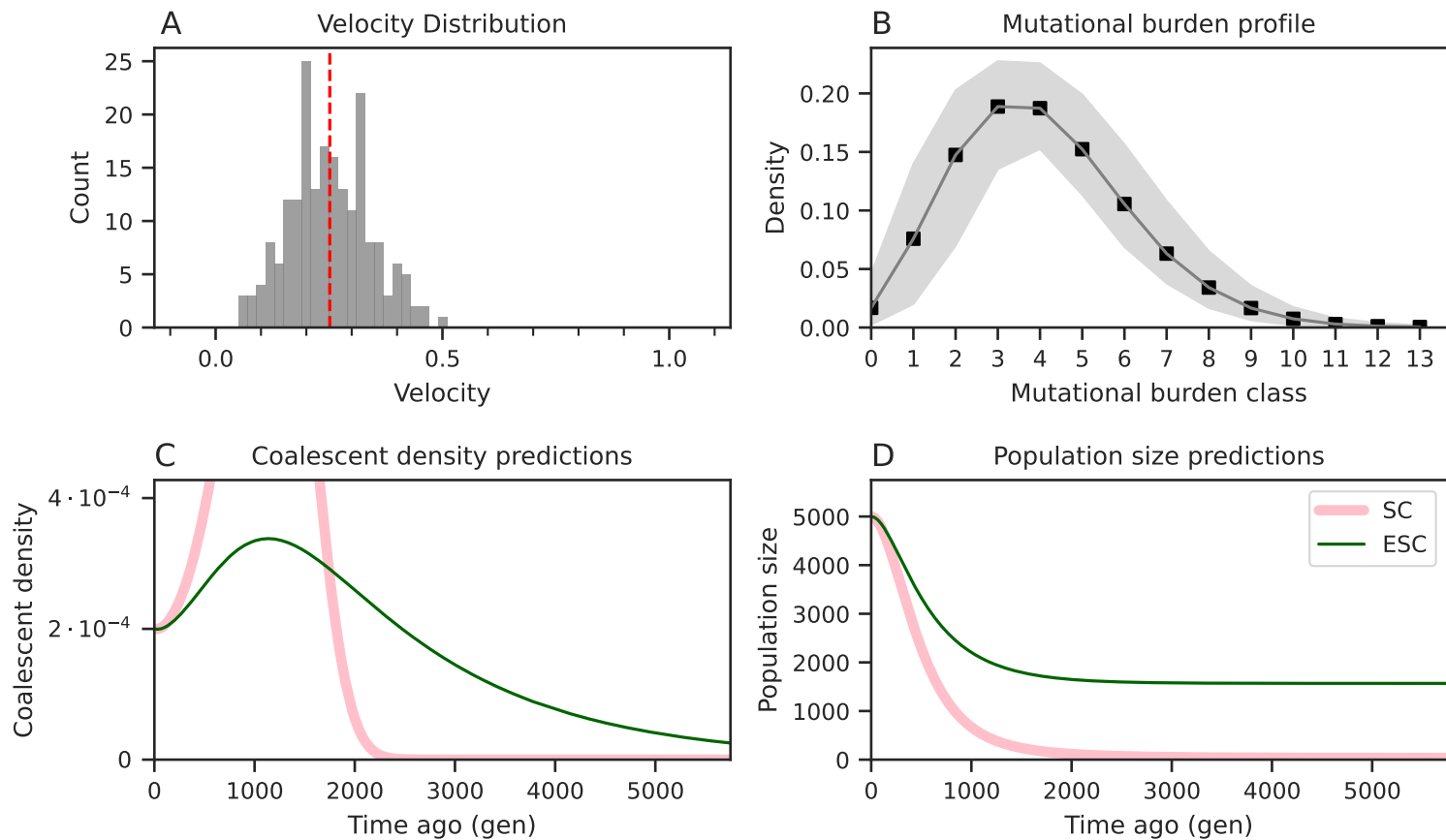
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0056$, $\phi = 0.0185$
Mean Velocity= 0.2587 ± 0.0829 , $n = 200$ simulations



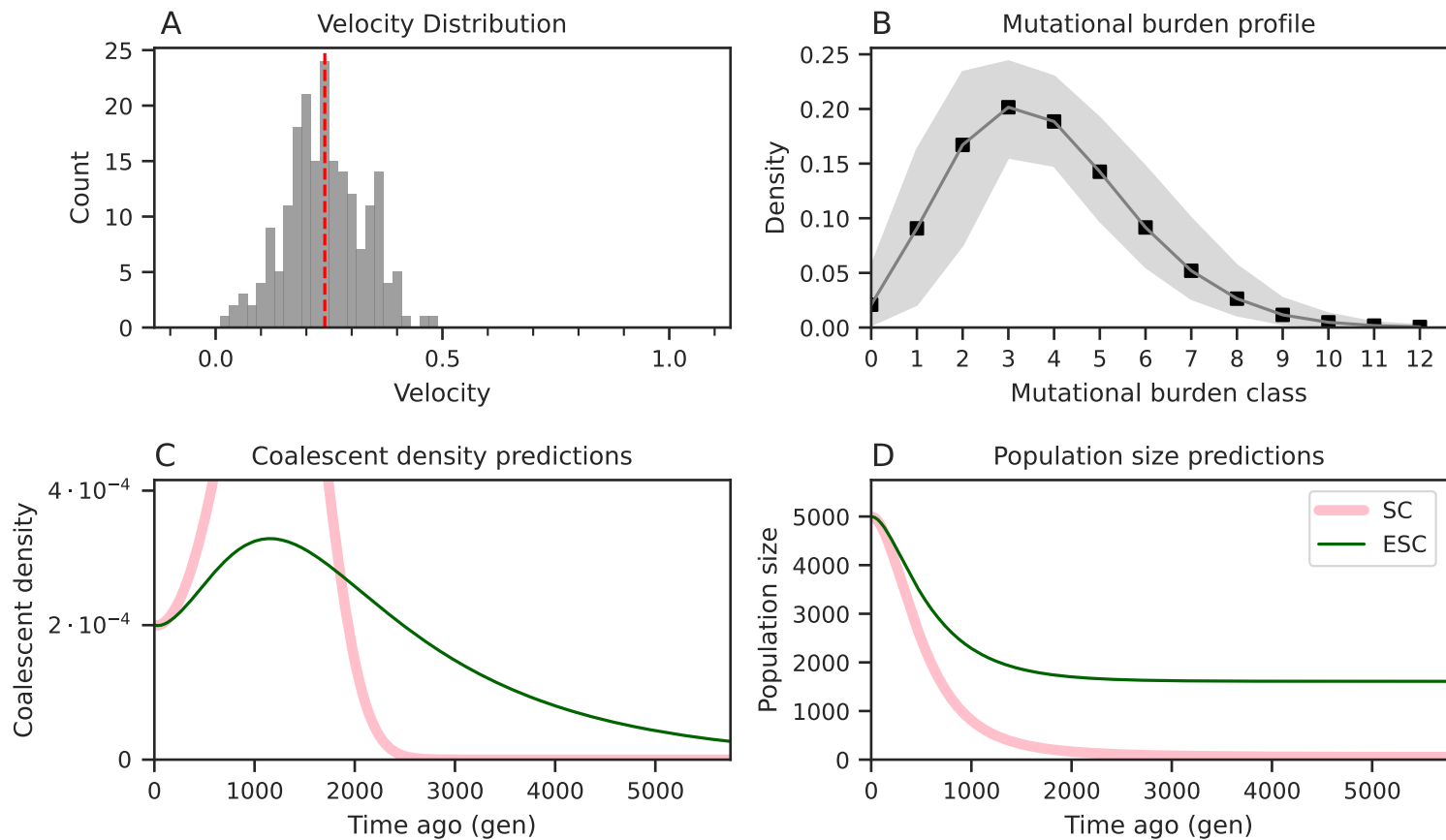
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00552$, $\phi = 0.02$
Mean Velocity= 0.2715 ± 0.0827 , $n = 200$ simulations



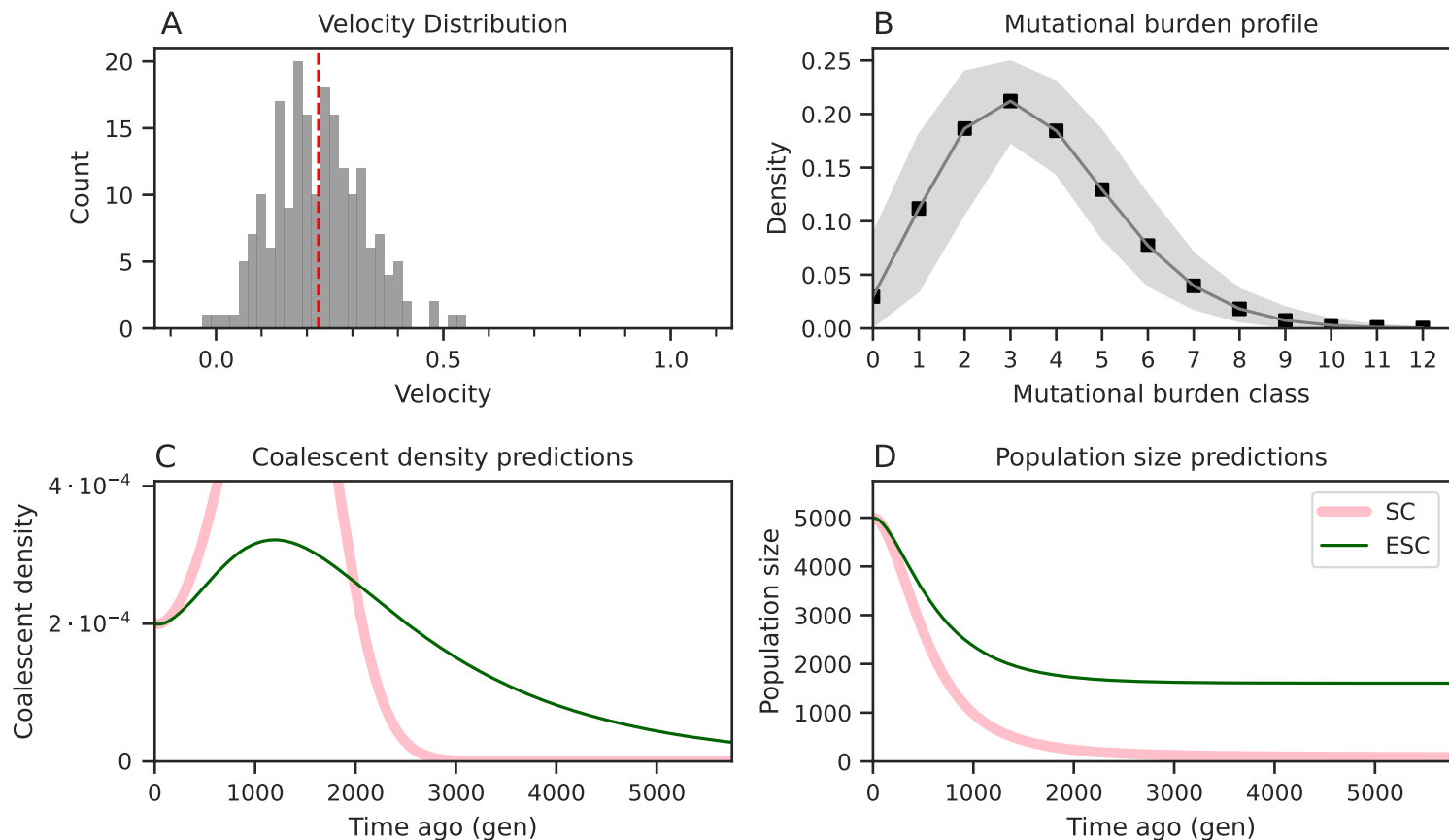
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00504$, $\phi = 0.0324$
Mean Velocity = 0.2516 ± 0.0893 , $n = 200$ simulations



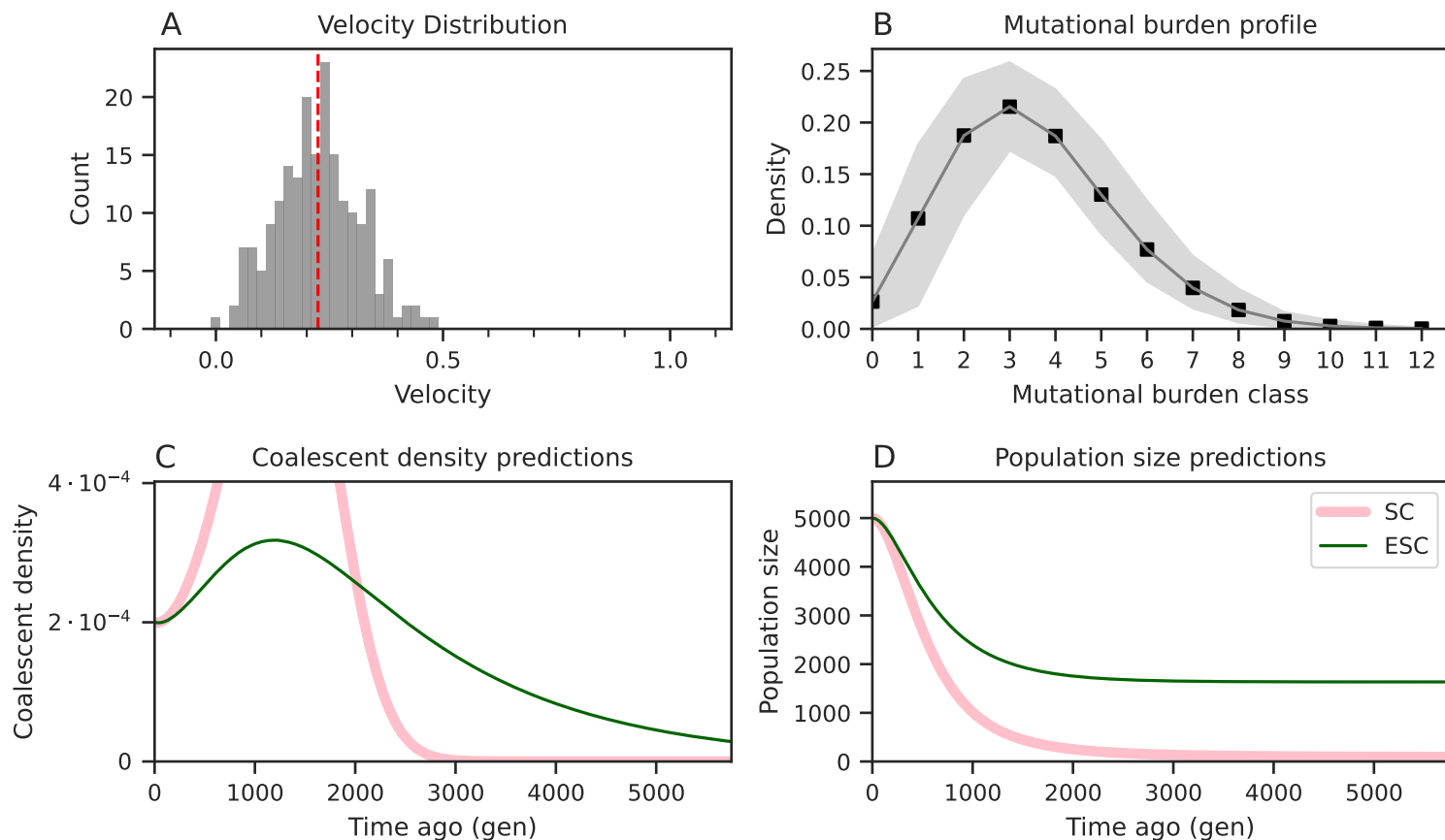
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00456$, $\phi = 0.0523$
Mean Velocity= 0.2407 ± 0.0864 , $n = 200$ simulations



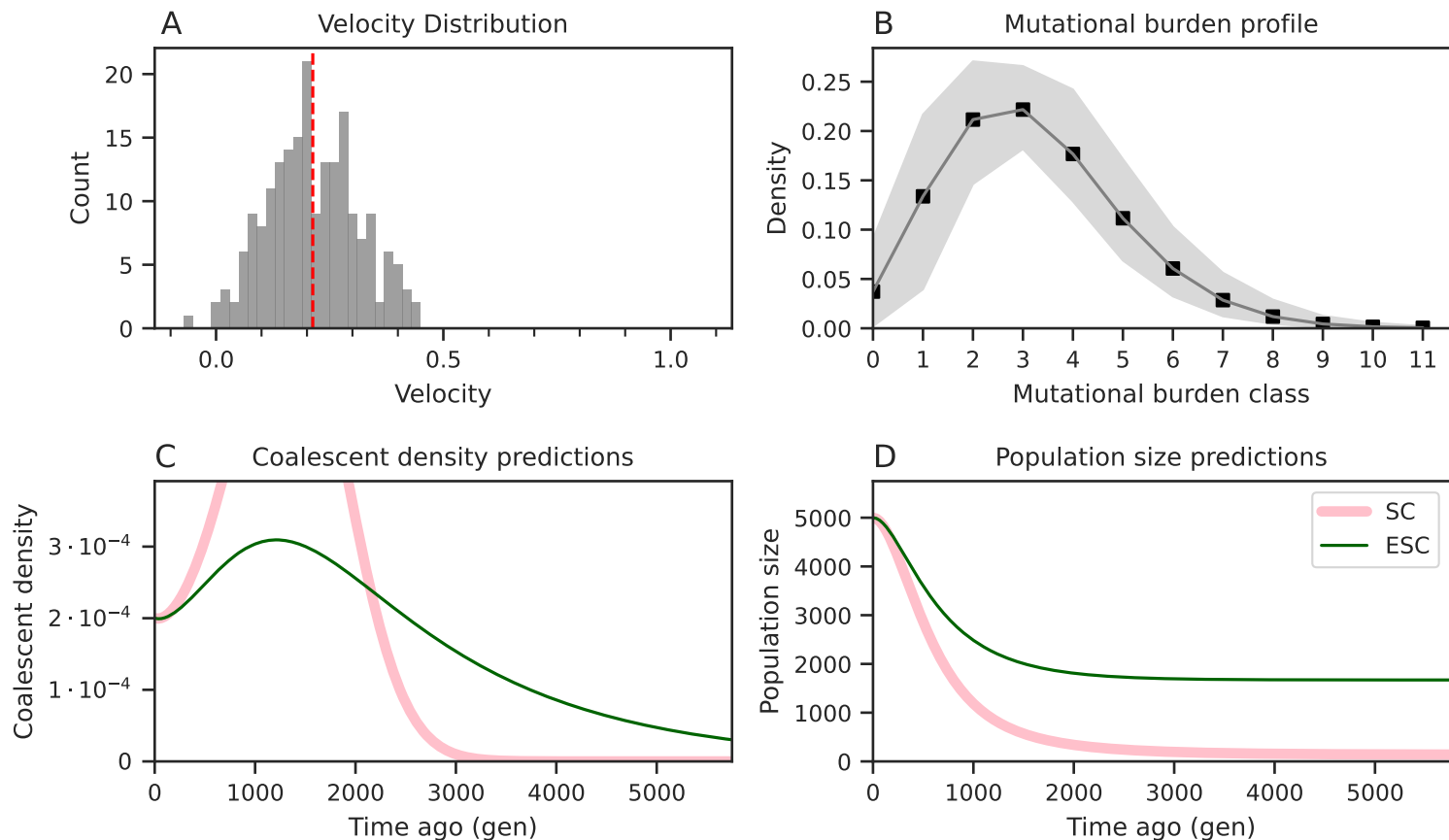
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00408$, $\phi = 0.0845$
Mean Velocity= 0.2255 ± 0.0991 , $n = 200$ simulations



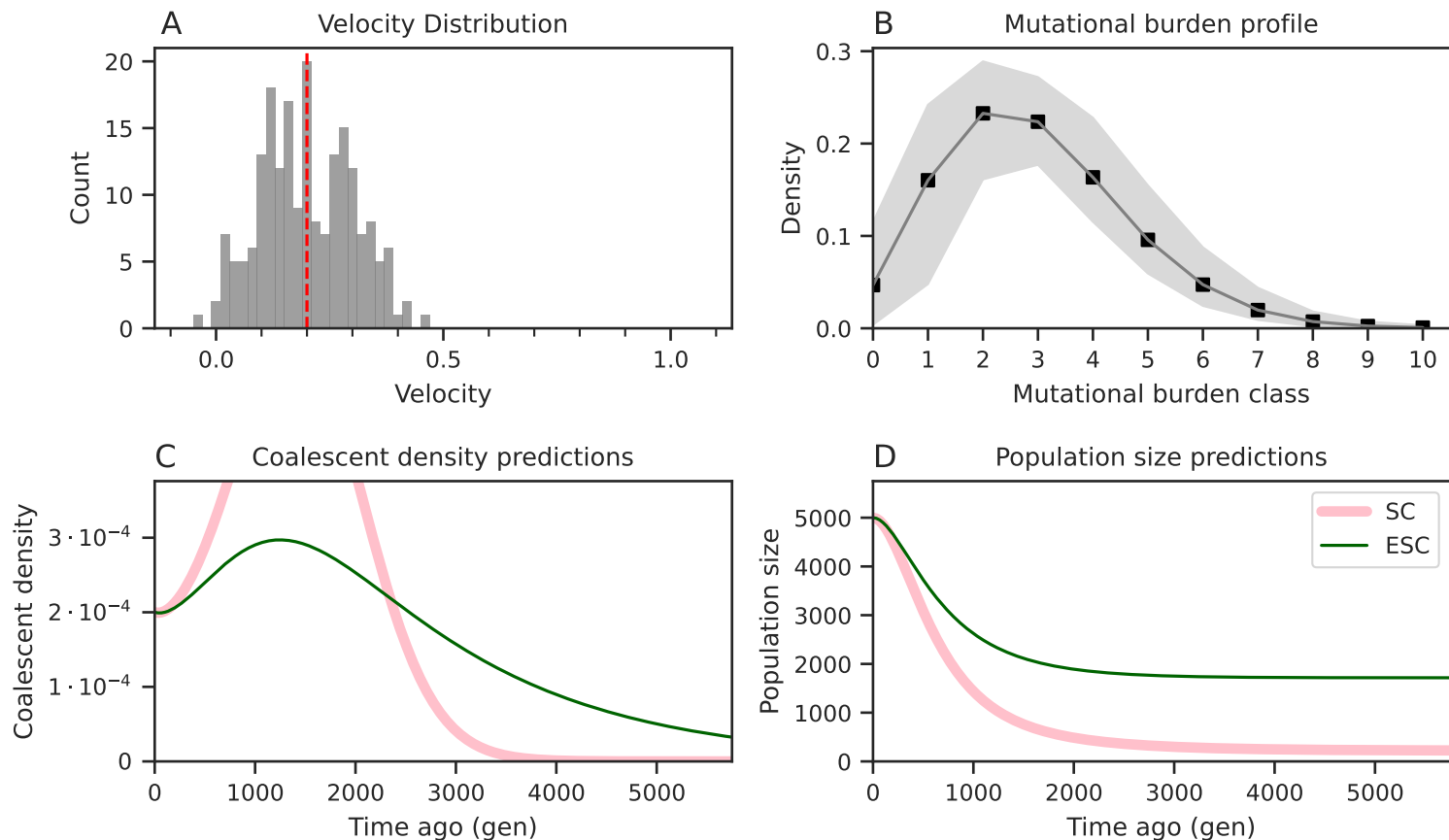
$N = 5e + 03$, $s = 0.001$, $U_d = 0.004$, $\phi = 0.0916$
Mean Velocity= 0.2249 ± 0.0917 , $n = 200$ simulations



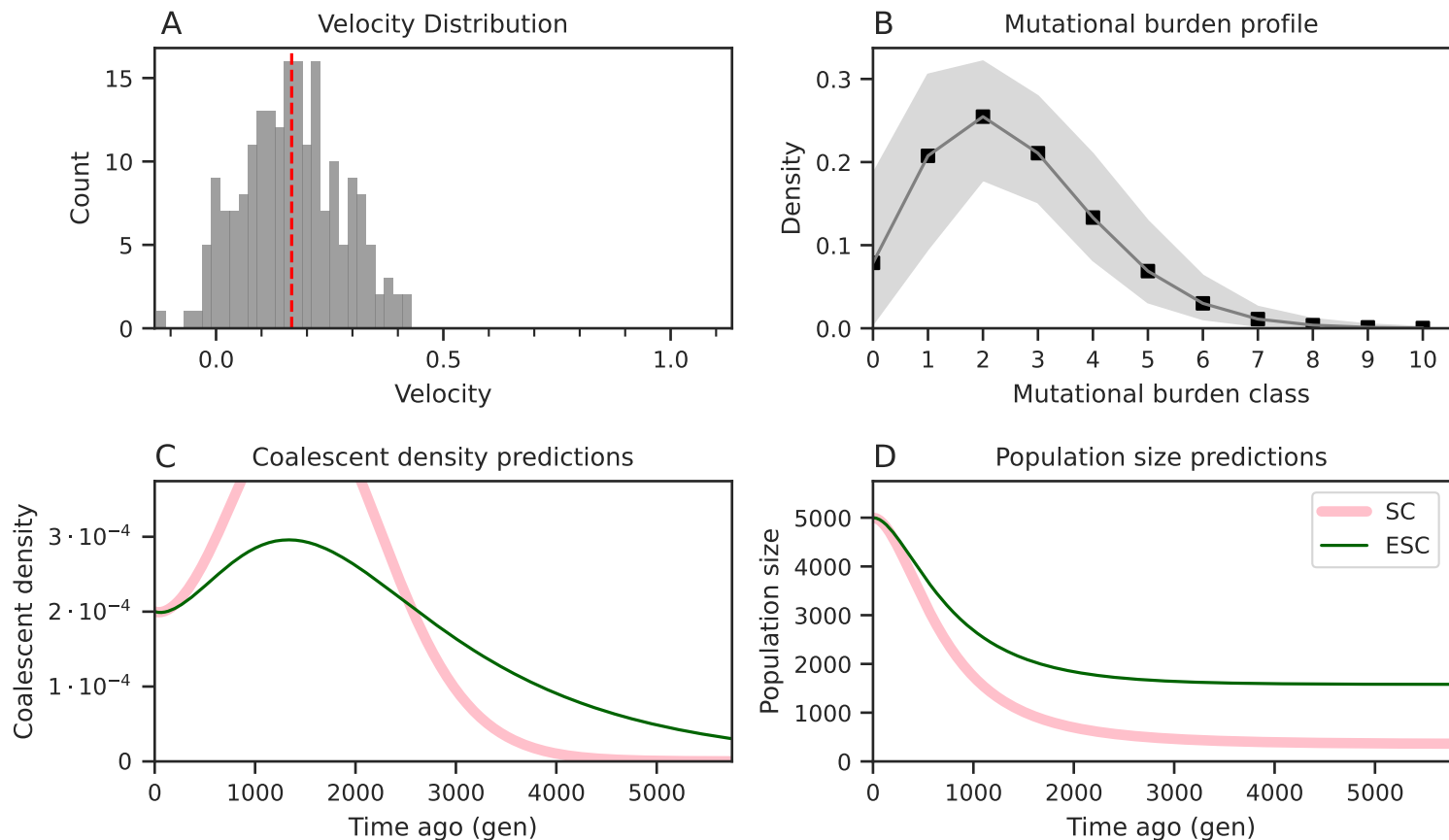
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0036$, $\phi = 0.137$
Mean Velocity= 0.2128 ± 0.0993 , $n = 200$ simulations



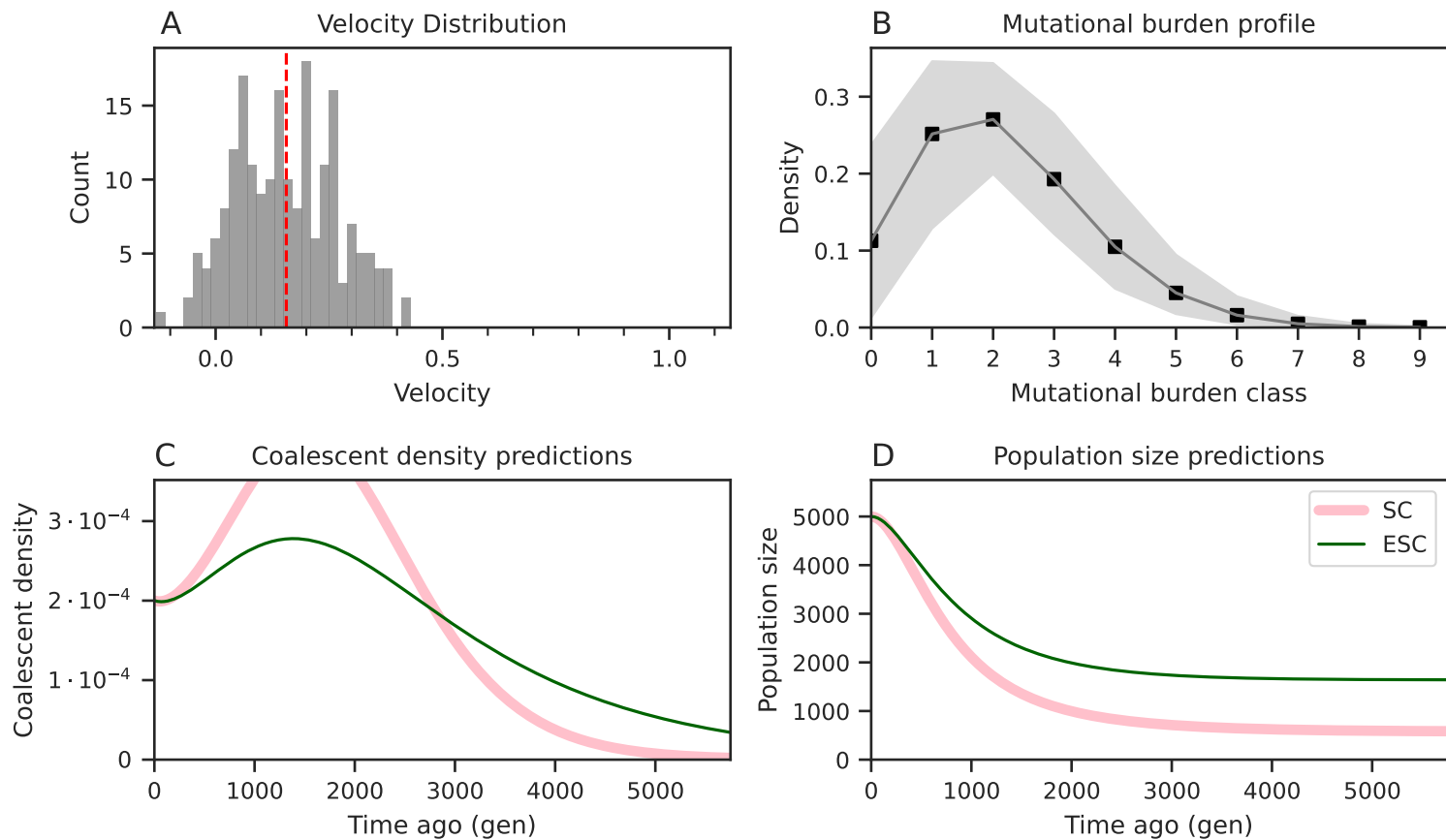
$N = 5e + 03$, $s = 0.001$, $U_d = 0.00312$, $\phi = 0.221$
Mean Velocity= 0.2000 ± 0.1011 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 0.00264$, $\phi = 0.357$
Mean Velocity= 0.1663 ± 0.1079 , $n = 200$ simulations

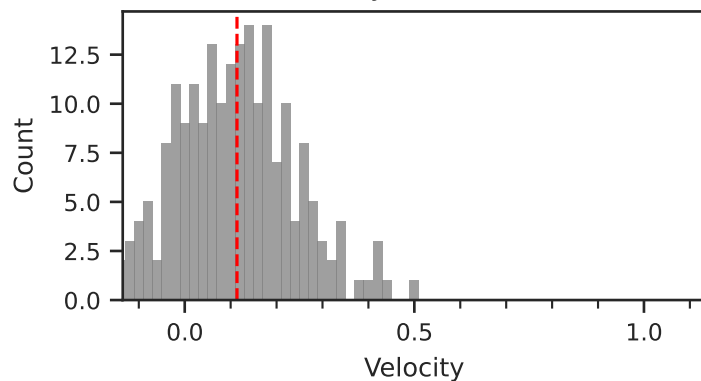


$N = 5e + 03$, $s = 0.001$, $U_d = 0.00216$, $\phi = 0.577$
Mean Velocity= 0.1559 ± 0.1125 , $n = 200$ simulations

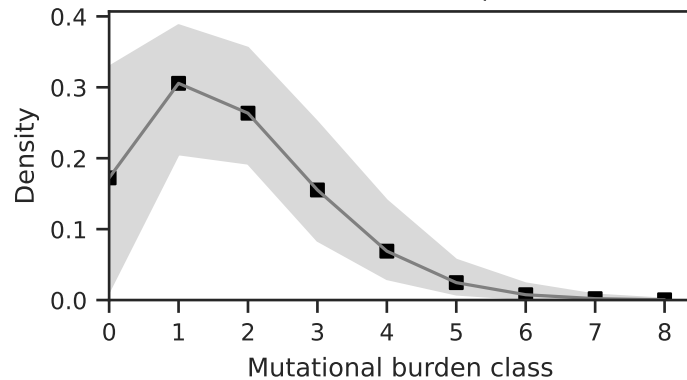


$N = 5e + 03$, $s = 0.001$, $U_d = 0.00168$, $\phi = 0.932$
Mean Velocity= 0.1138 ± 0.1246 , $n = 200$ simulations

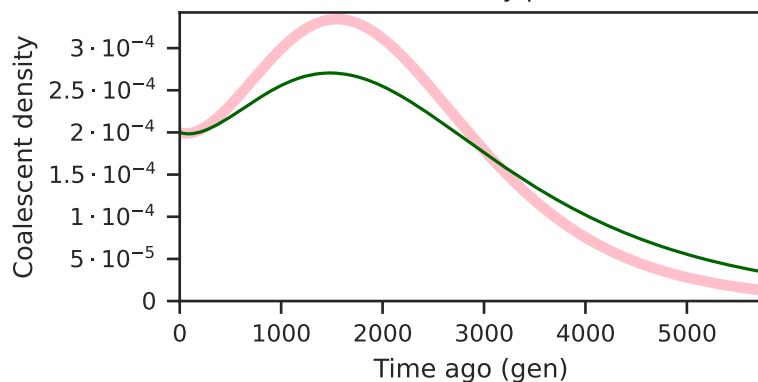
A Velocity Distribution



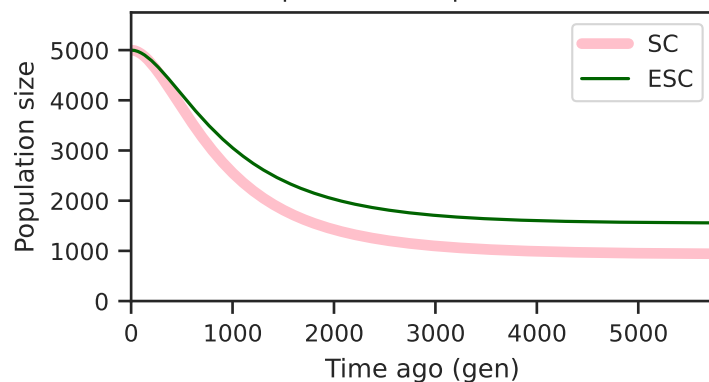
B Mutational burden profile



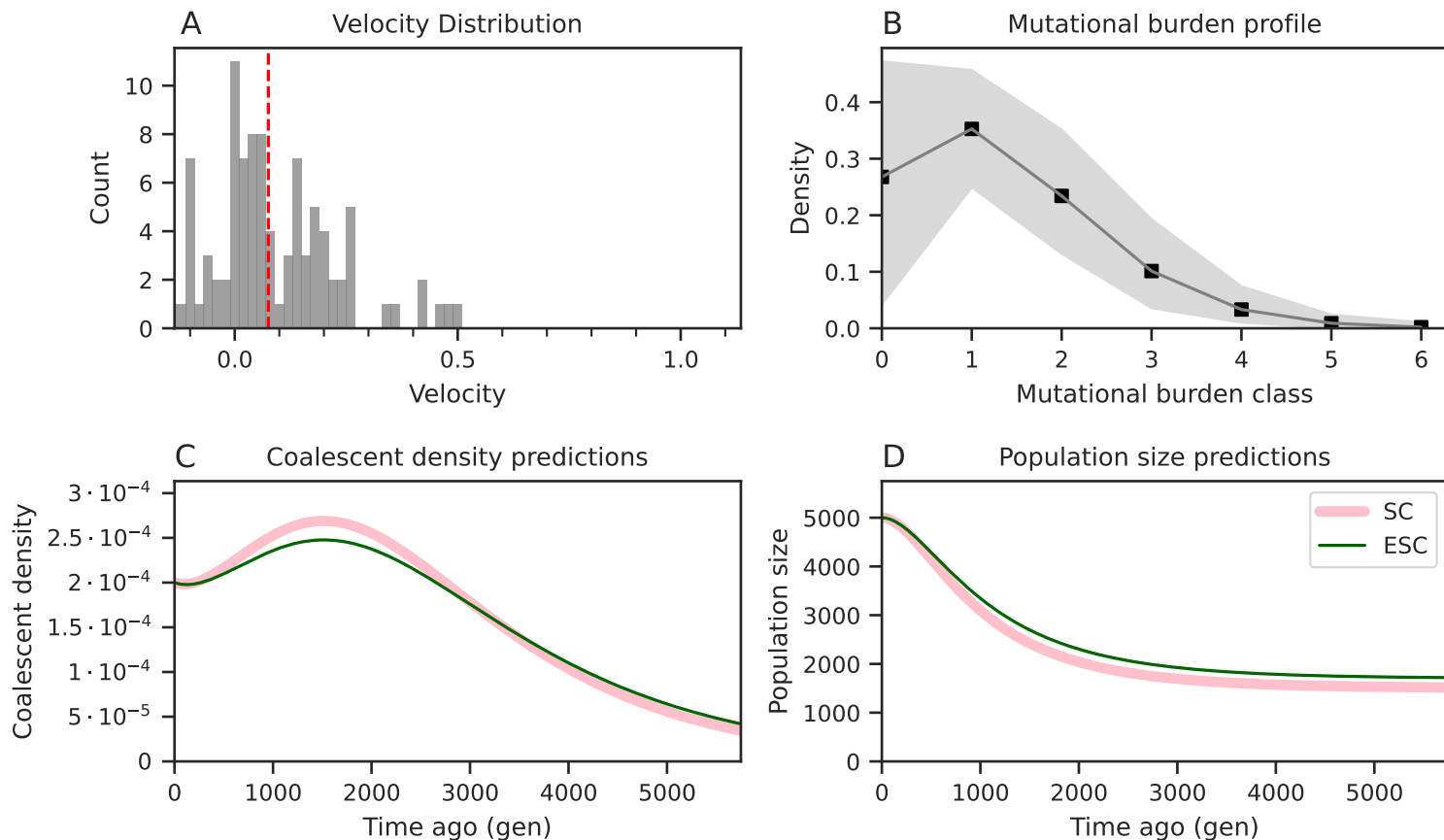
C Coalescent density predictions



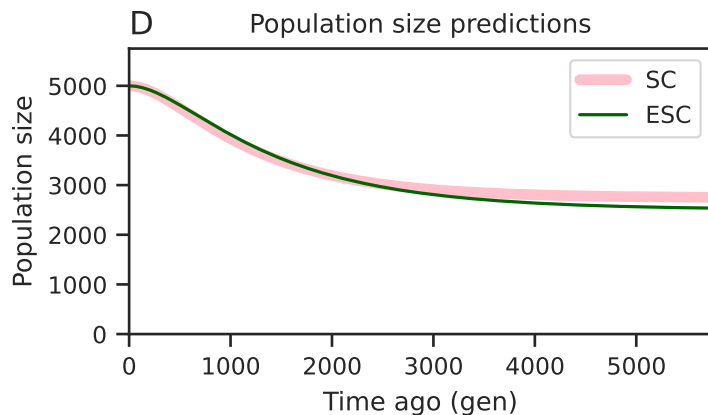
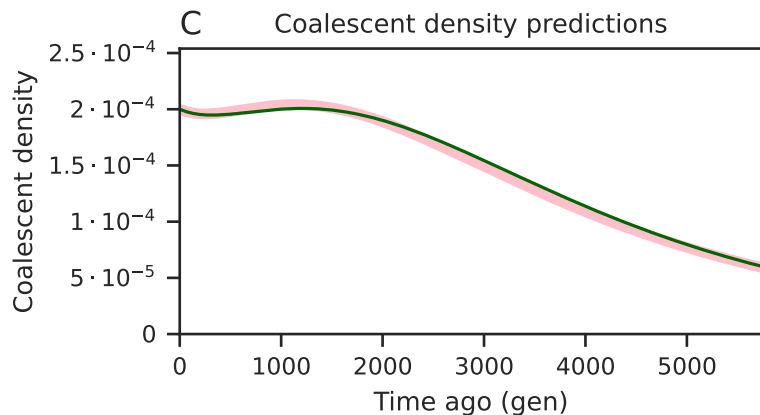
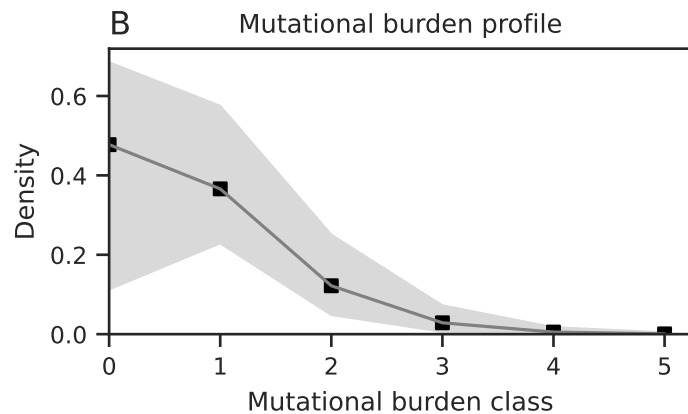
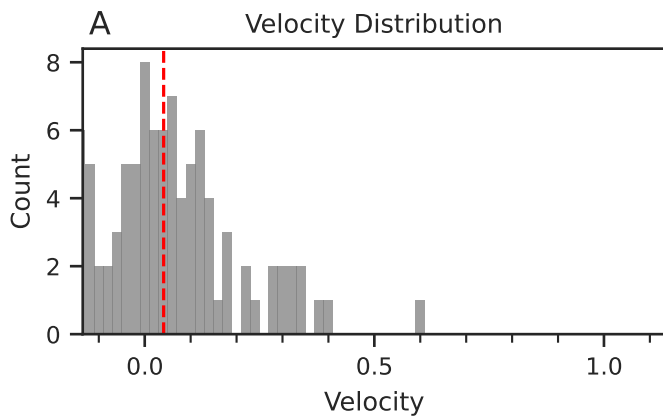
D Population size predictions



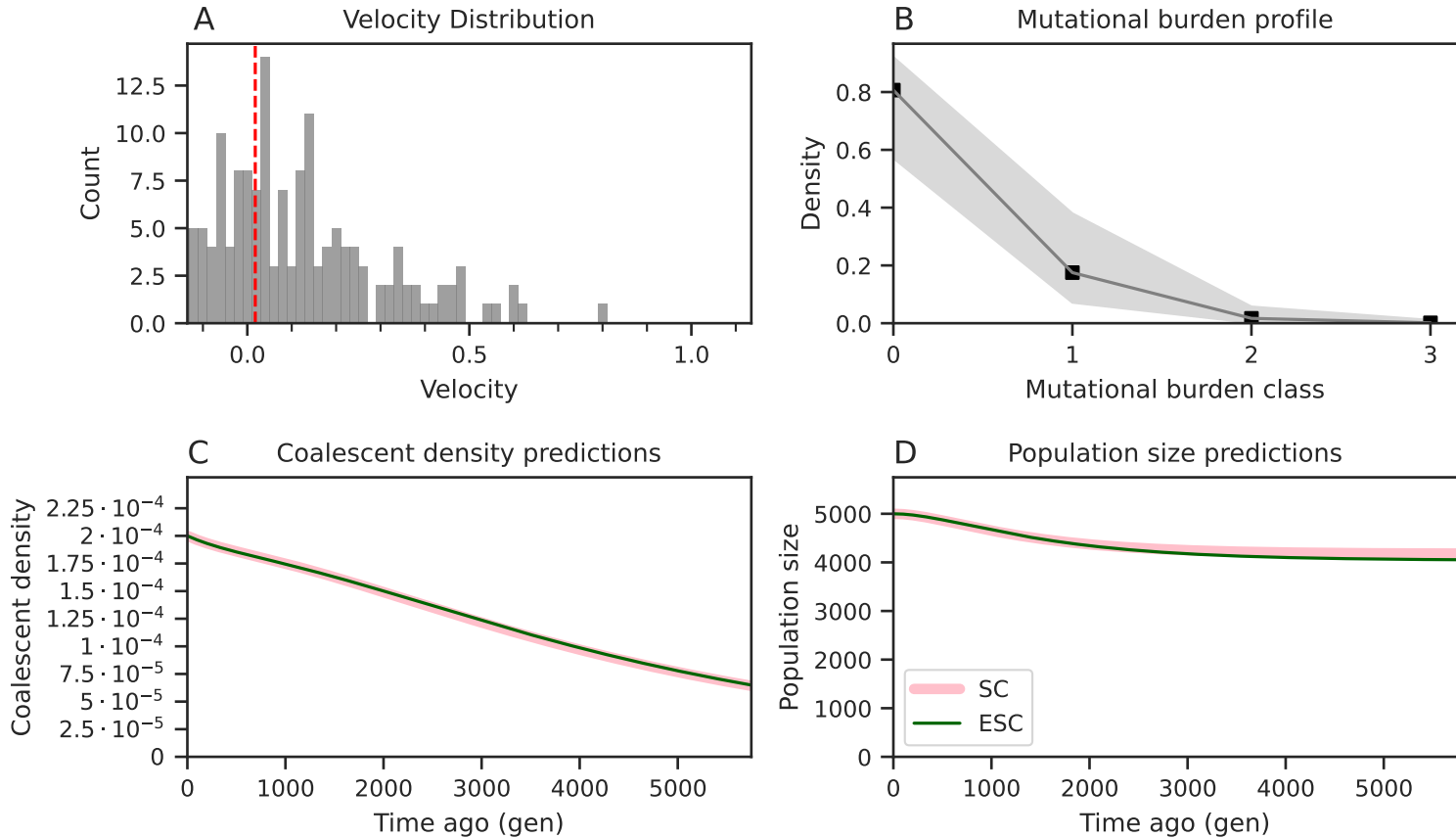
$N = 5e + 03$, $s = 0.001$, $U_d = 0.0012$, $\phi = 1.51$
Mean Velocity= 0.0754 ± 0.1516 , $n = 100$ simulations



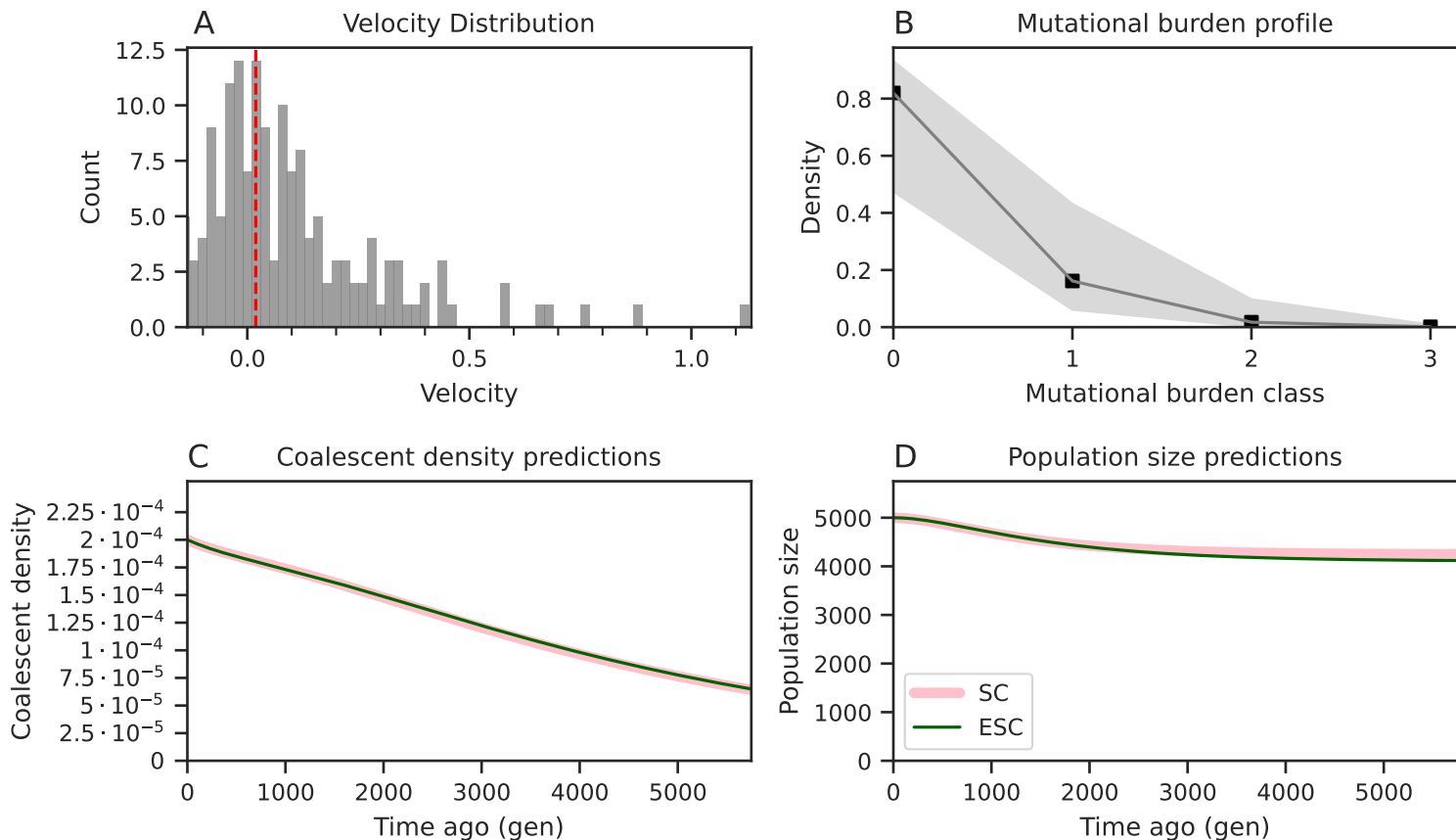
$N = 5e + 03$, $s = 0.001$, $U_d = 6 \cdot 10^{-4}$, $\phi = 2.74$
Mean Velocity= 0.0412 ± 0.1573 , $n = 100$ simulations



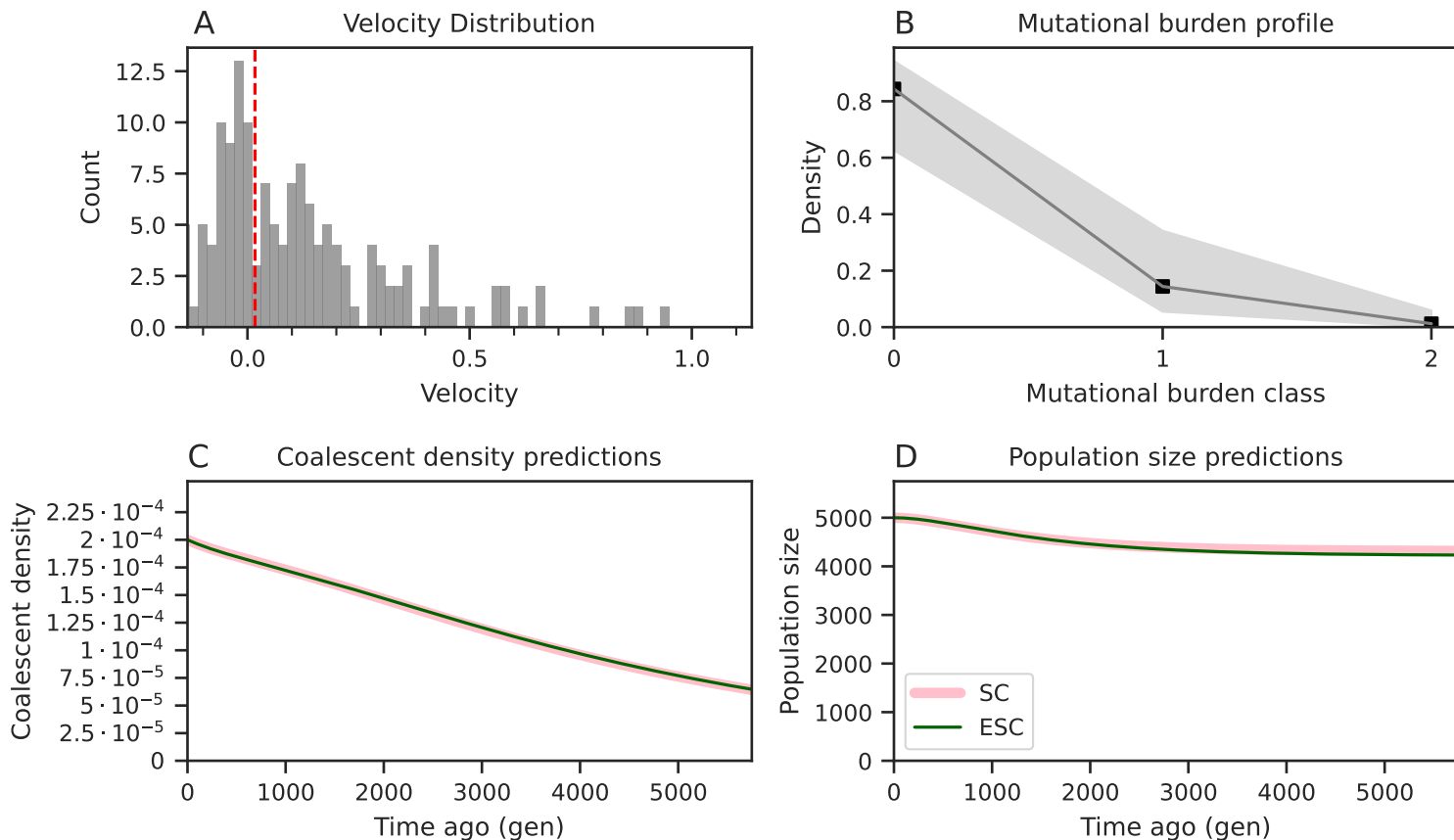
$N = 5e + 03$, $s = 0.001$, $U_d = 1.8 \cdot 10^{-4}$, $\phi = 4.18$
Mean Velocity= 0.0179 ± 0.2851 , $n = 200$ simulations



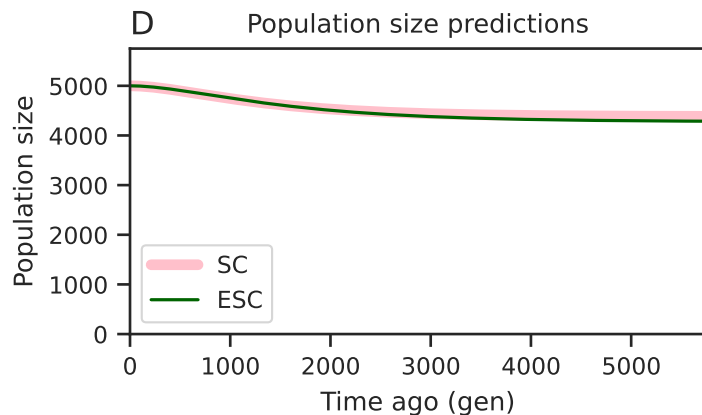
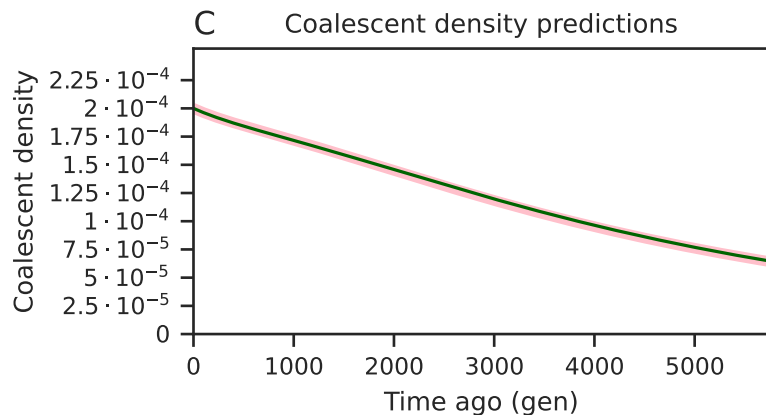
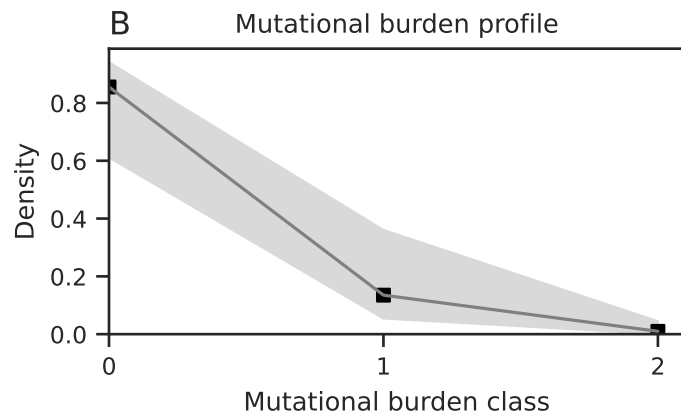
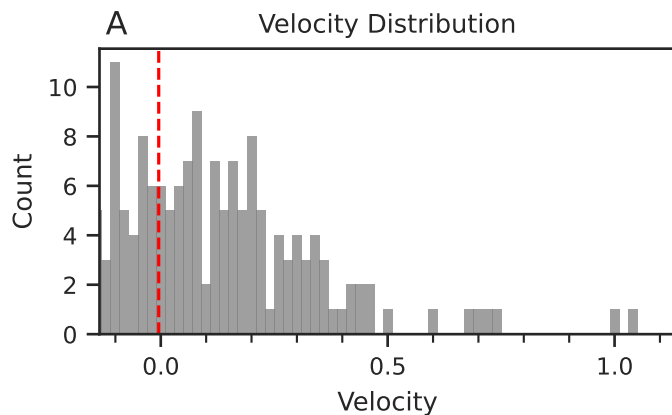
$N = 5e + 03$, $s = 0.001$, $U_d = 1.64 \cdot 10^{-4}$, $\phi = 4.24$
Mean Velocity = 0.0190 ± 0.2858 , $n = 200$ simulations



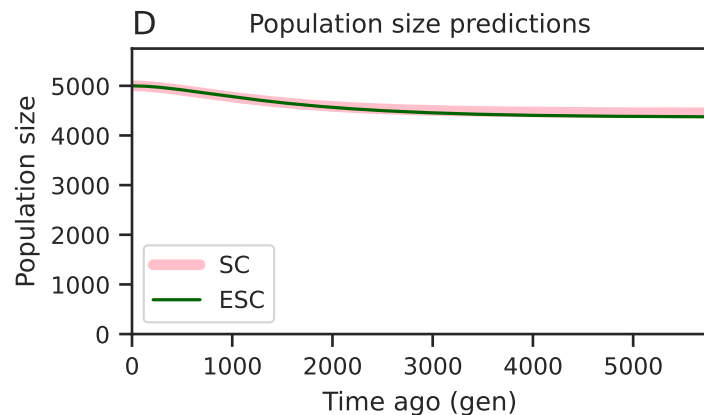
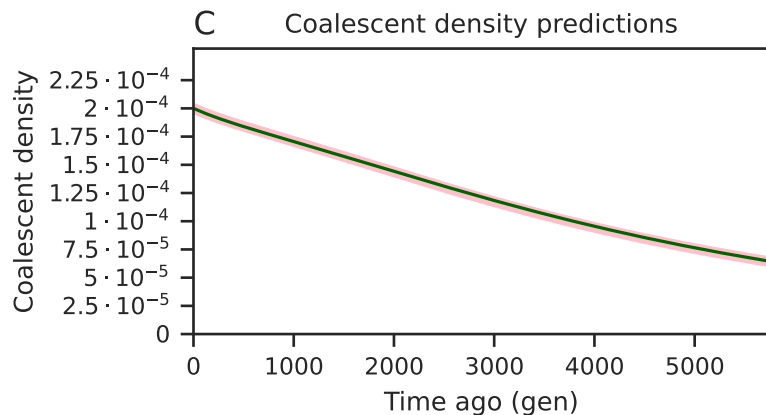
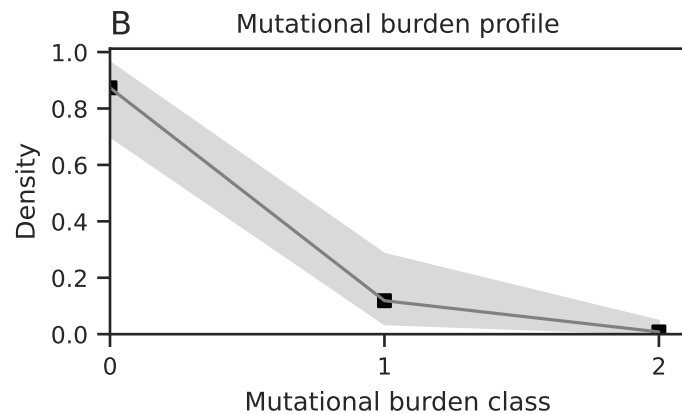
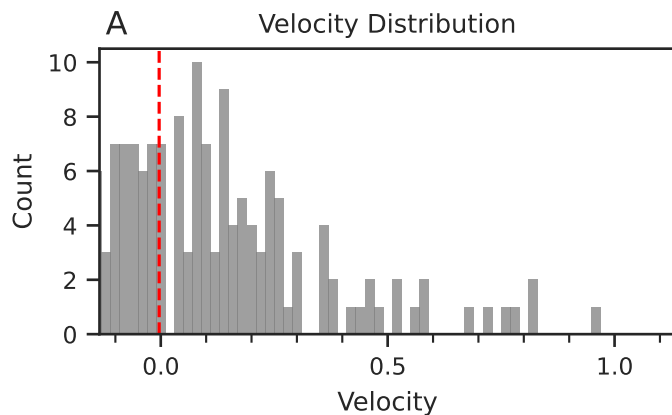
$N = 5e + 03$, $s = 0.001$, $U_d = 1.48 \cdot 10^{-4}$, $\phi = 4.31$
Mean Velocity= 0.0170 ± 0.3004 , $n = 200$ simulations



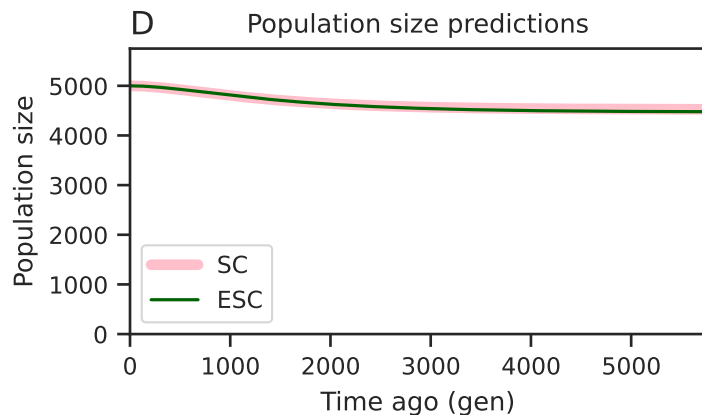
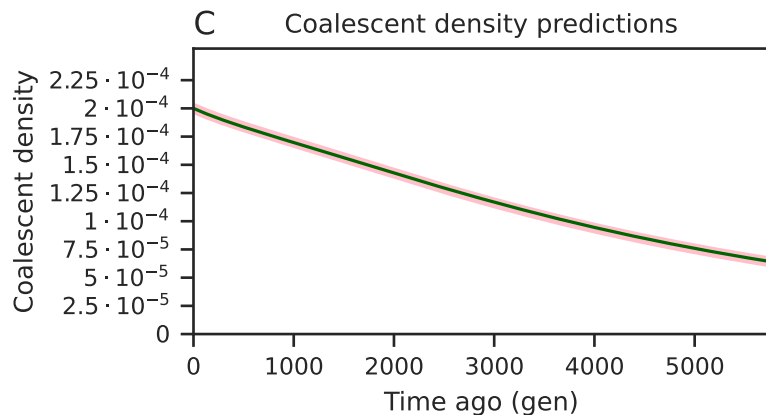
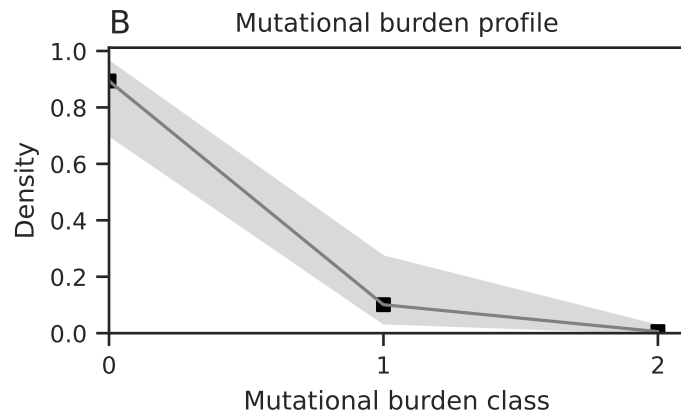
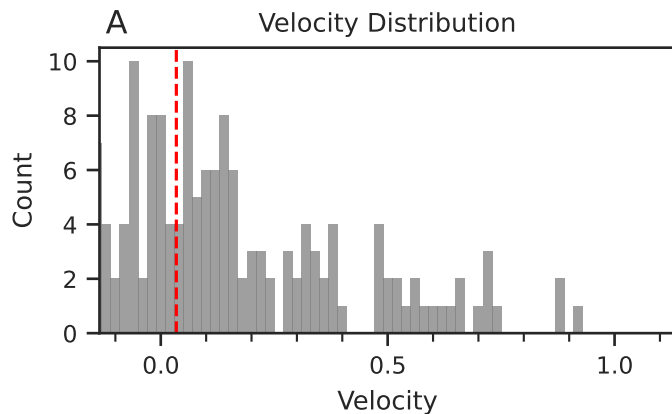
$N = 5e + 03$, $s = 0.001$, $U_d = 1.32 \cdot 10^{-4}$, $\phi = 4.38$
Mean Velocity = -0.0045 ± 0.3466 , $n = 200$ simulations



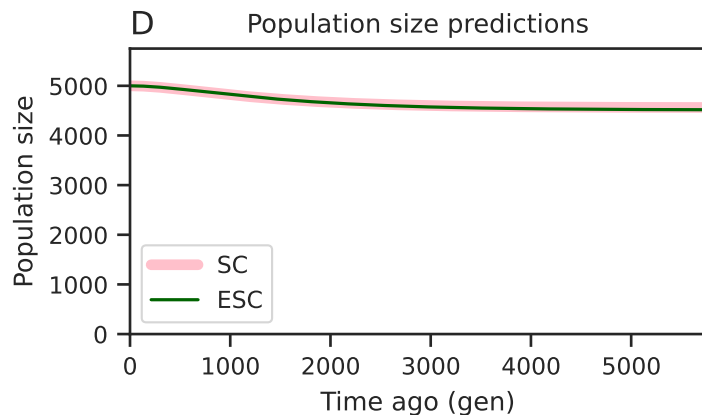
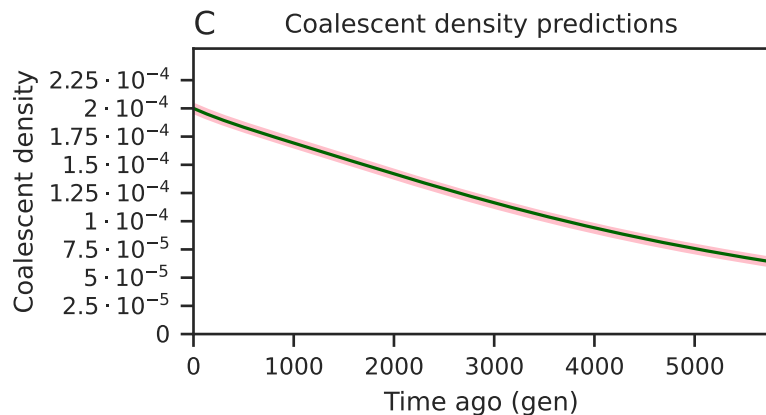
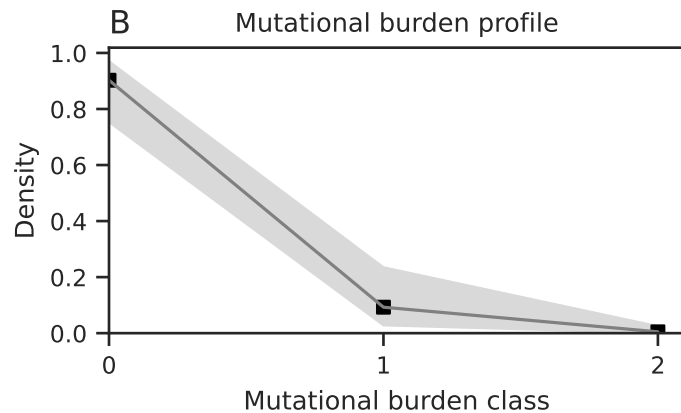
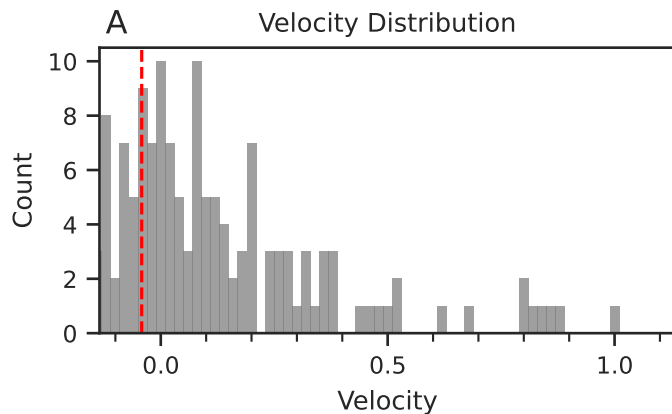
$N = 5e + 03$, $s = 0.001$, $U_d = 1.16 \cdot 10^{-4}$, $\phi = 4.45$
Mean Velocity = -0.0035 ± 0.3156 , $n = 200$ simulations



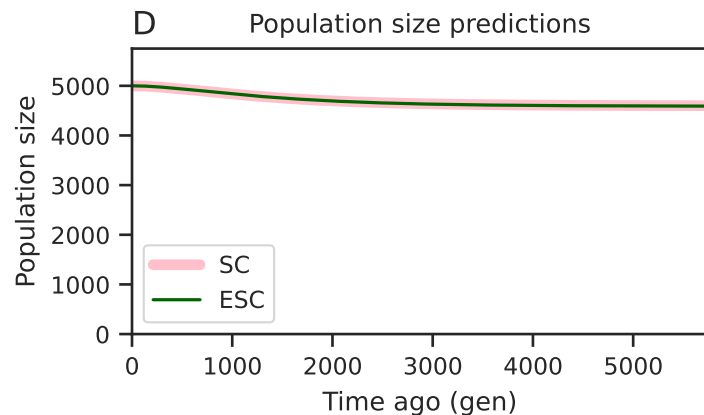
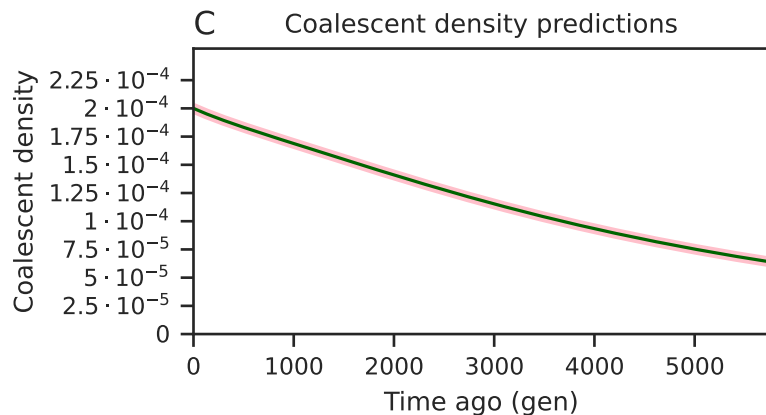
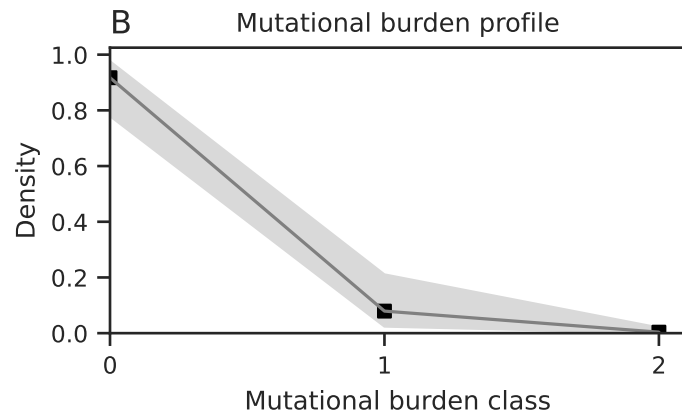
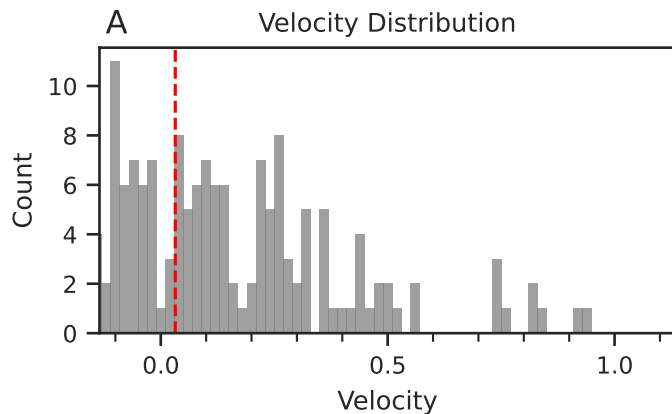
$N = 5e + 03$, $s = 0.001$, $U_d = 1 \cdot 10^{-4}$, $\phi = 4.52$
Mean Velocity= 0.0343 ± 0.3900 , $n = 200$ simulations



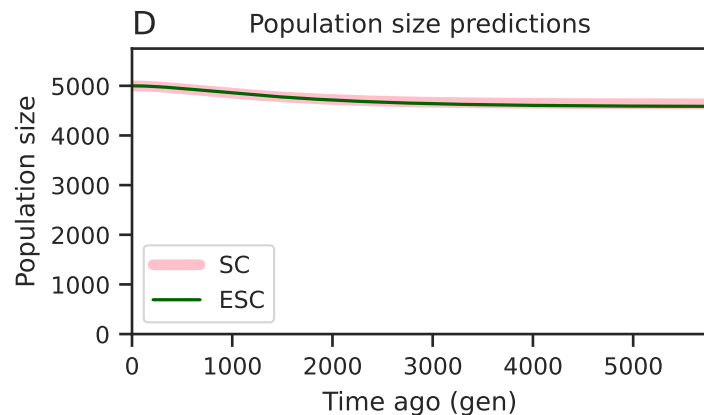
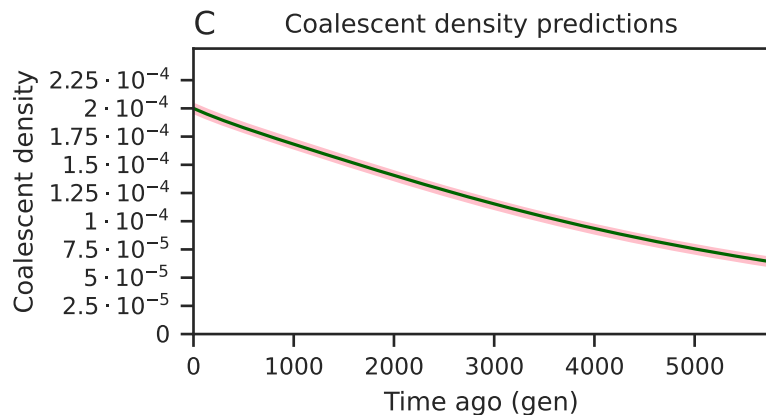
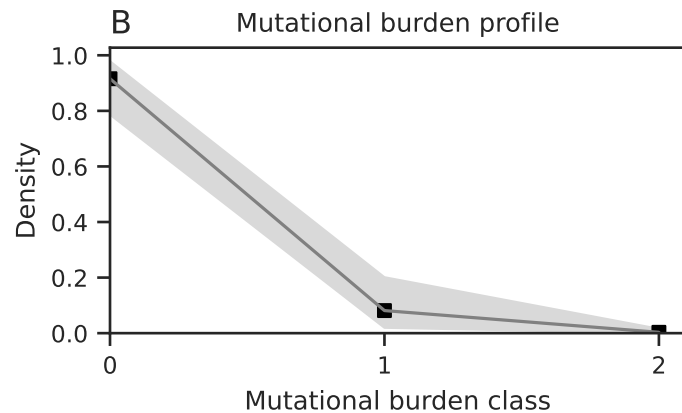
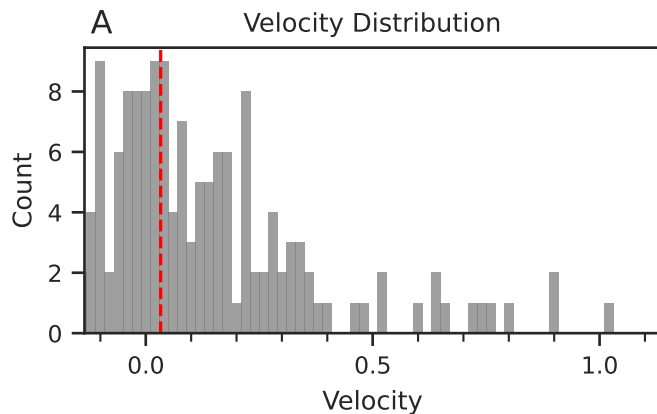
$N = 5e + 03$, $s = 0.001$, $U_d = 9.2 \cdot 10^{-5}$, $\phi = 4.56$
Mean Velocity = -0.0420 ± 0.3696 , $n = 200$ simulations



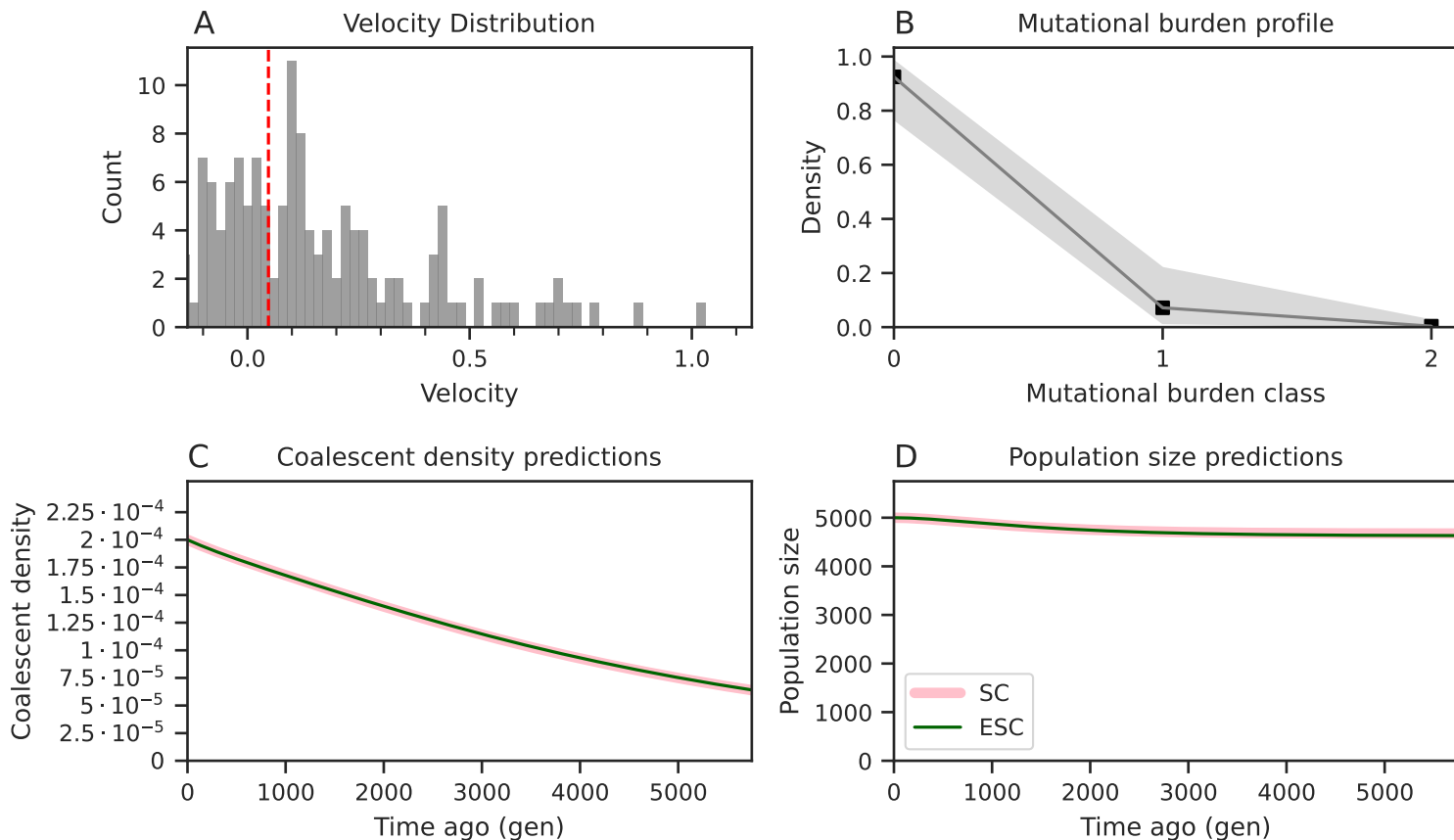
$N = 5e + 03$, $s = 0.001$, $U_d = 8.4 \cdot 10^{-5}$, $\phi = 4.6$
Mean Velocity= 0.0320 ± 0.3530 , $n = 200$ simulations



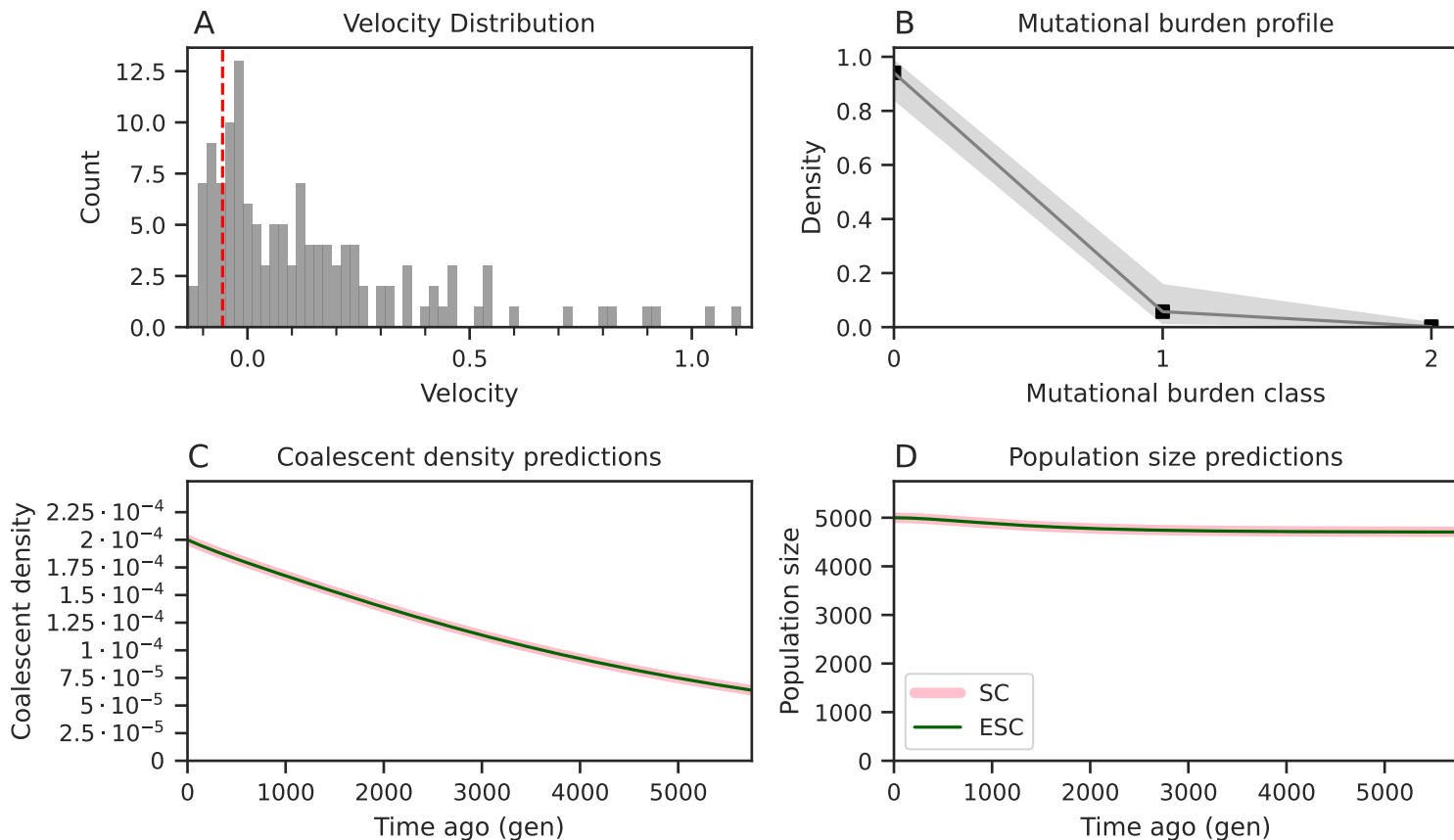
$N = 5e + 03$, $s = 0.001$, $U_d = 7.6 \cdot 10^{-5}$, $\phi = 4.63$
Mean Velocity= 0.0328 ± 0.4263 , $n = 200$ simulations



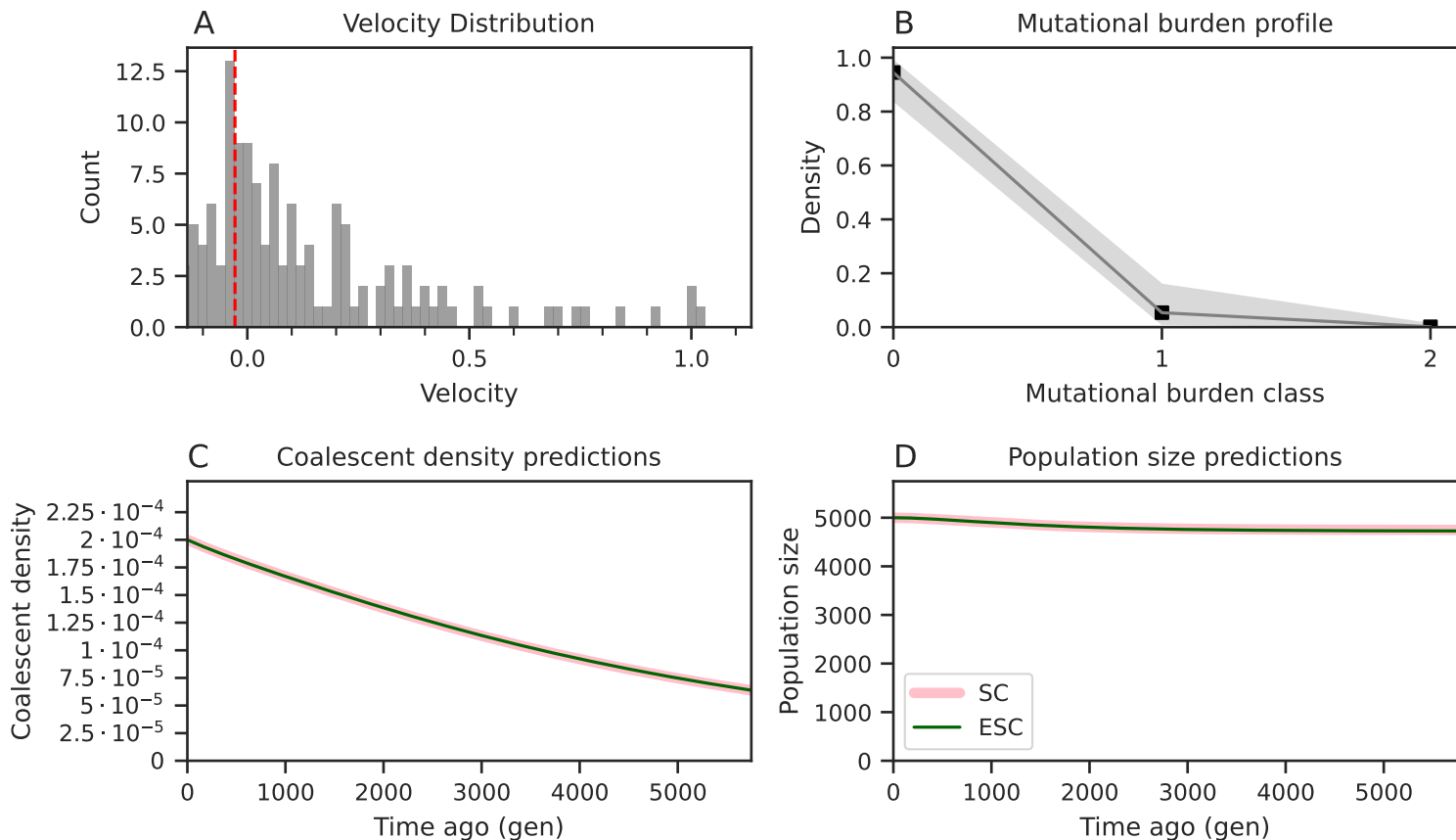
$N = 5e + 03$, $s = 0.001$, $U_d = 6.8 \cdot 10^{-5}$, $\phi = 4.67$
Mean Velocity= 0.0472 ± 0.4814 , $n = 200$ simulations



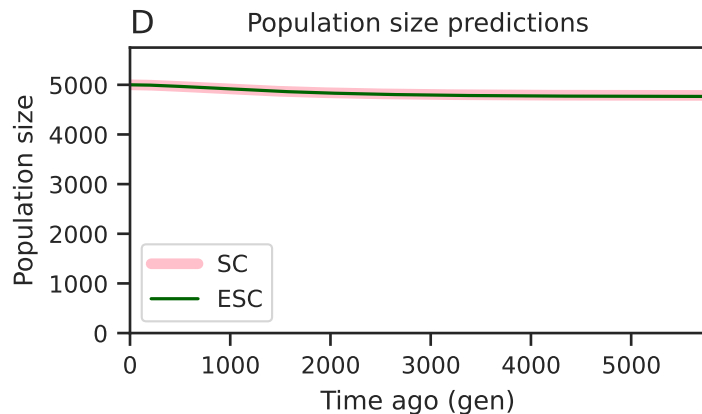
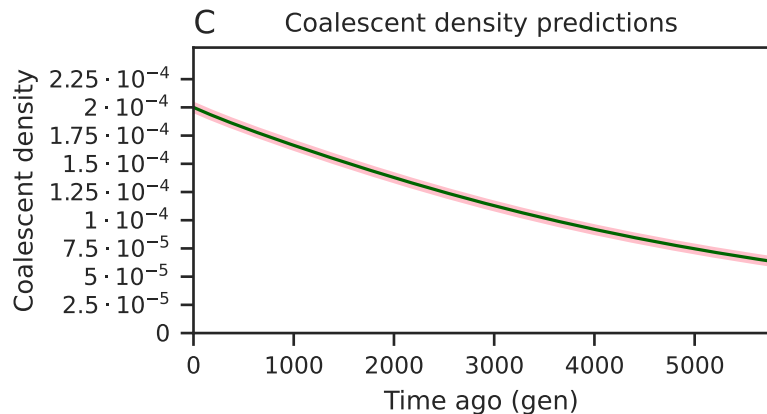
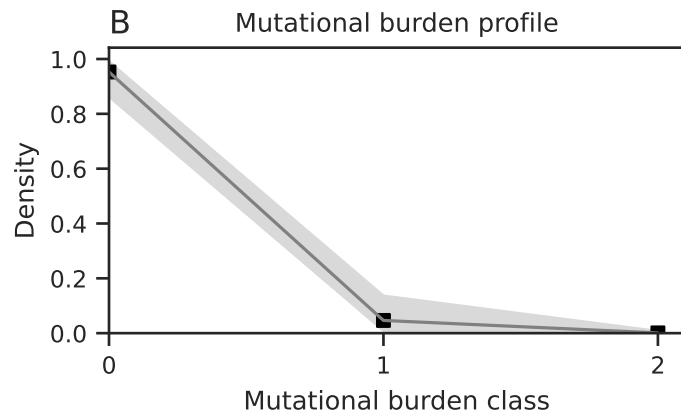
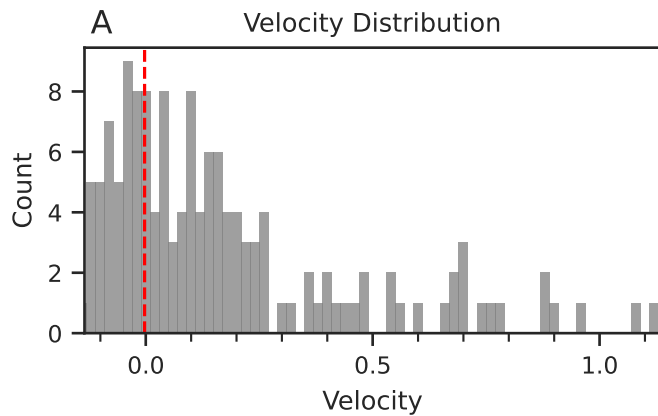
$N = 5e + 03$, $s = 0.001$, $U_d = 6 \cdot 10^{-5}$, $\phi = 4.71$
Mean Velocity = -0.0561 ± 0.4272 , $n = 200$ simulations



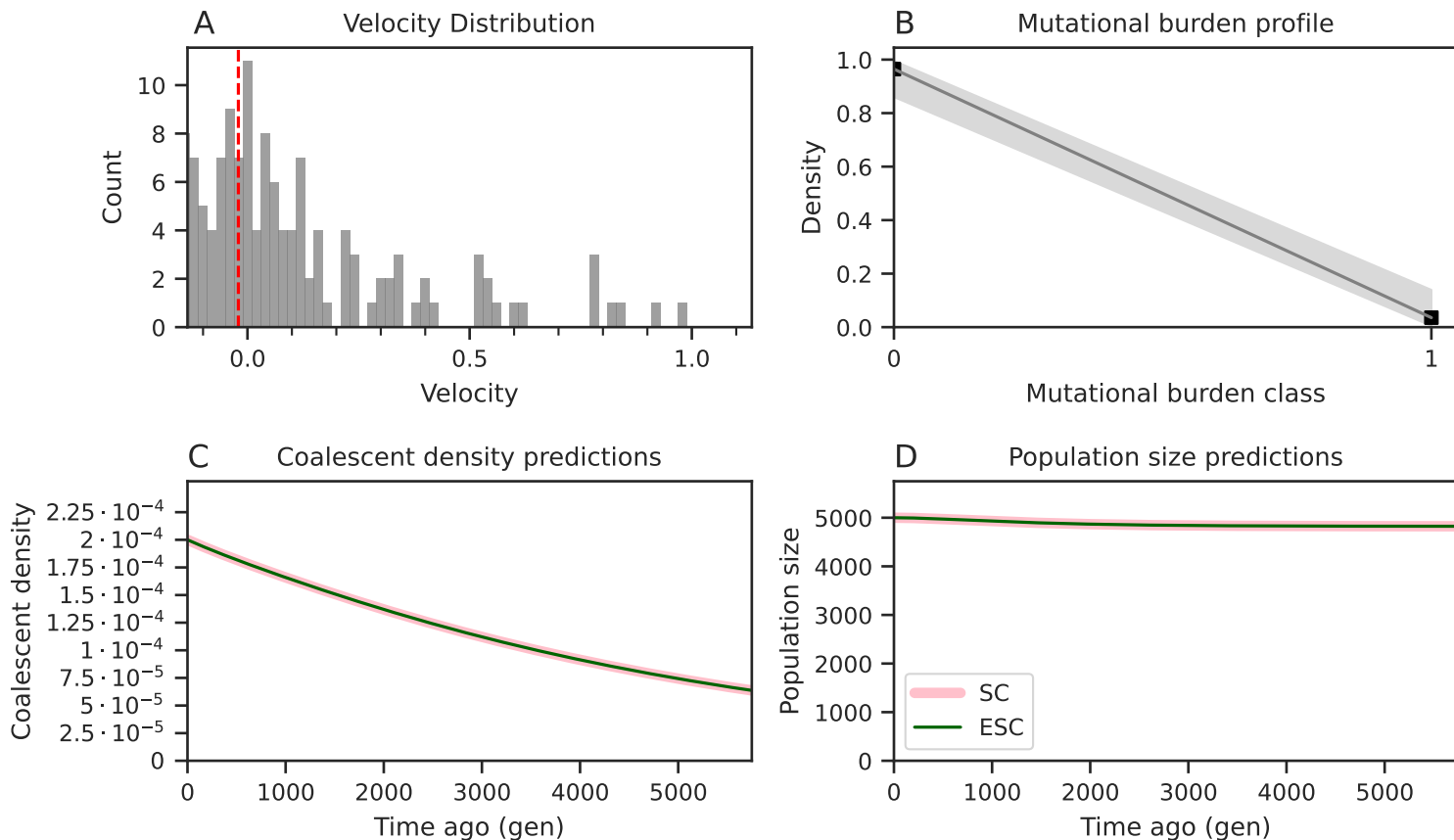
$N = 5e + 03$, $s = 0.001$, $U_d = 5.2 \cdot 10^{-5}$, $\phi = 4.75$
Mean Velocity = -0.0278 ± 0.5077 , $n = 200$ simulations



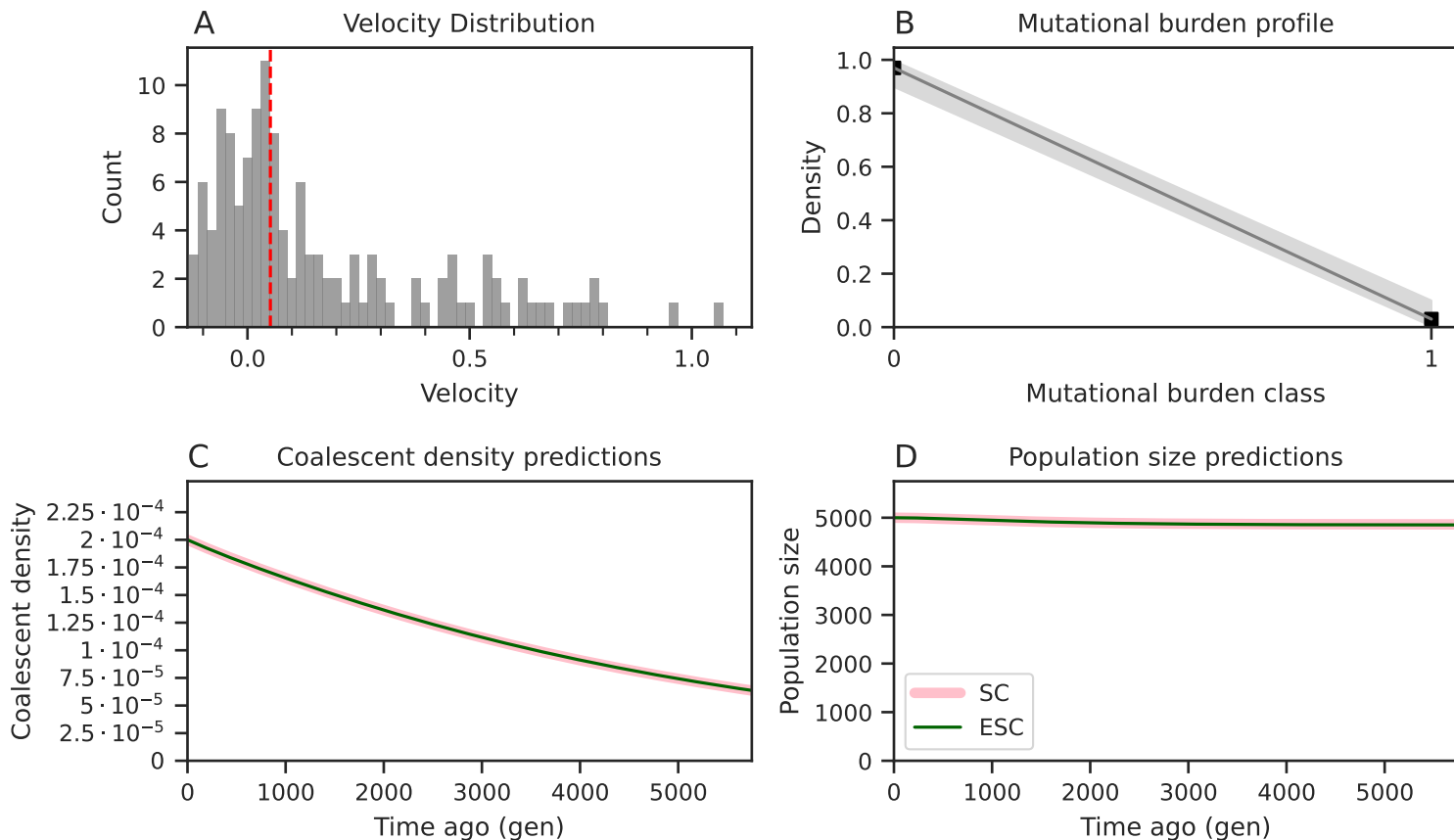
$N = 5e + 03$, $s = 0.001$, $U_d = 4.4 \cdot 10^{-5}$, $\phi = 4.78$
Mean Velocity = -0.0026 ± 0.5251 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 3.6 \cdot 10^{-5}$, $\phi = 4.82$
Mean Velocity = -0.0205 ± 0.5556 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 2.8 \cdot 10^{-5}$, $\phi = 4.86$
Mean Velocity= 0.0515 ± 0.7246 , $n = 200$ simulations



$N = 5e + 03$, $s = 0.001$, $U_d = 2 \cdot 10^{-5}$, $\phi = 4.9$
Mean Velocity = -0.0474 ± 0.7796 , $n = 200$ simulations

